

Planning exercises

worked step by step · Chapters 0–5

Every essential (★) and recommended (☆) exercise of the course planning for Chapters 0–5, written out in full: what is given, what has to be shown, the plan, every step with the rule or definition that justifies it, and a checklist to compare with your own attempt.

- **Source.** C. Newstead, *An Infinite Descent into Pure Mathematics*, Utrecht adaptation (29 July 2026). All numbers (Definition 0.36, Strategy 1.1.28, Theorem 0.42, ...) are the book's. Nothing is used that the book has not stated before the exercise, except the elementary algebra the book takes for granted.
- **Which exercises.** Exactly the exercises flagged ★ essential and ☆ recommended in the course planning (plus the three bonus ones), in book order. The companion *Workbook Ch. 0–5* contains the theory and shorter solutions of *all* exercises.
- **How to use it.** Do the exercise yourself first. Then read the solution top to bottom and tick the *Compare with your work* list: it names the sentences a complete answer must contain and the places where marks are usually lost.

How each solution is laid out

Given.	Every assumption, including the silent ones: what kind of object each letter is, and what the hypotheses mean once their definitions are unfolded.
To show.	The goal, often rewritten in symbols, with its <i>logical shape</i> (an implication, a “for all”, an “if and only if”, a negation, ...). The shape decides the first sentence of the proof.
Plan.	The strategy from the book that fits the shape, and the one idea that makes the proof work.
Step 1, Step 2, ...	The proof, one move per step. Grey text in brackets, like [Definition 0.36], names the definition, strategy, theorem or algebra rule that justifies the move. Variables are introduced the moment they appear.
Conclusion.	The last sentence, closing every open “suppose” and “let”, followed by $\square$.
Compare with your work.	What a complete answer must contain, and the typical mistake for this exercise.

Conventions of the book, used throughout

- $\mathbb{N} = \{0, 1, 2, \dots\}$ (Definition 0.6); $[n] = \{k \in \mathbb{N} \mid k < n\}$ (Definition 0.25); intervals consist of real numbers (Definition 0.28).
- “ b divides a ”, written $b \mid a$, means $a = qb$ for some $q \in \mathbb{Z}$ (Definition 0.36). It is a statement about multiplication: never write the fraction a/b in a divisibility proof.
- “Even” means divisible by 2; “odd” means not even (Definition 0.41). That an odd a equals $2b + 1$ is Proposition 1.1.58, to be quoted.
- Division theorem (Theorem 0.42): for $b \neq 0$ there is *exactly one* pair $q, r \in \mathbb{Z}$ with $a = qb + r$ and $0 \leq r < |b|$. The word “exactly one” is what lets you identify a remainder.
- Truth tables use $\checkmark$ and $\times$; $\perp$ is a contradiction; $\equiv$ is logical equivalence (a statement *about* formulae).

- In every proof the *reader* chooses the arbitrary element (“let $x \in X$ ”); you choose the witness of an existential (“define $n = \dots$ ”).

The flags

★ essential in the planning ★ recommended bonus = marked bonus in the planning. Quiz 1 (23 Sep): numbers and §1.1. Quiz 2 (7 Oct): sets and functions. Quiz 3 (21 Oct): relations and modular arithmetic. Exam 4 Nov.

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Chapter 0

Getting started

Contents

- A Propositions and number sets
- B Sets and set-builder notation
- C Division of integers
- D Irrational numbers
- E Chapter 0 exercises (0.E)

Planning

- ★ 0.2, 0.21, 0.24, 0.40; 0.E.3, 4, 22, 27, 28
 - ★ 0.14, 0.30, 0.38, 0.45, 0.51; 0.E.17, 18, 29, 30
- Tested in **Quiz 1** (23 Sep) together with §1.1.

A PROPOSITIONS AND NUMBER SETS

What you unfold in this section

Proposition (Definition 0.1): a statement that is either true or false. **Floor and ceiling** (Definition 0.10): for $x \in \mathbb{R}$, $\lfloor x \rfloor$ is the integer n with $n \leq x < n + 1$, and $\lceil x \rceil$ is the integer n with $n - 1 < x \leq n$. **Rational** (Definition 0.15): $x = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$. **Closure of $\mathbb{Q}$** (Theorem 0.16): sums, differences, products and quotients (by non-zero) of rationals are rational.

Exercise 0.2

essential

Think of an example of (a) a true proposition; (b) a false proposition; and (c) a proposition whose truth value you don't know.

Given. Nothing but Definition 0.1: a proposition is a statement that is either true or false (not both, not neither).

To show. Three statements of the required kinds, each with one sentence saying why it is a proposition of that kind.

Plan. Choose statements about numbers, so that the truth value is determined by a definition you can quote. For (c) pick a famous open problem: the statement is certainly true or false, but nobody knows which.

Step 1 (a) “2 is an even number.” It is a proposition (it is either true or false), and it is true because $2 = 1 \cdot 2$ [Definition 0.41: even means divisible by 2].

Step 2 (b) “2 is an odd number.” A proposition, and false, because 2 is even and odd means not even [Definition 0.41].

Step 3 (c) “Every even integer greater than 2 is the sum of two prime numbers.” This is Goldbach's conjecture. It is a proposition: for each even integer the statement is either true or false, so the whole statement is either true or false. Its truth value is unknown (it has been checked for enormous ranges but never proved).

Conclusion. Any three examples of these kinds are correct; the book's own answers are exactly (a)–(c) above. □

Compare with your work.

- ☐ Each example must be a *statement*, not a question (“is 2 even?”), not a command, and not an expression without a verb (“ $x + 1$ ”).
- ☐ For (c) do not give a statement whose truth value is merely *unknown to you*; the point is that it is unknown to everybody. “ π^π is rational” (Exercise 0.21(f)) is another valid example.

Exercise 0.14**recommended**

Suppose that x is a real number. Prove that $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

Given. $x \in \mathbb{R}$. Nothing else: x may or may not be an integer.

To show. Two equations between integers. Since $\lfloor y \rfloor$ is *the unique* integer n with $n \leq y < n + 1$, the equation $\lfloor -x \rfloor = m$ is proved by showing that the integer m satisfies $m \leq -x < m + 1$ [Definition 0.10]. Same for the ceiling with $m - 1 < y \leq m$.

Plan. Give the right-hand side a name, write down the inequality that defines it, multiply by -1 (which reverses inequalities), and recognise the inequality defining the left-hand side.

Step 1 *First equation.* Let $n = \lceil x \rceil$. Then $n \in \mathbb{Z}$ and

$$n - 1 < x \leq n \quad [\text{Definition 0.10, ceiling}].$$

Step 2 Multiply the inequalities by -1 ; this reverses them [algebra: $a < b \Leftrightarrow -a > -b$]:

$$-n \leq -x < -n + 1, \quad \text{i.e.} \quad (-n) \leq -x < (-n) + 1.$$

Step 3 $-n$ is an integer [closure of $\mathbb{Z}$ under negation, Assumption 0.9] satisfying $m \leq -x < m + 1$ with $m = -n$. By Definition 0.10 (floor), the only integer with this property is $\lfloor -x \rfloor$. Hence $\lfloor -x \rfloor = -n = -\lceil x \rceil$.

Step 4 *Second equation.* Let $k = \lfloor x \rfloor$, so $k \in \mathbb{Z}$ and $k \leq x < k + 1$ [Definition 0.10, floor].

Step 5 Multiply by -1 : $-k - 1 < -x \leq -k$, i.e. $(-k) - 1 < -x \leq (-k)$.

Step 6 $-k$ is an integer satisfying the defining inequality of $\lceil -x \rceil$ [Definition 0.10, ceiling, with $m = -k$], so $\lceil -x \rceil = -k = -\lfloor x \rfloor$.

Conclusion. Both equations hold for every real x . □

Compare with your work.

- ☐ The whole proof rests on the word “the” in Definition 0.10: you must say that the integer you found *satisfies the defining inequality* and therefore *is* the floor (or ceiling).
- ☐ Check the direction of every inequality after multiplying by -1 . The book’s solution starts from $n = \lceil x \rceil$ instead and works towards $\lceil x \rceil = -n$; both directions are fine.
- ☐ Do not split into “ x integer / not integer”: it is not needed.

Exercise 0.21**essential**

Determine, with justification, whether each of the following is true or false: (a) $2 \in \mathbb{N}$; (b) $-2 \in \mathbb{N}$; (c) $-2 \in \mathbb{Q}$; (d) $\frac{2}{3} \in \mathbb{Z}$; (e) $\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} \in \mathbb{Q}$; (f) $\pi^\pi \in \mathbb{Q}$; (g) $\sqrt{-7} \in \mathbb{R}$; (h) $\mathbb{N} \in \mathbb{Z}$.

Given. The definitions of the number sets: $\mathbb{N}$ = the numbers $0, 1, 2, \dots$ reached from 0 by unit steps to the right (Definition 0.6); $\mathbb{Z}$ = unit steps in both directions (Definition 0.8); $\mathbb{Q}$ = Definition 0.15; $\mathbb{R}$ = the points of the number line (Definition 0.4).

To show. For each item a truth value and a reason that quotes the relevant definition.

Plan. “ $x \in S$ ” is proved by checking the definition of S ; “ $x \notin S$ ” is proved by showing that the definition fails, usually because of a property (sign, size, not being a number at all) that every element of S has and x lacks.

Step 1 (a) **True.** 2 is reached from 0 by two unit steps to the right [Definition 0.6].

Step 2 (b) **False.** Every natural number is ≥ 0 , and $-2 < 0$ [Definition 0.6: steps to the right only].

Step 3 (c) **True.** $-2 = \frac{-2}{1}$ with $-2 \in \mathbb{Z}$, $1 \in \mathbb{Z}$ and $1 \neq 0$ [Definition 0.15].

Step 4 (d) **False.** $0 < \frac{2}{5} < 1$, and there is no integer strictly between the consecutive integers 0 and 1 [the number line, page 6].

Step 5 (e) **True.** Simplify first: $\sqrt{8} = \sqrt{4 \cdot 2} = 2\sqrt{2}$, so

$$\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} = \frac{2\sqrt{2} + \sqrt{2}}{2\sqrt{2} - \sqrt{2}} = \frac{3\sqrt{2}}{\sqrt{2}} = 3 = \frac{3}{1},$$

and $3, 1 \in \mathbb{Z}$ with $1 \neq 0$ [Definition 0.15].

Step 6 (f) The statement $\pi^\pi \in \mathbb{Q}$ is a proposition, but its truth value is **not known**: nobody has proved π^π rational or irrational. (So the honest answer is “unknown”; the book asks you to notice this.)

Step 7 (g) **False.** $\sqrt{-7}$ is not a point on the real number line: the square of every real number is ≥ 0 , so no real number squares to -7 [Definition 0.4; $\sqrt{-7} = \sqrt{7}i$ is complex, Definition 0.62].

Step 8 (h) **False.** The elements of $\mathbb{Z}$ are numbers; $\mathbb{N}$ is a *set*, not a number, so it is not an element of $\mathbb{Z}$ [Definition 0.8; see also page 71 of the book].

Conclusion. (a) T, (b) F, (c) T, (d) F, (e) T, (f) unknown, (g) F, (h) F. □

Compare with your work.

- ☐ Each “true” needs the explicit witness (the fraction in (c) and (e)); each “false” needs the property that fails.
- ☐ In (e) you must simplify: “the numerator and denominator are irrational” proves nothing, since a quotient of irrationals can be rational (this is the point of the question).
- ☐ In (h), confusing $\mathbb{N} \in \mathbb{Z}$ with $\mathbb{N} \subseteq \mathbb{Z}$ (which *is* true) is the standard error.

B SETS AND SET-BUILDER NOTATION

What you unfold in this section

Set-builder notation (Definition 0.19 and page 12): $a \in \{x \mid p(x)\}$ iff $p(a)$ is true; the two forms are $\{x \in X \mid p(x)\}$ (filter the elements of X) and $\{\text{expr}(x) \mid p(x)\}$ (all objects of the form $\text{expr}(x)$). **Intervals** (Definition 0.28): $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$, $(-\infty, b) = \{x \in \mathbb{R} \mid x < b\}$, etc.

Exercise 0.24

essential

A dyadic rational is a rational number that can be expressed as an integer divided by a power of 2. Express the set of all dyadic rationals using set-builder notation in at least two different ways.

Given. The definition in the statement: x is a dyadic rational iff $x \in \mathbb{Q}$ and $x = \frac{a}{2^n}$ for some integer a and some power 2^n ($n \in \mathbb{N}$).

To show. Two (or more) correct set-builder expressions, one in each form.

Plan. Translate the definition word by word. “Can be expressed as” is an existential quantifier: *there exist* $a \in \mathbb{Z}$ and $n \in \mathbb{N}$ with $x = a/2^n$.

Step 1 *Filter form.* Take $X = \mathbb{Q}$ and the property “ x can be written as an integer over a power of 2”:

$$D = \left\{ x \in \mathbb{Q} \mid \exists a \in \mathbb{Z}, \exists n \in \mathbb{N}, x = \frac{a}{2^n} \right\} \quad [\{x \in X \mid p(x)\} \text{ form}].$$

Step 2 *Expression form.* List all numbers of the shape $a/2^n$ directly:

$$D = \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\} \quad [\{\text{expr} \mid \dots\} \text{ form}].$$

(The condition $x \in \mathbb{Q}$ is now automatic: $a/2^n$ is an integer over a non-zero integer.)

Step 3 *A third way.* $x = a/2^n$ with $a \in \mathbb{Z}$ is the same as $2^n x \in \mathbb{Z}$:

$$D = \{x \in \mathbb{R} \mid \exists n \in \mathbb{N}, 2^n x \in \mathbb{Z}\}.$$

Conclusion. Any two of these are a complete answer. □

Compare with your work.

- ☐ “An integer divided by a power of 2” needs *two* bound variables, a and n ; writing $\{a/2^n \mid a \in \mathbb{Z}\}$ with n left free is a different set for each n .
- ☐ n ranges over $\mathbb{N}$ (so $2^0 = 1$ is allowed: every integer is dyadic). Writing $n \in \mathbb{Z}$ would also be acceptable, since $a/2^{-n} = a2^n$ is still an integer over 2^0 .

Exercise 0.30

recommended

Express each of the sets illustrated on the number line using the most appropriate notation. (The pictures show: (a) filled dot at -2 , hollow dot at 4 , the segment between them shaded; (b) filled dots at -2 and 4 , segment shaded; (c) filled dots at $-2, -1, 0, 1, 2, 3, 4$ only; (d) hollow dot at 4 , shading to the left without end; (e) filled dot at -2 , shading to the right without end; (f) the whole line shaded.)

Given. The conventions of the figures: a filled dot means the endpoint belongs to the set, a hollow dot means it does not, shading means every real number of that stretch belongs to the set, isolated dots mean only those points.

To show. One expression per picture, using Definition 0.28 for intervals and list notation otherwise.

Step 1 (a) All reals x with $-2 \leq x < 4$ (endpoint -2 in, 4 out): $[-2, 4)$ [Definition 0.28].

Step 2 (b) Both endpoints in: $[-2, 4]$.

Step 3 (c) Only seven isolated points, so this is *not* an interval: $\{-2, -1, 0, 1, 2, 3, 4\}$, or in set-builder form $\{n \in \mathbb{Z} \mid -2 \leq n \leq 4\}$.

Step 4 (d) All reals $x < 4$, no left end: $(-\infty, 4)$.

Step 5 (e) All reals $x \geq -2$: $[-2, \infty)$.

Step 6 (f) Every real number: $(-\infty, \infty) = \mathbb{R}$.

Conclusion. (a) $[-2, 4)$, (b) $[-2, 4]$, (c) $\{-2, \dots, 4\}$, (d) $(-\infty, 4)$, (e) $[-2, \infty)$, (f) $\mathbb{R}$. □

Compare with your work.

- ☐ Writing $[-2, 4]$ for (c) is wrong: an interval contains *all* real numbers between its endpoints (Definition 0.28), and $\frac{1}{2}$ is not in the pictured set.
- ☐ Infinity always gets a round bracket: $(-\infty, 4)$, never $[-\infty, 4)$.

C DIVISION OF INTEGERS

What you unfold in this section

Divides (Definition 0.36): $b \mid a$ iff $a = qb$ for some $q \in \mathbb{Z}$. **Division theorem** (Theorem 0.42): for $a, b \in \mathbb{Z}$, $b \neq 0$, there are unique $q, r \in \mathbb{Z}$ with $a = qb + r$ and $0 \leq r < |b|$. **The two moves:** to use $b \mid a$, write “ $a = qb$ for some $q \in \mathbb{Z}$ ” with a new letter; to prove $b \mid a$, display a as (an integer) $\cdot b$ and say why the factor is an integer.

Exercise 0.38

recommended

Prove that 1 divides every integer, and that every integer divides 0.

Given. Only Definition 0.36.

To show. Two universal statements: (i) $\forall a \in \mathbb{Z}, 1 \mid a$; (ii) $\forall b \in \mathbb{Z}, b \mid 0$. (shape: “for all”: start with “let”)

Plan. For each, let the integer be arbitrary and exhibit the integer q required by the definition.

Step 1 (i) Let $a \in \mathbb{Z}$ be arbitrary [to prove a “for all” statement, Strategy 1.2.10].

Step 2 $a = a \cdot 1$, and the factor a is an integer [$a \in \mathbb{Z}$ by assumption]. So there is $q \in \mathbb{Z}$ (namely $q = a$) with $a = q \cdot 1$.

Step 3 Hence $1 \mid a$ [Definition 0.36 with $b = 1$]. Since a was arbitrary, 1 divides every integer.

Step 4 (ii) Let $b \in \mathbb{Z}$ be arbitrary.

Step 5 $0 = 0 \cdot b$, and the factor 0 is an integer. So there is $q \in \mathbb{Z}$ (namely $q = 0$) with $0 = qb$.

Step 6 Hence $b \mid 0$ [Definition 0.36 with $a = 0$]. Since b was arbitrary, every integer divides 0.

Conclusion. Both statements hold. In particular $0 \mid 0$ (take $b = 0$ in (ii)), which the book points out is surprising but correct: divisibility is about multiplication, not about the fraction $0/0$. $\square$

Compare with your work.

- ☐ Your answer must name q explicitly ($q = a$, resp. $q = 0$) and say it is an integer.
- ☐ Do not write “ $a/1 = a$ is an integer” or “ $0/b = 0$ ”: the latter is even undefined for $b = 0$. The definition never divides.

Exercise 0.40

essential

Let $a, b, d \in \mathbb{Z}$. Suppose that d divides a and d divides b . Given integers u and v , prove that d divides $au + bv$.

Given. $a, b, d, u, v \in \mathbb{Z}$; $d \mid a$, i.e. $a = qd$ for some $q \in \mathbb{Z}$; $d \mid b$, i.e. $b = rd$ for some $r \in \mathbb{Z}$ [Definition 0.36, unfolded with two *different* new letters].

To show. $d \mid au + bv$, i.e. $au + bv = (\text{some integer}) \cdot d$.

Plan. Substitute the expressions for a and b into $au + bv$ and factor out d . Then say why the remaining factor is an integer.

Step 1 Since $d \mid a$, fix $q \in \mathbb{Z}$ with $a = qd$ [Definition 0.36; Strategy 1.2.30 for using an existential].

Step 2 Since $d \mid b$, fix $r \in \mathbb{Z}$ with $b = rd$ [same; a fresh letter because r may differ from q].

Step 3 Compute:

$$\begin{aligned} au + bv &= (qd)u + (rd)v \quad [\text{Steps 1, 2}] \\ &= qud + rvd \quad [\text{commutativity of multiplication}] \\ &= (qu + rv)d \quad [\text{distributivity}]. \end{aligned}$$

Step 4 $qu + rv \in \mathbb{Z}$ [$q, u, r, v \in \mathbb{Z}$ and $\mathbb{Z}$ is closed under $\times$ and $+$, Assumption 0.9].

Step 5 So $au + bv = kd$ with $k = qu + rv \in \mathbb{Z}$; by Definition 0.36, $d \mid au + bv$.

Conclusion. d divides $au + bv$ for all integers u, v . □

Compare with your work.

- Two *different* letters q, r : writing $a = qd$ and $b = qd$ silently assumes $a = b$.
- The sentence “ $qu + rv$ is an integer because ...” is part of the proof; without it Definition 0.36 has not been verified.
- This exercise is quoted in the digit tests (Example 1.1.46, Exercise 1.1.47) and in the Euclidean algorithm; learn its statement: *a common divisor of a and b divides every integer combination of a and b .*

Exercise 0.45

recommended

Prove that if an integer a leaves a remainder of r when divided by an integer b , then a leaves a remainder of r when divided by $-b$.

Given. $a, b \in \mathbb{Z}$ with $b \neq 0$ (otherwise remainders are not defined), and “ a leaves remainder r when divided by b ” means: $a = qb + r$ for some $q \in \mathbb{Z}$ with $0 \leq r < |b|$ [Theorem 0.42].

To show. “ a leaves remainder r when divided by $-b$ ”: there is $q' \in \mathbb{Z}$ with $a = q'(-b) + r$ and $0 \leq r < |-b|$. (shape: an implication: suppose the hypothesis, derive the goal)

Plan. The uniqueness clause of Theorem 0.42 says that *any* way of writing $a = q'(-b) + r'$ with $0 \leq r' < |-b|$ gives the remainder. So it suffices to produce one such way with $r' = r$.

Step 1 Suppose $a = qb + r$ with $q \in \mathbb{Z}$ and $0 \leq r < |b|$ [Strategy 1.1.28: assume the hypothesis, in unfolded form].

Step 2 Rewrite: $qb = (-q)(-b)$ [$(-q)(-b) = qb$, algebra], so $a = (-q)(-b) + r$.

Step 3 $-q \in \mathbb{Z}$ [closure of $\mathbb{Z}$], and $0 \leq r < |b| = |-b|$ [$|b| = |-b|$].

Step 4 So $a = q'(-b) + r$ with $q' = -q \in \mathbb{Z}$ and $0 \leq r < |-b|$. By the *uniqueness* part of Theorem 0.42 (applied with divisor $-b \neq 0$), q' is the quotient and r is the remainder of a on division by $-b$.

Conclusion. a leaves remainder r when divided by $-b$. □

Compare with your work.

- The key sentence is the appeal to *uniqueness* in Theorem 0.42: finding *some* expression $a = q'(-b) + r$ is not enough unless you check $0 \leq r < |-b|$ and then say that the remainder is therefore r .
- The quotient changes sign ($-q$); the remainder does not. Example: $7 = 2 \cdot 3 + 1$ and $7 = (-2)(-3) + 1$.

D IRRATIONAL NUMBERS**Exercise 0.51**

recommended

Let r be a rational number and a an irrational number. Prove that it is possible that ra be rational, and possible that ra be irrational.

Given. $\sqrt{2}$ is irrational (Assumption 0.49); Definition 0.48: irrational means real and not rational.

To show. Two existence statements: (i) there exist $r \in \mathbb{Q}$ and $a \notin \mathbb{Q}$ with $ra \in \mathbb{Q}$; (ii) there exist $r \in \mathbb{Q}$ and $a \notin \mathbb{Q}$ with $ra \notin \mathbb{Q}$. (shape: “it is possible that” = “there exist”)

Plan. For an existence statement, exhibit one example and verify it (Strategy 1.2.24).

Step 1 (i) Define $r = 0$ and $a = \sqrt{2}$. Then $r = \frac{0}{1} \in \mathbb{Q}$ [Definition 0.15] and a is irrational [Assumption 0.49].

Step 2 $ra = 0 \cdot \sqrt{2} = 0 = \frac{0}{1} \in \mathbb{Q}$. So ra is rational: (i) holds.

Step 3 (ii) Define $r = 1$ and $a = \sqrt{2}$. Then $r = \frac{1}{1} \in \mathbb{Q}$ and a is irrational.

Step 4 $ra = 1 \cdot \sqrt{2} = \sqrt{2}$, which is irrational [Assumption 0.49]. So (ii) holds.

Conclusion. Both possibilities occur. □

Compare with your work.

- “It is possible” asks for *examples*, not for a proof that ra is always this or that. Two concrete pairs (r, a) with the verification are a complete answer.
- You may only call a number irrational if the book has established it: $\sqrt{2}$ (Assumption 0.49) is the safe choice.

E CHAPTER 0 EXERCISES**The three answer formats of the chapter exercises**

Prove: a complete proof. **True or false:** decide, then either a proof (true) or one counterexample (false).

Always, sometimes or never: *always* needs a proof for all cases; *never* needs a proof that the conclusion fails for all cases; *sometimes* needs one example where it holds *and* one where it fails.

Exercise 0.E.3

essential

Suppose an integer m leaves a remainder of i when divided by 3, and an integer n leaves a remainder of j when divided by 3, where $0 \leq i, j < 3$. Prove that $m + n$ and $i + j$ leave the same remainder when divided by 3.

Given. $m, n \in \mathbb{Z}$; $m = 3q + i$ for some $q \in \mathbb{Z}$, and $n = 3q' + j$ for some $q' \in \mathbb{Z}$, with $0 \leq i < 3$ and $0 \leq j < 3$ [Theorem 0.42, unfolded].

To show. The remainder of $m + n$ on division by 3 equals the remainder of $i + j$ on division by 3. Note that $i + j$ can be up to 4, so $i + j$ is *not* in general its own remainder.

Plan. Let t be the remainder of $i + j$. Show that $m + n$ can be written as $3 \cdot (\text{integer}) + t$ with $0 \leq t < 3$; the uniqueness in Theorem 0.42 then says that the remainder of $m + n$ is t .

Step 1 Fix $q, q' \in \mathbb{Z}$ with $m = 3q + i$ and $n = 3q' + j$ [Theorem 0.42 applied to m and to n].

Step 2 Divide $i + j$ by 3: there are $s, t \in \mathbb{Z}$ with $i + j = 3s + t$ and $0 \leq t < 3$ [Theorem 0.42 applied to

$i + j$]. By definition, t is the remainder of $i + j$.

Step 3 Compute:

$$m + n = (3q + i) + (3q' + j) = 3q + 3q' + (i + j) = 3q + 3q' + 3s + t = 3(q + q' + s) + t.$$

Step 4 $q + q' + s \in \mathbb{Z}$ [closure of $\mathbb{Z}$] and $0 \leq t < 3$ [Step 2]. So $m + n = 3k + t$ with $k \in \mathbb{Z}$ and $0 \leq t < 3$.

Step 5 By the uniqueness clause of Theorem 0.42, the remainder of $m + n$ on division by 3 is t , which is also the remainder of $i + j$ [Step 2].

Conclusion. $m + n$ and $i + j$ leave the same remainder when divided by 3. $\square$

Compare with your work.

- ☐ Many attempts stop at $m + n = 3(q + q') + (i + j)$ and declare $i + j$ the remainder. That is only valid when $i + j < 3$. The extra division of $i + j$ (Step 2) is what makes the proof complete.
- ☐ Name the uniqueness in Theorem 0.42 explicitly: that is the theorem that identifies a remainder.

Exercise 0.E.4

essential

What are the possible remainders of n^2 when divided by 3, where $n \in \mathbb{Z}$?

Given. $n \in \mathbb{Z}$. By Theorem 0.42 (divisor 3), $n = 3k + r$ with $k \in \mathbb{Z}$ and $r \in \{0, 1, 2\}$.

To show. The set of remainders that n^2 can leave, with a proof that no other remainder occurs and that each listed one does occur.

Plan. Proof by cases on r (Strategy 1.1.19, the cases come from the division theorem). In each case write $n^2 = 3 \cdot (\text{integer}) + r'$ with $r' \in \{0, 1, 2\}$.

Step 1 Fix $k \in \mathbb{Z}$ and $r \in \{0, 1, 2\}$ with $n = 3k + r$ [Theorem 0.42].

Step 2 Case $r = 0$: $n = 3k$, so $n^2 = 9k^2 = 3(3k^2)$ with $3k^2 \in \mathbb{Z}$; remainder 0 [uniqueness in Theorem 0.42, since $0 \leq 0 < 3$].

Step 3 Case $r = 1$: $n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$; remainder 1.

Step 4 Case $r = 2$: $n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$ [$4 = 3 + 1$]; remainder 1.

Step 5 These cases are exhaustive [r must be 0, 1 or 2]. So the remainder is always 0 or 1; it is never 2.

Step 6 Both do occur: $n = 0$ gives $n^2 = 0$ (remainder 0), $n = 1$ gives $n^2 = 1$ (remainder 1).

Answer. The possible remainders are exactly 0 and 1. (This is Proposition 1.1.24 in the book.) $\square$

Compare with your work.

- ☐ All three cases must be written out; in the case $r = 2$ the constant 4 must be split as $3 + 1$ to read off the remainder.
- ☐ “Possible remainders” asks for both halves: no remainder outside $\{0, 1\}$, and each of 0, 1 actually occurs.

Exercise 0.E.17 and 0.E.18

recommended

True or false? **0.E.17** Every integer is a natural number. **0.E.18** Every integer is a rational number.

Given. Definitions 0.6 ($\mathbb{N}$), 0.8 ($\mathbb{Z}$), 0.15 ($\mathbb{Q}$).

To show. 0.E.17 is $\forall n \in \mathbb{Z}, n \in \mathbb{N}$; 0.E.18 is $\forall n \in \mathbb{Z}, n \in \mathbb{Q}$. A “for all” is refuted by one counterexample

(Strategy 1.3.30) and proved by “let $n \in \mathbb{Z}$ ”.

Step 1 0.E.17 is false. Counterexample: $-1 \in \mathbb{Z}$ [one unit step to the left of 0, Definition 0.8] but $-1 \notin \mathbb{N}$ [natural numbers are ≥ 0 , Definition 0.6].

Step 2 0.E.18 is true. Let $n \in \mathbb{Z}$ be arbitrary.

Step 3 $n = \frac{n}{1}$ with $n \in \mathbb{Z}$, $1 \in \mathbb{Z}$ and $1 \neq 0$; so $n \in \mathbb{Q}$ [Definition 0.15].

Step 4 Since n was arbitrary, every integer is rational.

Conclusion. 0.E.17 false, 0.E.18 true. □

Compare with your work.

- A false “for all” needs one explicit element and the reason it fails; a true one needs an arbitrary element, never a specific one.

Exercise 0.E.22

essential

True or false? The square root of every positive rational number is a rational number.

Given. Assumption 0.49: $\sqrt{2}$ is irrational.

To show. The statement is $\forall x \in \mathbb{Q}, (x > 0 \Rightarrow \sqrt{x} \in \mathbb{Q})$. To refute it, one counterexample suffices [Strategy 1.3.30].

Step 1 False. Take $x = 2$. Then $x \in \mathbb{Q}$ [$2 = \frac{2}{1}$] and $x > 0$, so x satisfies the hypothesis.

Step 2 $\sqrt{x} = \sqrt{2}$, which is not rational [Assumption 0.49]. So the conclusion fails for $x = 2$.

Conclusion. The statement is false. □

Compare with your work.

- The counterexample must satisfy *both* conditions (rational and positive) and must have a square root the book has shown to be irrational.

Exercise 0.E.27 and 0.E.28

essential

Let $a, b, c \in \mathbb{Z}$ and suppose that a divides c and b divides c . Always, sometimes or never: **0.E.27** ab divides c ; **0.E.28** ab divides c^2 .

Given. $a, b, c \in \mathbb{Z}$; $c = qa$ for some $q \in \mathbb{Z}$; $c = rb$ for some $r \in \mathbb{Z}$ [Definition 0.36].

To show. Decide for each conclusion: always (prove it), never (prove its negation), or sometimes (one example, one counterexample).

Step 1 0.E.27: sometimes. Example where it holds: $a = b = c = 1$. Then $a \mid c$ and $b \mid c$ ($1 = 1 \cdot 1$), and $ab = 1$ divides $c = 1$.

Step 2 Example where it fails: $a = b = c = 2$. Then $2 \mid 2$ [$2 = 1 \cdot 2$], so the hypotheses hold, but $ab = 4$ does not divide 2: if $2 = 4q$ with $q \in \mathbb{Z}$ then $4q = 2$ forces $q = \frac{1}{2} \notin \mathbb{Z}$ [no integer between 0 and 1].

Step 3 0.E.28: always. Fix $q, r \in \mathbb{Z}$ with $c = qa$ and $c = rb$ [Definition 0.36 twice, two letters].

Step 4 Compute: $c^2 = c \cdot c = (qa)(rb) = (qr)(ab)$ [commutativity and associativity of multiplication].

Step 5 $qr \in \mathbb{Z}$ [closure], so $c^2 = k \cdot (ab)$ with $k = qr \in \mathbb{Z}$; hence $ab \mid c^2$ [Definition 0.36].

Conclusion. 0.E.27 sometimes; 0.E.28 always. □

Compare with your work.

- “Sometimes” needs *two* explicit triples (a, b, c) , each checked against the hypotheses, one making the

conclusion true and one making it false.

- In 0.E.28 the trick is to use the *two different* expressions for c , one for each factor of $c \cdot c$.

Exercise 0.E.29 and 0.E.30

recommended

0.E.29 Let $x, y \in \mathbb{Q}$ and let $a, b, c, d \in \mathbb{Z}$ with $cy + d \neq 0$. Then $\frac{ax+b}{cy+d} \in \mathbb{Q}$. **0.E.30** Let $\frac{a}{b}$ be a rational number. Then $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$. (Always, sometimes or never?)

Given. Theorem 0.16 (closure of $\mathbb{Q}$ under $+$, $-$, $\times$ and $\div$ by non-zero) and 0.E.18 (integers are rational).

Step 1 0.E.29: always. $a, b, c, d \in \mathbb{Q}$ [0.E.18]. $ax \in \mathbb{Q}$ [Theorem 0.16, product], so $ax + b \in \mathbb{Q}$ [Theorem 0.16, sum].

Step 2 Likewise $cy + d \in \mathbb{Q}$, and $cy + d \neq 0$ by hypothesis.

Step 3 So $\frac{ax+b}{cy+d}$ is a quotient of rationals with non-zero denominator, hence rational [Theorem 0.16, quotient].

Step 4 0.E.30: sometimes. *Holds:* $\frac{1}{2}$ is rational and $a = 1, b = 2$ are integers.

Step 5 Fails: $\frac{\sqrt{2}}{\sqrt{2}} = 1$ is a rational number written as $\frac{a}{b}$ with $a = b = \sqrt{2}$, and $\sqrt{2} \notin \mathbb{Z}$ [it is not even rational, Assumption 0.49]. (Or $\frac{1/2}{1/4} = 2$.)

Conclusion. 0.E.29 always; 0.E.30 sometimes. □

Compare with your work.

- 0.E.30 tests reading: Definition 0.15 says a rational number *can be* written as an integer over an integer, not that every way of writing it as a fraction has integer parts.
- In 0.E.29 each use of Theorem 0.16 should be named (sum, product, quotient), and the hypothesis $cy + d \neq 0$ must be quoted where the quotient is formed.

Chapter 1

Logical structure

Contents

- A** Propositional logic (§1.1)
- B** Variables and quantifiers (§1.2)
- C** Logical equivalence (§1.3)
- D** Chapter 1 exercises (1.E)

Planning

★ 1.1: 21, 25, 32, 33, 38, 47, 56, 57 · 1.2: 6, 15, 18, 39 · 1.3: 5, 13, 14, 22, 27, 44ace · 1.E: 18
 ★ 1.1: 37, 43, 54, 55, 65, 66 · 1.2: 21, 22, 28, 37ab · 1.3: 32, 35 · 1.E: 11, 28, 29, 30
 Bonus: 1.2.14. §1.1 is in **Quiz 1**.

A PROPOSITIONAL LOGIC

The rules you quote in this section (Definitions 1.1.4, 1.1.15, 1.1.27, 1.1.50)

- $\wedge$: to prove $p \wedge q$, prove p and prove q (Strategy 1.1.7); from $p \wedge q$ you may use p and use q (Strategy 1.1.10).
- $\vee$: to prove $p \vee q$, prove one of them (Strategy 1.1.16); from $p \vee q$, argue by cases: derive the goal from p , then from q (Strategy 1.1.19).
- $\Rightarrow$: to prove $p \Rightarrow q$, assume p and derive q (Strategy 1.1.28); from $p \Rightarrow q$ and p , conclude q (modus ponens, Strategy 1.1.34).
- $\neg$: to prove $\neg p$, assume p and derive a contradiction (Strategy 1.1.51); from $\neg p$ and p you have a contradiction (rule $\neg E$, Strategy 1.1.59).
- $\Leftrightarrow$: $p \Leftrightarrow q$ means $(p \Rightarrow q) \wedge (q \Rightarrow p)$ (Definition 1.1.40).
- **LEM** (Axiom 1.1.60): $p \vee \neg p$ may always be assumed; **explosion** (Axiom 1.1.67): from a contradiction anything follows.

In the solutions below, each step records the current *assumptions* and *goal*, as in Example 1.1.1.

Exercise 1.1.21

essential

Let p , q and r be propositions, and suppose $q \vee r$ is true. Prove that $(p \vee q) \vee (p \vee r)$ is true, using Strategy 1.1.19 and Strategy 1.1.16.

Given. $q \vee r$ is true. (Nothing is known about p .)

To show. $(p \vee q) \vee (p \vee r)$. (shape: a disjunction whose two halves are disjunctions)

Plan. The only assumption is a disjunction, so split into cases (Strategy 1.1.19). In each case the goal is a disjunction, so it suffices to prove one half (Strategy 1.1.16), possibly twice.

Step 1 Assumption: $q \vee r$. By Strategy 1.1.19 it suffices to derive the goal in Case 1 (q true) and in Case 2 (r true).

Step 2 Case 1. Assume q . Then $p \vee q$ is true [Strategy 1.1.16, rule $\vee I_2$: the second disjunct q holds].

Step 3 Since $p \vee q$ is true, $(p \vee q) \vee (p \vee r)$ is true [Strategy 1.1.16, $\vee I_1$: the first disjunct holds]. Case 1 is done.

Step 4 Case 2. Assume r . Then $p \vee r$ is true $[\vee I_2]$.

Step 5 Hence $(p \vee q) \vee (p \vee r)$ is true $[\vee I_2]$: the second disjunct $p \vee r$ holds]. Case 2 is done.

Step 6 The goal holds in both cases, so it follows from $q \vee r$ [Strategy 1.1.19].

Conclusion. $(p \vee q) \vee (p \vee r)$ is true. □

Compare with your work.

- ☐ Both cases must be written; each must end with the full goal $(p \vee q) \vee (p \vee r)$, not just with $p \vee q$.
- ☐ Never try to decide whether p is true: nothing is known about p , and nothing needs to be.

Exercise 1.1.25

essential

Let n be an integer. Prove that n^2 leaves a remainder of 0, 1 or 4 when divided by 5.

Given. $n \in \mathbb{Z}$. Theorem 0.42 with $b = 5$: $n = 5k + r$ for some $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4\}$.

To show. The remainder of n^2 on division by 5 is 0, 1 or 4. (shape: a disjunction with three options; one of them must be proved in each case)

Plan. Proof by cases on the five possible values of r (Strategy 1.1.19). In each case write $n^2 = 5 \cdot (\text{integer}) + r'$ with $r' \in \{0, 1, 4\}$ and invoke uniqueness of the remainder.

Step 1 Fix $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4\}$ with $n = 5k + r$ [Theorem 0.42].

Step 2 Expand once for all cases:

$$n^2 = (5k + r)^2 = 25k^2 + 10kr + r^2 = 5(5k^2 + 2kr) + r^2 \quad [\text{algebra}].$$

So the remainder of n^2 equals the remainder of r^2 [as in 0.E.3: $5(5k^2 + 2kr)$ is a multiple of 5].

Step 3 Case $r = 0$: $r^2 = 0$, so $n^2 = 5(5k^2) + 0$: remainder 0.

Step 4 Case $r = 1$: $r^2 = 1$, so $n^2 = 5(5k^2 + 2k) + 1$: remainder 1.

Step 5 Case $r = 2$: $r^2 = 4$, so $n^2 = 5(5k^2 + 4k) + 4$: remainder 4.

Step 6 Case $r = 3$: $r^2 = 9 = 5 + 4$, so $n^2 = 5(5k^2 + 6k + 1) + 4$: remainder 4.

Step 7 Case $r = 4$: $r^2 = 16 = 15 + 1$, so $n^2 = 5(5k^2 + 8k + 3) + 1$: remainder 1.

Step 8 In every case the bracket is an integer and the remainder satisfies $0 \leq r' < 5$, so it is the remainder [uniqueness in Theorem 0.42]; and $r' \in \{0, 1, 4\}$. The cases are exhaustive [Step 1].

Conclusion. n^2 leaves remainder 0, 1 or 4 when divided by 5. □

Compare with your work.

- ☐ Five cases, each with the explicit integer in the bracket. The cases $r = 3, 4$ need the constant rewritten ($9 = 5 + 4$, $16 = 15 + 1$) before the remainder can be read off.
- ☐ Say why the cases are exhaustive (the division theorem) and why the number you read off is the remainder (uniqueness).

Exercise 1.1.32

essential

Let r and s be propositions. Prove that $r \Rightarrow (s \Rightarrow s)$ is true, using the strategies for implications.

Given. Nothing: no proposition is assumed true.

To show. $r \Rightarrow (s \Rightarrow s)$ (shape: an implication whose consequent is an implication).

Plan. Strategy 1.1.28 twice: assume the antecedent, the goal becomes the consequent.

Step 1 To prove $r \Rightarrow (s \Rightarrow s)$ it suffices to assume r and derive $s \Rightarrow s$ [Strategy 1.1.28]. So: assume r . Goal: $s \Rightarrow s$.

Step 2 To prove $s \Rightarrow s$ it suffices to assume s and derive s [Strategy 1.1.28]. So: assume s . Goal: s .

Step 3 s is true, because it is among the assumptions [Step 2].

Step 4 Hence $s \Rightarrow s$ is true [rule $\Rightarrow$ I: s was derived from the assumption s].

Step 5 Hence $r \Rightarrow (s \Rightarrow s)$ is true [$\Rightarrow$ I: $s \Rightarrow s$ was derived from the assumption r].

Conclusion. $r \Rightarrow (s \Rightarrow s)$ is true for all propositions r, s . □

Compare with your work.

- ☐ The assumption r is never used; that is fine. An implication with a true consequent is true.
- ☐ Each “assume” must be matched by a “hence $\dots \Rightarrow \dots$ ” that discharges it (Steps 4 and 5).

Exercise 1.1.33

essential

Let a, b and c be propositions. Prove that $(a \vee b) \Rightarrow (c \Rightarrow ((c \wedge a) \vee b))$ is true.

Given. Nothing.

To show. $(a \vee b) \Rightarrow (c \Rightarrow ((c \wedge a) \vee b))$ (shape: implication, then implication, then a disjunction).

Plan. Assume the two antecedents (Strategy 1.1.28 twice), then use the disjunction $a \vee b$ by cases (Strategy 1.1.19); in each case prove one half of the goal disjunction (Strategy 1.1.16).

Step 1 Assume $a \vee b$ [Strategy 1.1.28]. Goal: $c \Rightarrow ((c \wedge a) \vee b)$.

Step 2 Assume c [Strategy 1.1.28]. Goal: $(c \wedge a) \vee b$. Assumptions now: $a \vee b, c$.

Step 3 Split into cases on $a \vee b$ [Strategy 1.1.19].

Step 4 *Case 1.* Assume a . With c (Step 2) and a both true, $c \wedge a$ is true [Strategy 1.1.7, $\wedge$ I]. Hence $(c \wedge a) \vee b$ [Strategy 1.1.16, $\vee$ I₁].

Step 5 *Case 2.* Assume b . Then $(c \wedge a) \vee b$ [$\vee$ I₂].

Step 6 In both cases $(c \wedge a) \vee b$, so it follows from $a \vee b$ and c [Strategy 1.1.19].

Step 7 Hence $c \Rightarrow ((c \wedge a) \vee b)$ [$\Rightarrow$ I, discharging c], and hence $(a \vee b) \Rightarrow (c \Rightarrow ((c \wedge a) \vee b))$ [$\Rightarrow$ I, discharging $a \vee b$].

Conclusion. The formula is true. □

Compare with your work.

- ☐ Do the “assume” steps *before* the case split: in Case 1 you need c available to build $c \wedge a$.
- ☐ In Case 2 the assumption c is not needed; say so or just ignore it.

Exercise 1.1.37

recommended

Let p, q and r be propositions. Prove that $(p \Rightarrow (q \Rightarrow r)) \Rightarrow (q \Rightarrow (p \Rightarrow r))$ is true.

Given. Nothing.

To show. An implication whose consequent is $q \Rightarrow (p \Rightarrow r)$, again nested implications.

Plan. Assume all three antecedents in turn (Strategy 1.1.28 three times), then use modus ponens (Strategy 1.1.34) twice on the first assumption.

Step 1 Assume $p \Rightarrow (q \Rightarrow r)$. Goal: $q \Rightarrow (p \Rightarrow r)$.

Step 2 Assume q . Goal: $p \Rightarrow r$.

Step 3 Assume p . Goal: r . Assumptions: $p \Rightarrow (q \Rightarrow r)$, q , p .

Step 4 From $p \Rightarrow (q \Rightarrow r)$ and p , conclude $q \Rightarrow r$ [modus ponens, Strategy 1.1.34].

Step 5 From $q \Rightarrow r$ and q , conclude r [modus ponens]. The goal is reached.

Step 6 Discharge the assumptions in reverse order: $p \Rightarrow r$ [$\Rightarrow$ I]; then $q \Rightarrow (p \Rightarrow r)$; then the whole formula.

Conclusion. $(p \Rightarrow (q \Rightarrow r)) \Rightarrow (q \Rightarrow (p \Rightarrow r))$ is true. $\square$

Compare with your work.

- ☐ Three “assume”s, two modus ponens, three discharges. Nothing else is available, so nothing else should appear.
- ☐ The order of the assumptions q and p is dictated by the goal, not by the first assumption.

Exercise 1.1.38

essential

Let p , q and r be propositions. Prove that $(p \Rightarrow p \wedge q) \Rightarrow ((p \vee q) \Rightarrow q)$ is true.

Given. Nothing.

To show. $(p \Rightarrow (p \wedge q)) \Rightarrow ((p \vee q) \Rightarrow q)$.

Plan. Assume both antecedents; the second is a disjunction, so split into cases; in the case p , modus ponens gives $p \wedge q$, from which q follows.

Step 1 Assume $p \Rightarrow (p \wedge q)$ [Strategy 1.1.28]. Goal: $(p \vee q) \Rightarrow q$.

Step 2 Assume $p \vee q$ [Strategy 1.1.28]. Goal: q .

Step 3 Split into cases on $p \vee q$ [Strategy 1.1.19].

Step 4 *Case 1.* Assume p . From $p \Rightarrow (p \wedge q)$ and p , $p \wedge q$ is true [modus ponens]. Hence q is true [Strategy 1.1.10, $\wedge E_2$].

Step 5 *Case 2.* Assume q . Then q is true [it is assumed].

Step 6 In both cases q ; so q follows from the assumptions [Strategy 1.1.19].

Step 7 Hence $(p \vee q) \Rightarrow q$ [$\Rightarrow$ I], and hence the whole formula [$\Rightarrow$ I].

Conclusion. The formula is true. $\square$

Compare with your work.

- ☐ Case 2 is one line, but it must be present: a proof by cases with a missing case proves nothing.

Exercise 1.1.43

recommended

Let p and q be propositions. Prove that $(p \Leftrightarrow q) \Leftrightarrow (q \Leftrightarrow p)$ is true.

Given. Definition 1.1.40: $x \Leftrightarrow y$ is $(x \Rightarrow y) \wedge (y \Rightarrow x)$.

To show. A biconditional: by Definition 1.1.40 we must prove $(\Rightarrow) (p \Leftrightarrow q) \Rightarrow (q \Leftrightarrow p)$ and $(\Leftarrow) (q \Leftrightarrow p) \Rightarrow (p \Leftrightarrow q)$.

Plan. Unfold everything into conjunctions of implications; then the two halves of the conjunction only need to be swapped (Strategies 1.1.10 and 1.1.7).

Step 1 $(\Rightarrow)$ Assume $p \Leftrightarrow q$, i.e. $(p \Rightarrow q) \wedge (q \Rightarrow p)$ [Definition 1.1.40]. Goal: $q \Leftrightarrow p$, i.e. $(q \Rightarrow p) \wedge (p \Rightarrow q)$.

Step 2 From the assumed conjunction we may use $p \Rightarrow q$ and $q \Rightarrow p$ separately [Strategy 1.1.10].

Step 3 To prove $(q \Rightarrow p) \wedge (p \Rightarrow q)$ we prove each conjunct [Strategy 1.1.7]: $q \Rightarrow p$ holds by Step 2, and $p \Rightarrow q$ holds by Step 2. So $q \Leftrightarrow p$.

Step 4 Hence $(p \Leftrightarrow q) \Rightarrow (q \Leftrightarrow p)$ [$\Rightarrow$ I].

Step 5 ($\Leftarrow$) Assume $q \Leftrightarrow p$, i.e. $(q \Rightarrow p) \wedge (p \Rightarrow q)$. Goal: $(p \Rightarrow q) \wedge (q \Rightarrow p)$. Exactly as in Steps 2–3, both conjuncts are available, so $p \Leftrightarrow q$. Hence $(q \Leftrightarrow p) \Rightarrow (p \Leftrightarrow q)$.

Step 6 Both implications hold, so the biconditional holds [Definition 1.1.40, Strategy 1.1.7].

Conclusion. $(p \Leftrightarrow q) \Leftrightarrow (q \Leftrightarrow p)$ is true. □

Compare with your work.

- “Obviously symmetric” is not a proof. The proof is the unfolding of Definition 1.1.40 and the two conjunction rules, written out in both directions.

Exercise 1.1.47

essential

Prove that a natural number n is divisible by 3 if and only if the sum of its base-10 digits is divisible by 3.

Given. $n \in \mathbb{N}$ with base-10 expansion $d_r d_{r-1} \dots d_0$, so $n = \sum_{i=0}^r d_i 10^i$ with $0 \leq d_i \leq 9$ [Definition 0.31]. Let $s = \sum_{i=0}^r d_i$ be the digit sum. Exercise 0.40: a common divisor of two integers divides every integer combination of them.

To show. $3 \mid n \Leftrightarrow 3 \mid s$: two implications.

Plan. As in Example 1.1.46 (divisibility by 8): split $n = s + n''$ where $n'' = n - s$ is always divisible by 3. Then n and s are integer combinations of each other and n'' , and Exercise 0.40 transfers divisibility both ways.

Step 1 Define $n'' = n - s = \sum_{i=0}^r d_i 10^i - \sum_{i=0}^r d_i = \sum_{i=0}^r d_i (10^i - 1)$ [algebra].

Step 2 *Claim:* $3 \mid 10^i - 1$ for every $i \in \mathbb{N}$. For $i = 0$: $10^0 - 1 = 0 = 3 \cdot 0$. For $i \geq 1$: $10^i - 1 = (10 - 1)(10^{i-1} + 10^{i-2} + \dots + 10 + 1) = 9m_i$ with $m_i = 10^{i-1} + \dots + 1 \in \mathbb{N}$ [the factorisation $x^i - 1 = (x - 1)(x^{i-1} + \dots + 1)$], so $10^i - 1 = 3 \cdot (3m_i)$ with $3m_i \in \mathbb{Z}$. Hence $3 \mid 10^i - 1$ [Definition 0.36].

Step 3 *Claim:* $3 \mid n''$. By Step 2 write $10^i - 1 = 3c_i$ with $c_i \in \mathbb{Z}$. Then $n'' = \sum_i d_i \cdot 3c_i = 3 \sum_i d_i c_i$, and $\sum_i d_i c_i \in \mathbb{Z}$ [closure]. So $3 \mid n''$.

Step 4 ($\Rightarrow$) Assume $3 \mid n$ [Strategy 1.1.28]. Since also $3 \mid n''$ (Step 3), 3 divides $1 \cdot n + (-1) \cdot n''$ [Exercise 0.40 with $u = 1, v = -1$]. But $n - n'' = s$ [Step 1]. So $3 \mid s$.

Step 5 ($\Leftarrow$) Assume $3 \mid s$. Since $3 \mid n''$, 3 divides $1 \cdot s + 1 \cdot n'' = n$ [Exercise 0.40 with $u = v = 1$; $s + n'' = n$ by Step 1]. So $3 \mid n$.

Conclusion. $3 \mid n$ if and only if $3 \mid s$ [Definition 1.1.40]. □

Compare with your work.

- The proof has two labelled directions, and both quote Exercise 0.40 with explicit u, v .
- The fact $3 \mid 10^i - 1$ needs an argument (Step 2); “ $10^i - 1 = 99 \dots 9$ ” is the idea, the factorisation is the proof.
- Do not use congruences ($10 \equiv 1 \pmod{3}$): they appear only in Chapter 5.

Exercise 1.1.54

recommended

Let p and q be propositions. Prove that $(p \Rightarrow \neg q) \Rightarrow (q \Rightarrow \neg p)$ is true.

Given. Nothing.

To show. Nested implications ending in the negation $\neg p$.

Plan. Assume the antecedents; to prove $\neg p$, assume p and derive a contradiction (Strategy 1.1.51); the contradiction comes from $\neg q$ (by modus ponens) against q (rule $\neg E$).

Step 1 Assume $p \Rightarrow \neg q$ [Strategy 1.1.28]. Goal: $q \Rightarrow \neg p$.

Step 2 Assume q [Strategy 1.1.28]. Goal: $\neg p$.

Step 3 To prove $\neg p$, assume p and aim for a contradiction [Strategy 1.1.51]. Assumptions: $p \Rightarrow \neg q$, q , p . Goal: $\perp$.

Step 4 From $p \Rightarrow \neg q$ and p : $\neg q$ [modus ponens].

Step 5 $\neg q$ and q are both assumed/derived: contradiction [rule $\neg E$, Strategy 1.1.59].

Step 6 Hence $\neg p$ [$\neg I$: a contradiction was derived from p]. Hence $q \Rightarrow \neg p$ [$\Rightarrow I$], hence the whole formula [$\Rightarrow I$].

Conclusion. $(p \Rightarrow \neg q) \Rightarrow (q \Rightarrow \neg p)$ is true. □

Compare with your work.

- State explicitly which two formulae contradict each other (" $\neg q$ and q "); "contradiction" alone is not enough.

Exercise 1.1.55

recommended

Let p and q be propositions. Prove that $\neg p \Rightarrow (\neg q \Rightarrow \neg(p \vee q))$ is true.

Given. Nothing.

To show. Two implications, then the negation of a disjunction.

Plan. Assume $\neg p$ and $\neg q$; to prove $\neg(p \vee q)$ assume $p \vee q$ and derive a contradiction by cases.

Step 1 Assume $\neg p$. Goal: $\neg q \Rightarrow \neg(p \vee q)$.

Step 2 Assume $\neg q$. Goal: $\neg(p \vee q)$.

Step 3 Assume $p \vee q$, aiming for a contradiction [Strategy 1.1.51]. Split into cases [Strategy 1.1.19].

Step 4 *Case 1.* Assume p . Together with $\neg p$ (Step 1) this is a contradiction [$\neg E$].

Step 5 *Case 2.* Assume q . Together with $\neg q$ (Step 2) this is a contradiction [$\neg E$].

Step 6 In both cases a contradiction follows, so a contradiction follows from $p \vee q$ [Strategy 1.1.19 with goal $\perp$]. Hence $\neg(p \vee q)$ [$\neg I$].

Step 7 Discharge: $\neg q \Rightarrow \neg(p \vee q)$, then $\neg p \Rightarrow (\neg q \Rightarrow \neg(p \vee q))$ [$\Rightarrow I$ twice].

Conclusion. The formula is true. (It is one half of de Morgan's law $\neg(p \vee q) \equiv \neg p \wedge \neg q$.) □

Compare with your work.

- Proof by cases may have $\perp$ as its goal; each case must actually reach $\perp$ and name the clashing pair.

Exercise 1.1.56

essential

Let x be a real number. Prove by contradiction that if x is irrational then $-x$ and $\frac{1}{x}$ are irrational.

Given. $x \in \mathbb{R}$. "Irrational" = real and not rational (Definition 0.48), so " x is irrational" is the negation $\neg(x \in \mathbb{Q})$. Definition 0.15 for rational.

To show. x irrational $\Rightarrow (-x$ irrational $\wedge \frac{1}{x}$ irrational) (shape: implication, then a conjunction of two negations).

Plan. Assume x irrational. Prove each conjunct by Strategy 1.1.51: assume it is rational, write it as a

fraction, and manufacture a fraction for x , contradicting the assumption.

Step 1 Assume x is irrational [Strategy 1.1.28]. Note $x \neq 0$, because $0 = \frac{0}{1} \in \mathbb{Q}$. Goal: $-x$ irrational and $\frac{1}{x}$ irrational [Strategy 1.1.7: prove both].

Step 2 $-x$ is irrational. Suppose, towards a contradiction, that $-x$ is rational [Strategy 1.1.51]. Then $-x = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$ [Definition 0.15].

Step 3 Multiply by -1 : $x = \frac{-a}{b}$, and $-a \in \mathbb{Z}$, $b \in \mathbb{Z}$, $b \neq 0$. So $x \in \mathbb{Q}$ [Definition 0.15].

Step 4 This contradicts Step 1 (x irrational) $[\neg E]$. Hence $-x$ is irrational $[\neg I]$.

Step 5 $\frac{1}{x}$ is irrational. Suppose, towards a contradiction, that $\frac{1}{x} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$.

Step 6 Then $a \neq 0$: otherwise $\frac{1}{x} = 0$, but $\frac{1}{x} \cdot x = 1 \neq 0$ [algebra]. Hence we may invert: $x = \frac{b}{a}$ with $b, a \in \mathbb{Z}$ and $a \neq 0$, so $x \in \mathbb{Q}$ [Definition 0.15].

Step 7 Contradiction with Step 1. Hence $\frac{1}{x}$ is irrational.

Step 8 Both conjuncts hold [Strategy 1.1.7], and the implication follows $[\Rightarrow I]$.

Conclusion. If x is irrational then $-x$ and $\frac{1}{x}$ are irrational. □

Compare with your work.

- ☐ Each half starts with “suppose ... is rational, say $= \frac{a}{b}$ ” and ends with “so x is rational, contradiction”.
- ☐ The check $a \neq 0$ before inverting the fraction is a step markers look for; it is also why $x \neq 0$ must be noted.

Exercise 1.1.57

essential

Prove by contradiction that there is no least positive real number: that is, there is no positive real number a such that $a \leq b$ for all positive real numbers b .

Given. Only the arithmetic of real numbers.

To show. $\neg \exists a \in \mathbb{R}, (a > 0 \wedge \forall b \in \mathbb{R}, (b > 0 \Rightarrow a \leq b))$ (shape: a negation: assume the inside and derive a contradiction).

Plan. Assume such an a exists. The “for all b ” lets us choose any positive b we like (Strategy 1.2.19); choose $b = \frac{a}{2}$, which is positive and smaller than a .

Step 1 Suppose, towards a contradiction, that there is a positive real number a such that $a \leq b$ for all positive real b [Strategy 1.1.51].

Step 2 Define $b = \frac{a}{2}$. Then $b \in \mathbb{R}$ and $b > 0$ [$a > 0$ and $\frac{1}{2} > 0$].

Step 3 By the assumed property applied to this b [Strategy 1.2.19: $\forall$ may be specialised], $a \leq \frac{a}{2}$.

Step 4 Multiply by $2 > 0$: $2a \leq a$; subtract a : $a \leq 0$ [algebra, inequalities preserved by positive factors].

Step 5 $a \leq 0$ contradicts $a > 0$ $[\neg E]$.

Conclusion. The assumption is impossible, so no least positive real number exists $[\neg I]$. □

Compare with your work.

- ☐ The proof must say what the contradiction is: $a \leq 0$ versus $a > 0$.
- ☐ Any b with $0 < b < a$ works ($b = \frac{a}{2}$, $b = \frac{a}{3}$, ...); you must check *both* $b > 0$ (so the property applies) and $b < a$ (so it contradicts).

Exercise 1.1.65**recommended**

Let p and q be propositions. Prove that $(\neg q \Rightarrow \neg p) \Rightarrow (p \Rightarrow q)$ is true.

Given. Nothing. Available: LEM (Axiom 1.1.60, Strategy 1.1.61) and the principle of explosion (Axiom 1.1.67).

To show. Nested implications ending in q , which has no logical structure of its own, so no introduction rule applies: this is where LEM comes in.

Plan. Assume $\neg q \Rightarrow \neg p$ and p . Split on $q \vee \neg q$. In the case $\neg q$, modus ponens gives $\neg p$, contradicting p ; explosion then yields q .

Step 1 Assume $\neg q \Rightarrow \neg p$. Goal: $p \Rightarrow q$.

Step 2 Assume p . Goal: q .

Step 3 By the law of excluded middle, $q \vee \neg q$ is true [Axiom 1.1.60]; split into cases [Strategy 1.1.61].

Step 4 *Case 1.* q is true. The goal holds.

Step 5 *Case 2.* $\neg q$ is true. From $\neg q \Rightarrow \neg p$ and $\neg q$: $\neg p$ [modus ponens]. With p (Step 2) this is a contradiction [$\neg E$]. From a contradiction, q follows [principle of explosion, Axiom 1.1.67].

Step 6 In both cases q , so q holds [Strategy 1.1.19]. Discharge: $p \Rightarrow q$, then the whole formula [$\Rightarrow I$ twice].

Conclusion. $(\neg q \Rightarrow \neg p) \Rightarrow (p \Rightarrow q)$ is true. (This is the law of contraposition in one direction; see Theorem 1.3.19.) $\square$

Compare with your work.

- ☐ Without LEM there is no way to start on the goal q : say explicitly “by LEM, q or $\neg q$ ”.
- ☐ In Case 2 the final move is explosion, not “so q must be true”; name the axiom.

Exercise 1.1.66**recommended**

Let a and b be irrational numbers. By considering $\sqrt{2}^{\sqrt{2}}$, prove that it is possible that a^b be rational.

Given. $\sqrt{2}$ is irrational (Assumption 0.49). The exponent rule $(x^y)^z = x^{yz}$ for positive x .

To show. $\exists a, b \in \mathbb{R}, (a \notin \mathbb{Q} \wedge b \notin \mathbb{Q} \wedge a^b \in \mathbb{Q})$ (shape: an existence statement), but we cannot decide whether $\sqrt{2}^{\sqrt{2}}$ is rational, so we use LEM instead of exhibiting a definite pair.

Plan. Case split on “ $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q}$ ” (Strategy 1.1.61). In each case exhibit a suitable pair.

Step 1 By LEM, either $\sqrt{2}^{\sqrt{2}}$ is rational or it is not [Axiom 1.1.60].

Step 2 *Case 1:* $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q}$. Define $a = b = \sqrt{2}$. Both are irrational [Assumption 0.49], and $a^b = \sqrt{2}^{\sqrt{2}}$ is rational by the case assumption. So the required pair exists.

Step 3 *Case 2:* $\sqrt{2}^{\sqrt{2}} \notin \mathbb{Q}$. Define $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$. Both are irrational [case assumption; Assumption 0.49].

Step 4 Compute $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$ [exponent rule; $\sqrt{2} \cdot \sqrt{2} = 2$], and $2 = \frac{2}{1} \in \mathbb{Q}$. So the required pair exists.

Step 5 In both cases there are irrational a, b with a^b rational [Strategy 1.1.19].

Conclusion. It is possible that a^b is rational for irrational a, b . We do not learn which case holds: the proof is *non-constructive*. $\square$

Compare with your work.

- A correct answer has two cases and never claims to know whether $\sqrt{2}^{\sqrt{2}}$ is rational.
- Check that *both* numbers of the pair are irrational in each case; in Case 2 that is exactly the case assumption.

B VARIABLES AND QUANTIFIERS

The rules you quote in this section (Definitions 1.2.9, 1.2.23, 1.2.32)

- $\forall$: to prove $\forall x \in X, p(x)$: “Let $x \in X$ (arbitrary)” and prove $p(x)$ (Strategy 1.2.10). From $\forall x \in X, p(x)$ you may use $p(a)$ for any particular $a \in X$ (Strategy 1.2.19).
- $\exists$: to prove $\exists x \in X, p(x)$: “Define $x = \dots$ ”, check $x \in X$, prove $p(x)$ (Strategy 1.2.24). From $\exists x \in X, p(x)$: “Take $a \in X$ with $p(a)$ ” (Strategy 1.2.30).
- $\exists!$: existence as above, then uniqueness: “let $a, b \in X$ with $p(a)$ and $p(b)$... so $a = b$ ” (Strategy 1.2.35).
- Translating: “ n is even” is $\exists k \in \mathbb{Z}, n = 2k$; “ $b \mid a$ ” is $\exists q \in \mathbb{Z}, a = qb$; restrictions inside $\forall$ become implications, inside $\exists$ conjunctions.

Exercise 1.2.6

essential

Find logical formulae that represent each of the following English statements: (a) there is an integer that is divisible by every integer; (b) there is no greatest odd integer; (c) between any two distinct rational numbers is a third distinct rational number; (d) if an integer has a rational square root, then that root is an integer.

Given. The translation rules of Examples 1.2.5 and 1.2.7, and Definition 0.36 for “divides”.

To show. One formula per sentence, all variables bound by a quantifier with a domain.

Plan. Introduce a variable for each noun phrase, decide the quantifier (“there is” = $\exists$, “every/any” = $\forall$, “no” = $\neg\exists$), then unfold “divisible”, “odd”, “greatest”, “between”.

Step 1 (a) “There is an integer n ” gives $\exists n \in \mathbb{Z}$; “divisible by every integer m ” gives $\forall m \in \mathbb{Z}, (m \mid n)$; unfold $m \mid n$ as $\exists q \in \mathbb{Z}, n = qm$ [Definition 0.36]:

$$\exists n \in \mathbb{Z}, \forall m \in \mathbb{Z}, \exists q \in \mathbb{Z}, n = qm.$$

(It is true: $n = 0$ works, Exercise 0.38.)

Step 2 (b) “ n is a greatest odd integer” means: n is odd, and every odd m satisfies $m \leq n$. “ n is odd” is $\exists k \in \mathbb{Z}, n = 2k + 1$ [Proposition 1.1.58]. So

$$\neg\exists n \in \mathbb{Z}, \left([\exists k \in \mathbb{Z}, n = 2k + 1] \wedge \forall m \in \mathbb{Z}, ([\exists \ell \in \mathbb{Z}, m = 2\ell + 1] \Rightarrow m \leq n) \right).$$

Equivalent positive form (de Morgan): $\forall n \in \mathbb{Z}, (n \text{ odd} \Rightarrow \exists m \in \mathbb{Z}, (m \text{ odd} \wedge m > n))$.

Step 3 (c) “Any two distinct rationals a, b ” gives $\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}$ with the restriction $a \neq b$ as an implication; “between” is $a < c < b$ (or $b < c < a$). Taking $a < b$ without loss of generality:

$$\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}, (a < b \Rightarrow \exists c \in \mathbb{Q}, (a < c \wedge c < b)).$$

If you keep “distinct” literally: $\forall a, b \in \mathbb{Q}, (a \neq b \Rightarrow \exists c \in \mathbb{Q}, ((a < c \wedge c < b) \vee (b < c \wedge c < a)))$.

Step 4 (d) “An integer n has a rational square root x ”: $x \in \mathbb{Q}$ and $x^2 = n$. “That root is an integer”: $x \in \mathbb{Z}$. The “if ... then” is an implication under both quantifiers:

$$\forall n \in \mathbb{Z}, \forall x \in \mathbb{Q}, (x^2 = n \Rightarrow x \in \mathbb{Z}).$$

Conclusion. The four formulae above. □

Compare with your work.

- ☐ Every variable must be bound and have a domain ($n \in \mathbb{Z}, c \in \mathbb{Q}, \dots$).
- ☐ Inside $\forall$ a restriction is written with $\Rightarrow$ (" $a < b \Rightarrow \dots$ "); inside $\exists$ with $\wedge$ (" $a < c \wedge c < b$ "). Swapping these is the most common error.
- ☐ In (d) the quantifier over the root x must be $\forall$ ("that root" refers to any rational root), not $\exists$.

Exercise 1.2.14

bonus

Prove that $\forall x \in \mathbb{R}, x = x$.

Given. Nothing.

To show. A universal statement (shape: "let x ").

Step 1 Let $x \in \mathbb{R}$ be arbitrary [Strategy 1.2.10].

Step 2 $x = x$ holds, since every object is equal to itself [the meaning of $=$; the book treats this as a basic fact].

Step 3 Since x was arbitrary, $\forall x \in \mathbb{R}, x = x$ [$\forall I$].

Conclusion. Done. □

Compare with your work.

- ☐ The point of this exercise is the *form*: "let x ", then the statement, then "since x was arbitrary". Do not pick a value ("say $x = 17$ "), see the Common error on page 54.

Exercise 1.2.15

essential

Prove that for every integer n , n is even if and only if n^2 is even.

Given. Definition 0.41: n even means $2 \mid n$, i.e. $n = 2k$ for some $k \in \mathbb{Z}$; odd means not even. Theorem 0.42: $n = 2k$ or $n = 2k + 1$. Proposition 1.2.11: the square of an odd integer is odd.

To show. $\forall n \in \mathbb{Z}, (n \text{ even} \Leftrightarrow n^2 \text{ even})$ (shape: universal, then a biconditional: "let n ", then two implications).

Plan. $(\Rightarrow)$ is direct from the definition. $(\Leftarrow)$ is proved by contradiction: if n were odd, n^2 would be odd by Proposition 1.2.11.

Step 1 Let $n \in \mathbb{Z}$ [Strategy 1.2.10]. We prove both directions [Definition 1.1.40].

Step 2 $(\Rightarrow)$ Suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$ [Definition 0.41 and 0.36; take such a k , Strategy 1.2.30].

Step 3 $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, and $2k^2 \in \mathbb{Z}$ [closure]. So $2 \mid n^2$: n^2 is even [Definition 0.41].

Step 4 $(\Leftarrow)$ Suppose n^2 is even. Suppose, towards a contradiction, that n is not even, i.e. n is odd [Strategy 1.3.16/1.1.51; Definition 0.41].

Step 5 Then n^2 is odd [Proposition 1.2.11], i.e. n^2 is not even. This contradicts Step 4 [$\neg E$].

Step 6 Hence n is even [double negation, Theorem 1.3.15: "not not even" is "even"].

Step 7 Both implications hold, so n is even iff n^2 is even; since n was arbitrary, this holds for every integer [$\forall I$].

Conclusion. For every integer n : n is even $\Leftrightarrow n^2$ is even. □

Compare with your work.

- ☐ The $(\Leftarrow)$ direction cannot be done directly from " $n^2 = 2k$ "; it needs contradiction (or contraposition,

Exercise 1.3.22) via Proposition 1.2.11. If you re-prove 1.2.11 inline, write $n = 2k + 1$ and compute $n^2 = 2(2k^2 + 2k) + 1$.

□ “Odd” may be replaced by “ $n = 2k + 1$ ” only by quoting Proposition 1.1.58 (or Theorem 0.42).

Exercise 1.2.18

essential

Prove that for all real numbers x and y , if x is irrational, then $x + y$ and $x - y$ are not both rational.

Given. Theorem 0.16 (closure of $\mathbb{Q}$). “Irrational” = not rational.

To show. $\forall x, y \in \mathbb{R}, (x \notin \mathbb{Q} \Rightarrow \neg(x + y \in \mathbb{Q} \wedge x - y \in \mathbb{Q}))$ (shape: universal, implication, then a negation).

Plan. Let x, y ; assume x irrational; to prove the negation assume both sums are rational and derive a contradiction: their sum is $2x$, so x would be rational.

Step 1 Let $x, y \in \mathbb{R}$ [Strategy 1.2.10] and suppose x is irrational [Strategy 1.1.28].

Step 2 Suppose, towards a contradiction, that $x + y \in \mathbb{Q}$ and $x - y \in \mathbb{Q}$ [Strategy 1.1.51, then Strategy 1.1.10 to use both].

Step 3 Then $(x + y) + (x - y) \in \mathbb{Q}$ [Theorem 0.16, sums of rationals], and $(x + y) + (x - y) = 2x$ [algebra]. So $2x \in \mathbb{Q}$.

Step 4 Then $x = \frac{2x}{2} \in \mathbb{Q}$ [Theorem 0.16: quotient of rationals, $2 = \frac{2}{1} \in \mathbb{Q}, 2 \neq 0$].

Step 5 This contradicts “ x is irrational” [$\neg E$]. Hence $x + y$ and $x - y$ are not both rational [$\neg I$].

Step 6 Since x, y were arbitrary and the implication was derived, the statement holds [$\Rightarrow I, \forall I$].

Conclusion. For all real x, y : if x is irrational then $x + y, x - y$ are not both rational. □

Compare with your work.

- “Not both rational” is $\neg(p \wedge q)$: the natural proof assumes $p \wedge q$ and derives $\perp$. Do not try to prove “ $x + y$ is irrational” (false in general: $y = -x$ gives $x + y = 0$).
- Quote Theorem 0.16 for the sum and for the division by 2.

Exercise 1.2.21 and 1.2.22

recommended

1.2.21 Suppose $\forall x \in \mathbb{R}, P(x)$ is true. Show that we can conclude $P(21)$. **1.2.22** Let $P(x)$ and $Q(x)$ be predicates with $x \in \mathbb{R}$. Prove that $(\forall x \in \mathbb{R}, (P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x \in \mathbb{R}, P(x)) \Rightarrow (\forall x \in \mathbb{R}, Q(x)))$.

Given. The rules for $\forall$ (Definition 1.2.9).

Step 1 1.2.21. Assumption: $\forall x \in \mathbb{R}, P(x)$. Since $21 \in \mathbb{R}$, the rule $\forall E$ gives $P(21)$ [Strategy 1.2.19: a universal statement may be applied to any particular element of the domain]. □

Step 2 1.2.22. Assume $\forall x \in \mathbb{R}, (P(x) \Rightarrow Q(x))$ [Strategy 1.1.28]. Goal: $(\forall x \in \mathbb{R}, P(x)) \Rightarrow (\forall x \in \mathbb{R}, Q(x))$.

Step 3 Assume $\forall x \in \mathbb{R}, P(x)$ [Strategy 1.1.28]. Goal: $\forall x \in \mathbb{R}, Q(x)$.

Step 4 Let $x \in \mathbb{R}$ be arbitrary [Strategy 1.2.10]. Goal: $Q(x)$.

Step 5 From the first assumption, $P(x) \Rightarrow Q(x)$ [$\forall E$ at this x]. From the second, $P(x)$ [$\forall E$ at this x].

Step 6 Hence $Q(x)$ [modus ponens]. Since x was arbitrary, $\forall x \in \mathbb{R}, Q(x)$ [$\forall I$].

Step 7 Discharge the two assumptions [$\Rightarrow I$ twice].

Conclusion. Both statements are proved. □

Compare with your work.

- In 1.2.22 the two universal assumptions must be specialised to the *same* arbitrary x before modus

ponents can be used.

Exercise 1.2.28

recommended

Prove that there is an integer which is divisible by zero.

Given. Definition 0.36: $0 \mid n$ means $n = q \cdot 0$ for some $q \in \mathbb{Z}$.

To show. $\exists n \in \mathbb{Z}, \exists q \in \mathbb{Z}, n = q \cdot 0$ (shape: existence: define a witness and verify).

Step 1 Define $n = 0$. Then $n \in \mathbb{Z}$ [Strategy 1.2.24: the witness must be in the domain].

Step 2 $n = 0 = 0 \cdot 0$, so $q = 0 \in \mathbb{Z}$ satisfies $n = q \cdot 0$; hence $0 \mid n$ [Definition 0.36].

Step 3 So there is an integer divisible by zero $[\exists!]$.

Conclusion. 0 is divisible by 0 (and it is the only such integer: $q \cdot 0 = 0$ for every q). $\square$

Compare with your work.

$\square$ “Divisible by zero” is about multiplication by 0, not division by 0; your answer must display $0 = 0 \cdot 0$.

Exercise 1.2.37 (a), (b)

recommended

Write each statement as a logical formula using $\exists!$ and prove it: (a) for each real number a , the equation $x^2 + 2ax + a^2 = 0$ has exactly one real solution x ; (b) there is a unique real number a for which the equation $x^2 + a^2 = 0$ has a real solution x .

Given. Definition 1.2.32: $\exists!x \in X, p(x)$ means “ $\exists x \in X, p(x)$ and $\forall y, z \in X, (p(y) \wedge p(z)) \Rightarrow y = z$ ”. Strategy 1.2.35: prove existence, then uniqueness.

Step 1 (a) Formula: $\forall a \in \mathbb{R}, \exists!x \in \mathbb{R}, x^2 + 2ax + a^2 = 0$.

Step 2 Let $a \in \mathbb{R}$ [Strategy 1.2.10]. *Existence:* define $x = -a \in \mathbb{R}$; then $x^2 + 2ax + a^2 = a^2 - 2a^2 + a^2 = 0$ [Strategy 1.2.24].

Step 3 *Uniqueness:* let $y, z \in \mathbb{R}$ with $y^2 + 2ay + a^2 = 0$ and $z^2 + 2az + a^2 = 0$ [Strategy 1.2.35]. Since $y^2 + 2ay + a^2 = (y + a)^2$ [algebra], we get $(y + a)^2 = 0$, hence $y + a = 0$ [a real number whose square is 0 is 0], hence $y = -a$. Likewise $z = -a$. So $y = z$.

Step 4 Existence and uniqueness give $\exists!x \in \mathbb{R}, x^2 + 2ax + a^2 = 0$; since a was arbitrary, (a) holds. $\square$

Step 5 (b) Formula: $\exists!a \in \mathbb{R}, \exists x \in \mathbb{R}, x^2 + a^2 = 0$. (Note the inner quantifier is a plain $\exists$.)

Step 6 *Existence:* define $a = 0$; then $x = 0$ satisfies $0^2 + 0^2 = 0$, so $\exists x \in \mathbb{R}, x^2 + a^2 = 0$ holds for $a = 0$.

Step 7 *Uniqueness:* let $a, b \in \mathbb{R}$ both have the property. Take $x \in \mathbb{R}$ with $x^2 + a^2 = 0$ [Strategy 1.2.30]. Since $x^2 \geq 0$ and $a^2 \geq 0$, their sum is 0 only if both are 0 [if $a^2 > 0$ then $x^2 + a^2 > 0$]; so $a^2 = 0$ and $a = 0$. The same argument gives $b = 0$. Hence $a = b$.

Step 8 So $\exists!a \in \mathbb{R}, \exists x \in \mathbb{R}, x^2 + a^2 = 0$. $\square$

Compare with your work.

- $\square$ A unique-existence proof has two labelled parts. In uniqueness you take *two* objects with the property and show they are equal (or show any object with the property equals your witness).
- $\square$ In (a), the uniqueness argument must not just say “the quadratic has a double root”: show that any solution equals $-a$.

Exercise 1.2.39**essential**

Let $p(x, y)$ be the statement “ $x + y$ is even”, with $x, y \in \mathbb{Z}$. Prove that $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, p(x, y)$ is true, and that $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, p(x, y)$ is false.

Given. Definition 0.41 (even). The order of quantifiers matters (Theorem 1.2.40 and the discussion after it): in $\forall x \exists y$ the y may depend on x ; in $\exists y \forall x$ one y must work for every x .

To show. (i) prove $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, (x + y \text{ even})$; (ii) prove $\neg \exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, (x + y \text{ even})$.

Step 1 (i) Let $x \in \mathbb{Z}$ [Strategy 1.2.10]. Define $y = x$; then $y \in \mathbb{Z}$ [Strategy 1.2.24].

Step 2 $x + y = 2x$ with $x \in \mathbb{Z}$, so $2 \mid x + y$: $x + y$ is even [Definition 0.36, 0.41]. Hence $\exists y \in \mathbb{Z}, p(x, y)$, and as x was arbitrary, (i) holds [$\exists$ I, $\forall$ I].

Step 3 (ii) Suppose, towards a contradiction, that there is $y \in \mathbb{Z}$ such that $x + y$ is even for every $x \in \mathbb{Z}$ [Strategy 1.1.51; fix such a y , Strategy 1.2.30].

Step 4 Apply the “for every x ” to $x = 1 - y$, which is an integer [Strategy 1.2.19]: then $x + y = 1$ is even, i.e. $1 = 2k$ for some $k \in \mathbb{Z}$ [Definition 0.41].

Step 5 But then $k = \frac{1}{2}$, which is not an integer [no integer between 0 and 1]: contradiction.

Step 6 Hence no such y exists: the second statement is false [$\neg$ I].

Conclusion. $\forall x \exists y, p(x, y)$ is true; $\exists y \forall x, p(x, y)$ is false. $\square$

Compare with your work.

- ☐ In (i) the chosen y depends on x (“ $y = x$ ” or “ $y = -x$ ” or “ $y = x + 2$ ”); that dependence is exactly what (ii) forbids.
- ☐ In (ii), after fixing the supposed y , you must choose an x *in terms of* y that breaks the property; “ $x = 1 - y$ ” or “ $x = y + 1$ ” both work.

C LOGICAL EQUIVALENCE**What you quote in this section**

Definition 1.3.2: $p \equiv q$ iff each can be derived from the other. **Strategy 1.3.11:** for propositional formulae, $p \equiv q$ iff their truth-table columns agree. **Theorem 1.3.15:** $p \equiv \neg \neg p$. **Theorem 1.3.19:** $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$ (contraposition). **Theorems 1.3.24, 1.3.29 (de Morgan):** $\neg(p \wedge q) \equiv \neg p \vee \neg q$, $\neg(p \vee q) \equiv \neg p \wedge \neg q$, $\neg \forall x, p(x) \equiv \exists x, \neg p(x)$, $\neg \exists x, p(x) \equiv \forall x, \neg p(x)$. **Exercise 1.3.13:** $p \Rightarrow q \equiv \neg p \vee q$. **Exercise 1.3.27:** $\neg(p \Rightarrow q) \equiv p \wedge \neg q$. **Theorem 1.3.45:** $p \equiv q$ iff $p \Leftrightarrow q$ is a tautology; q can be derived from p iff $p \Rightarrow q$ is a tautology.

Exercise 1.3.5**essential**

Let p, q and r be propositional variables. Prove that $(p \vee q) \Rightarrow r$ is logically equivalent to $(p \Rightarrow r) \wedge (q \Rightarrow r)$.

Given. Definition 1.3.2 only (no truth tables yet in the book at this point).

To show. Two derivations: from $(p \vee q) \Rightarrow r$ derive $(p \Rightarrow r) \wedge (q \Rightarrow r)$; and from $(p \Rightarrow r) \wedge (q \Rightarrow r)$ derive $(p \vee q) \Rightarrow r$.

Step 1 *First derivation.* Assume $(p \vee q) \Rightarrow r$. Goal: $(p \Rightarrow r) \wedge (q \Rightarrow r)$; prove both conjuncts [Strategy 1.1.7].

Step 2 $p \Rightarrow r$: assume p [Strategy 1.1.28]. Then $p \vee q$ [$\vee$ I₁]. From $(p \vee q) \Rightarrow r$ and $p \vee q$: r [modus ponens]. Hence $p \Rightarrow r$.

Step 3 $q \Rightarrow r$: assume q . Then $p \vee q$ [$\vee I_2$], so r [modus ponens]. Hence $q \Rightarrow r$.

Step 4 So $(p \Rightarrow r) \wedge (q \Rightarrow r)$ [$\wedge I$].

Step 5 *Second derivation.* Assume $(p \Rightarrow r) \wedge (q \Rightarrow r)$; so $p \Rightarrow r$ and $q \Rightarrow r$ [$\wedge E$]. Goal: $(p \vee q) \Rightarrow r$.

Step 6 Assume $p \vee q$ [Strategy 1.1.28]; split into cases [Strategy 1.1.19]. Case p : r by modus ponens with $p \Rightarrow r$. Case q : r by modus ponens with $q \Rightarrow r$. In both cases r .

Step 7 Hence $(p \vee q) \Rightarrow r$ [$\Rightarrow I$].

Conclusion. Each formula is derivable from the other, so they are logically equivalent [Definition 1.3.2]. $\square$

Compare with your work.

- ☐ Two separate derivations, each beginning “assume ...” with the *other* formula as goal.
- ☐ Exercise 1.3.14 proves the same equivalence with a truth table; here a truth table is not what is asked.

Exercise 1.3.13

essential

Use a truth table to prove that $p \Rightarrow q \equiv (\neg p) \vee q$.

Given. Basic truth tables (Definition 1.3.7): $p \Rightarrow q$ is $\times$ only in the row $p = \checkmark, q = \times$; $\neg p$ flips; $p \vee q$ is $\times$ only when both are $\times$.

To show. Identical columns for the two formulae [Strategy 1.3.11].

Step 1 Two variables, so four rows. Columns: p , q , then $p \Rightarrow q$, then the subformula $\neg p$, then $(\neg p) \vee q$.

Step 2 Fill in:

p	q	$p \Rightarrow q$	$\neg p$	$(\neg p) \vee q$
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
$\checkmark$	$\times$	$\times$	$\times$	$\times$
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$

Row 2: $\neg p = \times$ and $q = \times$, so $(\neg p) \vee q = \times$; in rows 3–4 $\neg p = \checkmark$ makes the disjunction $\checkmark$; in row 1 $q = \checkmark$ does.

Step 3 The columns of $p \Rightarrow q$ and $(\neg p) \vee q$ agree in all four rows.

Conclusion. $p \Rightarrow q \equiv (\neg p) \vee q$ [Strategy 1.3.11]. $\square$

Compare with your work.

- ☐ A column for the subformula $\neg p$ and a final sentence “the columns agree, hence equivalent” are both required.

Exercise 1.3.14

essential

Use a truth table to prove that $(p \vee q) \Rightarrow r \equiv (p \Rightarrow r) \wedge (q \Rightarrow r)$.

To show. Identical columns, now with three variables (8 rows).

Step 1 Layout: p four $\checkmark$ then four $\times$; q alternating in pairs; r alternating. Subformula columns: $p \vee q, p \Rightarrow r, q \Rightarrow r$.

Step 2

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$	$p \Rightarrow r$	$q \Rightarrow r$	$(p \Rightarrow r) \wedge (q \Rightarrow r)$
✓	✓	✓	✓	✓	✓	✓	✓
✓	✓	×	✓	×	×	×	×
✓	×	✓	✓	✓	✓	✓	✓
✓	×	×	✓	×	×	✓	×
×	✓	✓	✓	✓	✓	✓	✓
×	✓	×	✓	×	✓	×	×
×	×	✓	×	✓	✓	✓	✓
×	×	×	×	✓	✓	✓	✓

Step 3 The fifth column and the last column agree in every row.

Conclusion. The formulae are logically equivalent [Strategy 1.3.11]. □

Compare with your work.

- Eight rows, no row skipped. The only rows where the formulas are $\times$ are those with $r = \times$ and at least one of p, q true (rows 2, 4, 6).

Exercise 1.3.22

essential

Use the law of contraposition to prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$. Use the technique this suggests to prove that an integer is even if and only if its square is even.

Given. Definition 1.1.40; Theorem 1.3.19: $a \Rightarrow b \equiv \neg b \Rightarrow \neg a$; Proposition 1.2.11 (odd squared is odd).

Step 1 $p \Leftrightarrow q$ is $(p \Rightarrow q) \wedge (q \Rightarrow p)$ [Definition 1.1.40].

Step 2 Apply contraposition to the second conjunct: $q \Rightarrow p \equiv (\neg p) \Rightarrow (\neg q)$ [Theorem 1.3.19 with $a = q, b = p$].

Step 3 Replacing a subformula by an equivalent one gives an equivalent formula, so $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$.

Step 4 *Technique.* To prove “ p iff q ”: prove $p \Rightarrow q$, then prove $\neg p \Rightarrow \neg q$ (instead of $q \Rightarrow p$).

Step 5 *Application.* Let $n \in \mathbb{Z}$; p : “ n is even”, q : “ n^2 is even”.

Step 6 $p \Rightarrow q$: suppose n is even, $n = 2k$ with $k \in \mathbb{Z}$ [Definition 0.41]. Then $n^2 = 2(2k^2)$ is even.

Step 7 $\neg p \Rightarrow \neg q$: suppose n is not even, i.e. odd [Definition 0.41]. Then n^2 is odd [Proposition 1.2.11], i.e. not even.

Step 8 By Step 3, n is even iff n^2 is even.

Conclusion. Both parts proved. □

Compare with your work.

- Compare with Exercise 1.2.15: the same mathematics, but the second half is now a direct proof of $\neg p \Rightarrow \neg q$ instead of a proof by contradiction. Both are acceptable; this one is shorter.

Exercise 1.3.27

essential

Prove that $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ in two ways: using truth tables, and using Exercise 1.3.13 together with de Morgan’s laws and double negation.

Step 1 Truth table.

p	q	$p \Rightarrow q$	$\neg(p \Rightarrow q)$	$\neg q$	$p \wedge (\neg q)$
✓	✓	✓	×	×	×
✓	×	×	✓	✓	✓
×	✓	✓	×	×	×
×	×	✓	×	✓	×

The columns of $\neg(p \Rightarrow q)$ and $p \wedge (\neg q)$ agree, so the formulae are equivalent [Strategy 1.3.11].

Step 2 Algebraically.

$$\begin{aligned}
 \neg(p \Rightarrow q) &\equiv \neg((\neg p) \vee q) && \text{[Exercise 1.3.13]} \\
 &\equiv (\neg\neg p) \wedge (\neg q) && \text{[de Morgan, Theorem 1.3.24(b)]} \\
 &\equiv p \wedge (\neg q) && \text{[double negation, Theorem 1.3.15].}
 \end{aligned}$$

Conclusion. $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$: an implication is false exactly when its antecedent is true and its consequent false. $\square$

Compare with your work.

- $\square$ Every line of the chain must cite the law used; three laws, three citations.

Exercise 1.3.32

recommended

Prove by counterexample that it is not the case that every rational number can be expressed as $\frac{a}{b}$, where a is an even integer and b is an odd integer.

Given. Strategy 1.3.30: to refute $\forall x \in X, p(x)$, give $a \in X$ with $\neg p(a)$. Here $p(x)$ is “ $\exists a, b \in \mathbb{Z}, (a \text{ even} \wedge b \text{ odd} \wedge x = \frac{a}{b})$ ”, so $\neg p(x)$ is “for all even a and odd $b, x \neq \frac{a}{b}$ ”.

Plan. Pick $x = \frac{1}{3}$ and prove $\neg p(\frac{1}{3})$ by contradiction: if $\frac{1}{3} = \frac{a}{b}$ then $b = 3a$ is even, but b is odd.

Step 1 Counterexample candidate: $x = \frac{1}{3} \in \mathbb{Q}$.

Step 2 Suppose, towards a contradiction, that $\frac{1}{3} = \frac{a}{b}$ with $a, b \in \mathbb{Z}, a \text{ even}, b \text{ odd}$ [Strategy 1.1.51].

Step 3 Cross-multiplying ($b \neq 0$ since b is odd): $b = 3a$.

Step 4 a is even, so $a = 2k$ for some $k \in \mathbb{Z}$ [Definition 0.41]; then $b = 3 \cdot 2k = 2(3k)$ is even [Definition 0.41].

Step 5 But b is odd, i.e. not even [Definition 0.41]: contradiction. So $\frac{1}{3}$ has no such expression.

Conclusion. $\frac{1}{3}$ is a counterexample, so not every rational number can be written as (even)/(odd) [Strategy 1.3.30]. $\square$

Compare with your work.

- $\square$ A counterexample must come with a *proof* that it fails the property; “ $\frac{1}{3}$ has odd numerator” is not enough, because $\frac{1}{3} = \frac{2}{6} = \dots$ has many representations.

Exercise 1.3.35

recommended

Which of the following formulae are maximally negated? (a) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, \neg(r(x, y) \wedge s(x, y))$; (b) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, (\neg r(x, y)) \vee (\neg s(x, y))$; (c) $\forall x \in \mathbb{R}, [x > 1 \Rightarrow (\exists y \in \mathbb{R}, [x < y \wedge \neg(x^2 \leq y)])]$; (d) $\neg \exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$.

Given. Definition 1.3.33: a formula is maximally negated if every $\neg$ appears immediately before a

predicate that contains no logical operators or quantifiers.

Plan. For each formula, locate every $\neg$ and look at what follows it.

Step 1 (a) **No.** The second negation is $\neg(r(x,y) \wedge s(x,y))$: it stands in front of a conjunction, not a bare predicate.

Step 2 (b) **Yes.** The negations are $\neg p(x)$, $\neg r(x,y)$, $\neg s(x,y)$, each directly before a predicate. (This is (a) after one de Morgan step.)

Step 3 (c) **Yes.** The only negation is $\neg(x^2 \leq y)$, in front of the predicate $x^2 \leq y$, which contains no operators or quantifiers. (One may also write it as $x^2 > y$, but it is already maximally negated.)

Step 4 (d) **No.** The negation is in front of $\exists x \in \mathbb{R}, \dots$, a quantified formula.

Conclusion. (a) no, (b) yes, (c) yes, (d) no. □

Compare with your work.

- The test is purely syntactic: “what stands right after each $\neg$ ”. The inequality $x^2 \leq y$ counts as a predicate.

Exercise 1.3.44 (a), (c), (e)

essential

Prove that each of the following formulae is a tautology, and say how it can be interpreted: (a) $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$; (c) $\exists y \in Y, \forall x \in X, p(x,y) \Rightarrow \forall x \in X, \exists y \in Y, p(x,y)$; (e) $(\neg \forall x \in X, p(x)) \Leftrightarrow (\exists x \in X, \neg p(x))$.

Given. Definition 1.3.40 (tautology: true under every assignment); Theorem 1.3.45: (a) q derivable from p iff $p \Rightarrow q$ is a tautology; (b) $p \equiv q$ iff $p \Leftrightarrow q$ is a tautology.

Plan. For an implication, derive the consequent from the antecedent (then Theorem 1.3.45(a)); for a biconditional, show the two sides are equivalent (then Theorem 1.3.45(b)).

Step 1 (a) Assume $(p \Rightarrow q) \wedge (q \Rightarrow r)$; so $p \Rightarrow q$ and $q \Rightarrow r$ [$\wedge E$]. To prove $p \Rightarrow r$, assume p . Then q [modus ponens], then r [modus ponens]. Hence $p \Rightarrow r$ [$\Rightarrow I$].

Step 2 This derivation uses no assumptions beyond the antecedent, so the implication is a tautology [Theorem 1.3.45(a)]. *Interpretation:* implications chain (transitivity): having proved $p \Rightarrow q$ and $q \Rightarrow r$ you may conclude $p \Rightarrow r$.

Step 3 (c) The consequent was derived from the antecedent in Theorem 1.2.40: assume $\exists y \in Y, \forall x \in X, p(x,y)$; take such a y_0 ; let $x \in X$ be arbitrary; then $p(x, y_0)$, so $\exists y \in Y, p(x,y)$; since x was arbitrary, $\forall x \in X, \exists y \in Y, p(x,y)$.

Step 4 So the implication is a tautology [Theorem 1.3.45(a)]. *Interpretation:* a single y that works for all x in particular gives, for each x , some y (the converse fails: the y in $\forall x \exists y$ may depend on x).

Step 5 (e) $\neg \forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$ [de Morgan for quantifiers, Theorem 1.3.29(a)].

Step 6 Hence the biconditional is a tautology [Theorem 1.3.45(b)]. *Interpretation:* to disprove a “for all” statement, exhibit a counterexample (Strategy 1.3.30).

Conclusion. (a), (c), (e) are tautologies with the stated meanings. □

Compare with your work.

- A truth table proves (a) as well (8 rows, all $\checkmark$), but (c) and (e) contain quantifiers, so only a derivation or a cited theorem can prove them.
- Each part needs the citation of Theorem 1.3.45 turning “derivable” into “tautology”.

D CHAPTER 1 EXERCISES**Exercise 1.E.11****recommended**

Prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$. How does this help to prove a statement of the form “ p if and only if q ”?

Step 1 $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$ [Definition 1.1.40].

Step 2 $q \Rightarrow p \equiv (\neg p) \Rightarrow (\neg q)$ [contraposition, Theorem 1.3.19].

Step 3 Substituting the equivalent second conjunct: $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$.

Step 4 *Use.* To prove “ p iff q ” you may prove $p \Rightarrow q$ and then $\neg p \Rightarrow \neg q$, i.e. assume p is false and show q is false. This is convenient when $\neg p$ is a more concrete hypothesis than q (e.g. “ n is odd” is easier to compute with than “ n^2 is even”, Exercise 1.3.22).

Conclusion. As stated. (Same content as Exercise 1.3.22.) □

Exercise 1.E.18**essential**

Let X be $\mathbb{Z}$ or $\mathbb{Q}$, and let p be the statement $\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])$. Write down $\neg p$ in maximally negated form. Prove that p is true when $X = \mathbb{Z}$ and false when $X = \mathbb{Q}$.

Given. The negation rules (Theorems 1.3.24, 1.3.29, 1.3.15, Exercise 1.3.27); Exercise 1.2.29 (between two rationals lies a rational); there is no integer strictly between n and $n + 1$ (page 6).

To show. (i) the maximally negated $\neg p$; (ii) a proof of p for $X = \mathbb{Z}$; (iii) a proof of $\neg p$ for $X = \mathbb{Q}$.

Plan. Read p first: “every x has a next element y : something bigger with nothing strictly in between”. True in $\mathbb{Z}$ ($y = x + 1$), false in $\mathbb{Q}$ (between x and any $y > x$ lies $\frac{x+y}{2}$).

Step 1 *Negation, one operator at a time:*

$$\begin{aligned} \neg p &\equiv \exists x \in X, \neg \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)]) && [\neg \forall \rightarrow \exists \neg] \\ &\equiv \exists x \in X, \forall y \in X, \neg(x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)]) && [\neg \exists \rightarrow \forall \neg] \\ &\equiv \exists x \in X, \forall y \in X, (\neg(x < y) \vee \neg[\forall z \in X, \neg(x < z \wedge z < y)]) && [\text{de Morgan (a)}] \\ &\equiv \exists x \in X, \forall y \in X, (x \geq y \vee [\exists z \in X, \neg \neg(x < z \wedge z < y)]) && [\neg \forall \rightarrow \exists \neg; \neg(x < y) \text{ is } x \geq y] \\ &\equiv \exists x \in X, \forall y \in X, (x \geq y \vee [\exists z \in X, (x < z \wedge z < y)]) && [\text{double negation}]. \end{aligned}$$

Step 2 $X = \mathbb{Z}$: p is true. Let $x \in \mathbb{Z}$ [Strategy 1.2.10]. Define $y = x + 1$; then $y \in \mathbb{Z}$ [closure] and $x < y$.

Step 3 Let $z \in \mathbb{Z}$ be arbitrary. If $x < z$ and $z < y = x + 1$, then z is an integer strictly between x and $x + 1$, which is impossible [consecutive integers, page 6]. So $\neg(x < z \wedge z < y)$ [$\neg I$]. Since z was arbitrary, $\forall z \in \mathbb{Z}, \neg(x < z \wedge z < y)$.

Step 4 So y satisfies $x < y \wedge [\dots]$ [$\wedge I$]; hence $\exists y \in \mathbb{Z}, \dots$ [$\exists I$]; since x was arbitrary, p holds for $X = \mathbb{Z}$.

Step 5 $X = \mathbb{Q}$: p is false. We prove $\neg p$ in the form of Step 1. Define $x = 0 \in \mathbb{Q}$ [Strategy 1.2.24; any rational works].

Step 6 Let $y \in \mathbb{Q}$ be arbitrary [Strategy 1.2.10]. We must prove $x \geq y \vee \exists z \in \mathbb{Q}, (x < z \wedge z < y)$. Split: either $y \leq 0$ or $y > 0$ [LEM].

Step 7 If $y \leq 0$: then $x = 0 \geq y$, the first disjunct holds [$\vee I_1$].

Step 8 If $y > 0$: define $z = \frac{0+y}{2} = \frac{y}{2}$. Then $z \in \mathbb{Q}$ [Theorem 0.16] and $0 < z < y$ [Exercise 1.2.29, or directly: $0 < \frac{y}{2}$ and $\frac{y}{2} < y$ since $y > 0$]. So the second disjunct holds [$\exists I, \vee I_2$].

Step 9 In both cases the disjunction holds; y was arbitrary; so $\neg p$ holds for $X = \mathbb{Q}$, i.e. p is false.

Conclusion. $\neg p \equiv \exists x \in X, \forall y \in X, (x \geq y \vee \exists z \in X, (x < z \wedge z < y))$; p is true in $\mathbb{Z}$ and false in $\mathbb{Q}$. $\square$

Compare with your work.

- ☐ Negate *one* outermost operator per line and name the rule; never change the domains $x \in X, y \in X, z \in X$.
- ☐ For $X = \mathbb{Q}$ the proof must follow the shape of $\neg p$: choose x , let y be arbitrary, produce z depending on y . The case $y \leq x$ must not be forgotten.

Exercise 1.E.28, 1.E.29, 1.E.30

recommended

True or false? Let φ be the formula $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$. **1.E.28** $\varphi \equiv \forall x < 0, \exists y \notin \mathbb{R}, y^2 \neq x$. **1.E.29** $\varphi \equiv \exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$. **1.E.30** $\varphi \equiv \exists x < 0, \forall y \in \mathbb{R}, y^2 = x$.

Given. “ $\forall x \geq 0, \dots$ ” abbreviates $\forall x \in \mathbb{R}, (x \geq 0 \Rightarrow \dots)$ and “ $\exists x \geq 0, \dots$ ” abbreviates $\exists x \in \mathbb{R}, (x \geq 0 \wedge \dots)$. The negation rules.

Plan. Compute the maximal negation of φ once; compare each proposed formula with it. A proposed formula that is not the maximal negation can still be equivalent by accident, so to say “false” we show the two formulae have different truth values (or different logical form in a way that matters).

Step 1 Maximally negate: $\varphi \equiv \exists x \in \mathbb{R}, \neg(x \geq 0 \Rightarrow \exists y \in \mathbb{R}, y^2 = x)$ $[\neg \forall \rightarrow \exists \neg] \equiv \exists x \in \mathbb{R}, (x \geq 0 \wedge \neg \exists y \in \mathbb{R}, y^2 = x)$ [Exercise 1.3.27] $\equiv \exists x \in \mathbb{R}, (x \geq 0 \wedge \forall y \in \mathbb{R}, y^2 \neq x)$ $[\neg \exists \rightarrow \forall \neg]$, i.e. $\exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$.

Step 2 1.E.29: true. The proposed formula is exactly the maximal negation of Step 1.

Step 3 1.E.28: false. Negation never changes the domain of a bound variable, but the proposed formula has $x < 0$ and $y \notin \mathbb{R}$. To be sure they are not equivalent, compare truth values: φ is *false* (every $x \geq 0$ has the real square root $y = \sqrt{x}$, so the un-negated statement is true), whereas the proposed formula is *true*: for $x < 0$ take $y = 2i \notin \mathbb{R}$ if $x \neq -4$ (then $y^2 = -4 \neq x$) and $y = i$ if $x = -4$ (then $y^2 = -1 \neq x$). Different truth values, so not equivalent.

Step 4 1.E.30: false. It negates the domain condition ($x < 0$ instead of $x \geq 0$) and leaves $y^2 = x$ un-negated. Compare truth values: φ is false (Step 3). The proposed formula says some negative x equals y^2 for *every* real y ; this is false too (for $y = 0$ and $y = 1$ we would need $x = 0$ and $x = 1$). Both false, so here truth values do not separate them; but equivalence of *formulae* must hold for every reading of the predicate (Definition 1.3.2 treats $y^2 = x$ as an abstract $q(x, y)$). Reading $q(x, y)$ as “ $y = y$ ” (always true) and $x \geq 0$ as “true”, $x < 0$ as “false”: the left side $\exists x, (\text{true} \wedge \forall y, \neg q)$ is false while the right side $\exists x, (\text{false} \wedge \dots)$ is also false; reading q as always false: left side true, right side false. So they are not equivalent.

Conclusion. 1.E.28 false, 1.E.29 true, 1.E.30 false. $\square$

Compare with your work.

- ☐ The maximal negation is computed by three rule applications; write them.
- ☐ For a “false”, the cleanest argument is a pair of truth values that differ; when both happen to be false (1.E.30), point to the rule that was violated (domain negated, predicate not negated).

Chapter 2

Sets

Contents

- A Sets (§2.1)
- B Set operations (§2.2)
- C Products (§2.2)
- D Chapter 2 exercises (2.E)

Planning

★ 2.1: 10, 27, 31, 35 · 2.2: 5, 24, 26, 28, 35, 36, 37, 40, 51 · 2.E: 6, 33, 38
 ★ 2.1: 17, 18, 23, 24, 32, 36 · 2.2: 4, 6, 9, 25, 41, 45, 53 · 2.E: 4, 5, 16, 28, 32, 41
 Bonus: 2.E.18. Tested in **Quiz 2** (7 Oct).

A SETS

What you unfold in this section

Membership in set-builder notation (Definition 2.1.4): $a \in \{x \in X \mid p(x)\}$ iff $a \in X$ and $p(a)$. **Subset** (Definition 2.1.6): $U \subseteq X$ iff $\forall a \in U, a \in X$; proved by “let $a \in U$... so $a \in X$ ” (Strategy 2.1.7). **Equality** (Axiom 2.1.14): $X = Y$ iff $X \subseteq Y$ and $Y \subseteq X$ (double containment, Strategy 2.1.15), equivalently $\forall a, (a \in X \Leftrightarrow a \in Y)$. **Empty** (Definition 2.1.19, Strategy 2.1.20): to prove X empty, assume $a \in X$ and derive a contradiction; to prove non-empty, exhibit an element. **Power set** (Definition 2.1.29): $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$.

Exercise 2.1.10

essential

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that $[a, b] \subseteq (c, d]$ if and only if $a > c$ and $b \leq d$.

Given. $a < b, c < d$ (so both intervals are non-empty). $x \in [a, b]$ iff $a \leq x < b$; $x \in (c, d]$ iff $c < x \leq d$ [Definition 0.28].

To show. A biconditional. ($\Leftarrow$): from $a > c, b \leq d$ prove the containment by Strategy 2.1.7. ($\Rightarrow$): from the containment prove $a > c$ (test the element a) and $b \leq d$ (by contradiction, test an element just below b).

Step 1 ($\Leftarrow$) Suppose $a > c$ and $b \leq d$ [Strategy 1.1.28]. Let $x \in [a, b]$, so $a \leq x < b$ [Strategy 2.1.7, Definition 0.28].

Step 2 $c < a \leq x$ gives $c < x$; $x < b \leq d$ gives $x < d$, hence $x \leq d$ [transitivity of $<, \leq$]. So $c < x \leq d$, i.e. $x \in (c, d]$.

Step 3 Since x was arbitrary, $[a, b] \subseteq (c, d]$ [Definition 2.1.6].

Step 4 ($\Rightarrow$) Suppose $[a, b] \subseteq (c, d]$. First $a > c$: since $a \leq a < b, a \in [a, b]$ [here $a < b$ is used]. By the containment, $a \in (c, d]$, so $c < a \leq d$; in particular $a > c$.

Step 5 Then $b \leq d$: suppose, towards a contradiction, that $b > d$ [Strategy 1.3.16]. Define $x = \frac{d+b}{2}$. Then $d < x < b$ [midpoint of $d < b$].

Step 6 $x \geq a$: because $x > d \geq a$ [$a \leq d$ from Step 4]. So $a \leq x < b$: $x \in [a, b]$. By the containment, $x \in (c, d]$, so $x \leq d$. But $x > d$ [Step 5]: contradiction.

Step 7 Hence $b \leq d$ [$\neg I$ and double negation]. Both $a > c$ and $b \leq d$ hold [$\wedge I$].

Conclusion. $[a, b] \subseteq (c, d]$ iff $a > c$ and $b \leq d$. □

Compare with your work.

- $(\Rightarrow)$ cannot be done by “let $x \in [a, b]$ ”: you must *apply* the containment to well-chosen elements (a itself; a point between d and b).
- Note where $a < b$ is used (Step 4: so that $a \in [a, b)$). Without it, $[a, b) = \emptyset$ is contained in everything and the statement fails.

Exercise 2.1.17

recommended

Prove that $\{x \in \mathbb{R} \mid x^2 < x\} = (0, 1)$.

Given. $a \in \{x \in \mathbb{R} \mid x^2 < x\}$ iff $a \in \mathbb{R}$ and $a^2 < a$ [Definition 2.1.4]; $a \in (0, 1)$ iff $0 < a < 1$.

To show. Set equality: double containment [Strategy 2.1.15].

Plan. $(\subseteq)$ $a^2 < a$ means $a(a - 1) < 0$: a product is negative iff the factors have opposite signs. $(\supseteq)$ multiply $a < 1$ by the positive number a .

Step 1 $(\subseteq)$ Let $a \in \{x \in \mathbb{R} \mid x^2 < x\}$: so $a \in \mathbb{R}$ and $a^2 < a$. Then $a^2 - a < 0$, i.e. $a(a - 1) < 0$ [algebra].

Step 2 A product of two reals is negative only if one factor is positive and the other negative. *Case 1:* $a > 0$ and $a - 1 < 0$; then $0 < a < 1$. *Case 2:* $a < 0$ and $a - 1 > 0$; then $a < 0$ and $a > 1$, impossible. So $0 < a < 1$, i.e. $a \in (0, 1)$.

Step 3 $(\supseteq)$ Let $a \in (0, 1)$, so $0 < a < 1$. Multiplying $a < 1$ by $a > 0$ preserves the inequality: $a \cdot a < 1 \cdot a$, i.e. $a^2 < a$ [inequalities and positive factors]. Also $a \in \mathbb{R}$. So $a \in \{x \in \mathbb{R} \mid x^2 < x\}$.

Conclusion. The two sets are equal [Axiom 2.1.14]. □

Compare with your work.

- Two labelled containments, each starting with “let $a \in \dots$ ”.
- Dividing $a^2 < a$ by a requires knowing $a > 0$ first; the factorisation $a(a - 1) < 0$ avoids the circularity.

Exercise 2.1.18

recommended

Prove by double containment that $\{0, 1\} = \{1, 0\}$ and $\{0, 0\} = \{0\}$.

Given. $a \in \{x_1, \dots, x_n\}$ iff $a = x_1$ or $\dots$ or $a = x_n$ (list notation).

Step 1 $\{0, 1\} \subseteq \{1, 0\}$: let $a \in \{0, 1\}$; then $a = 0$ or $a = 1$ [list notation]. In either case a is one of the listed elements $1, 0$, so $a \in \{1, 0\}$.

Step 2 $\{1, 0\} \subseteq \{0, 1\}$: let $a \in \{1, 0\}$; then $a = 1$ or $a = 0$, so $a \in \{0, 1\}$. Hence $\{0, 1\} = \{1, 0\}$ [Axiom 2.1.14].

Step 3 $\{0, 0\} \subseteq \{0\}$: let $a \in \{0, 0\}$; then $a = 0$ or $a = 0$, so $a = 0$ [$p \vee p$ gives p by cases], so $a \in \{0\}$.

Step 4 $\{0\} \subseteq \{0, 0\}$: let $a \in \{0\}$; then $a = 0$, so $a \in \{0, 0\}$. Hence $\{0, 0\} = \{0\}$.

Conclusion. Order and repetition in list notation do not matter, as a consequence of extensionality. □

Exercise 2.1.23

recommended

Let $a, b \in \mathbb{R}$. Prove that $[a, b]$ is empty if and only if $a > b$, and that (a, b) is empty if and only if $a \geq b$.

Given. Strategy 2.1.20; Definition 0.28; Exercise 1.2.29 (a midpoint lies strictly between).

To show. Two biconditionals. For “empty $\Rightarrow$ inequality” we argue by contraposition: if the inequality fails, exhibit an element.

Step 1 $[a, b]$, $(\Leftarrow)$: suppose $a > b$. Suppose $x \in [a, b]$; then $a \leq x \leq b$, so $a \leq b$ [transitivity], contradicting $a > b$. Hence $[a, b]$ has no element: it is empty [Strategy 2.1.20].

Step 2 $[a, b]$, $(\Rightarrow)$: by contraposition [Strategy 1.3.20] suppose $a \leq b$ (the negation of $a > b$). Then $a \leq a \leq b$, so $a \in [a, b]$, and $[a, b]$ is non-empty. Hence if $[a, b]$ is empty then $a > b$.

Step 3 (a, b) , $(\Leftarrow)$: suppose $a \geq b$. If $x \in (a, b)$ then $a < x < b$, so $a < b$, contradicting $a \geq b$. So (a, b) is empty.

Step 4 (a, b) , $(\Rightarrow)$: by contraposition suppose $a < b$. Define $x = \frac{a+b}{2}$; then $a < x < b$ [Exercise 1.2.29’s computation: $2a < a + b < 2b$], so $x \in (a, b)$ and (a, b) is non-empty.

Conclusion. $[a, b] = \emptyset \Leftrightarrow a > b$ and $(a, b) = \emptyset \Leftrightarrow a \geq b$. □

Compare with your work.

- Four parts. The two “ $\Rightarrow$ ” parts are easiest by contraposition; say “by contraposition” and negate correctly ($\neg(a > b)$ is $a \leq b$; $\neg(a \geq b)$ is $a < b$).

Exercise 2.1.24

recommended

Let E be an empty set and let $p(x)$ be a predicate with domain E . Show that $\forall x \in E, p(x)$ is true and $\exists x \in E, p(x)$ is false. What does $\forall x \in E, x \neq x$ mean? Is it true?

Given. Definition 2.1.2: $\forall x \in E, p(x)$ means $\forall x, (x \in E \Rightarrow p(x))$ and $\exists x \in E, p(x)$ means $\exists x, (x \in E \wedge p(x))$. E empty: $\neg \exists a, a \in E$. Principle of explosion (Axiom 1.1.67).

Step 1 $\forall x \in E, p(x)$ is true. Let x be arbitrary [Strategy 1.2.10]; we prove $x \in E \Rightarrow p(x)$. Assume $x \in E$ [Strategy 1.1.28].

Step 2 Since E is empty, $x \in E$ is a contradiction [$\neg E$ with $\neg \exists a, a \in E$]. From a contradiction, $p(x)$ follows [explosion]. Hence $x \in E \Rightarrow p(x)$ [$\Rightarrow I$]; since x was arbitrary, the universal statement is true (“vacuously”).

Step 3 $\exists x \in E, p(x)$ is false. Suppose it were true; take x with $x \in E \wedge p(x)$ [Strategy 1.2.30]. Then $x \in E$ [$\wedge E$], contradicting emptiness. So the statement is false [$\neg I$].

Step 4 $\forall x \in E, x \neq x$ means: every element of the empty set is different from itself. By Step 2 with $p(x)$: “ $x \neq x$ ”, it is **true**.

Conclusion. Universal statements over the empty set are true, existential ones false. □

Compare with your work.

- The proof of the universal statement must go through the implication form and explosion; “there is nothing to check” is the idea, not the proof.

Exercise 2.1.27

essential

Is it true that $\emptyset \in X$ for all sets X ? Is it true that $\emptyset \subseteq X$ for all sets X ?

Given. Definition 2.1.6 and Exercise 2.1.24.

Step 1 $\emptyset \in X$ for all X : **false**. Counterexample $X = \emptyset$ [Strategy 1.3.30]: $\emptyset$ has no elements, so $\emptyset \notin \emptyset$. (Also $X = \{1\}$: its only element is $1 \neq \emptyset$.)

Step 2 $\emptyset \subseteq X$ for all X : **true**. Let X be any set. $\emptyset \subseteq X$ means $\forall a \in \emptyset, a \in X$ [Definition 2.1.6], a universal statement over the empty set, which is true [Exercise 2.1.24].

Conclusion. $\emptyset \in X$ fails for some X ; $\emptyset \subseteq X$ holds for every X . □

Compare with your work.

□ $\in$ and $\subseteq$ are different relations; this exercise is the standard test of that distinction.

Exercise 2.1.31

essential

Write out the elements of $\mathcal{P}(\{1, 2, 3\})$.

Given. $\mathcal{P}(X) = \{U \mid U \subseteq X\}$ [Definition 2.1.29].

Plan. List subsets by size: 0, 1, 2, 3 elements.

Step 1 Size 0: $\emptyset$ [Exercise 2.1.27]. Size 1: $\{1\}, \{2\}, \{3\}$. Size 2: $\{1, 2\}, \{1, 3\}, \{2, 3\}$. Size 3: $\{1, 2, 3\}$ [$X \subseteq X$, Example 2.1.8].

Step 2 So

$$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Conclusion. Eight subsets (2^3 : each of 1, 2, 3 is in or out). □

Compare with your work.

□ $\emptyset$ and $\{1, 2, 3\}$ itself are the two most often forgotten; the elements are *sets*, so $1 \in \mathcal{P}(\{1, 2, 3\})$ is wrong, $\{1\} \in \mathcal{P}(\{1, 2, 3\})$ is right.

Exercise 2.1.32

recommended

Let X be a set. Show that $\emptyset \in \mathcal{P}(X)$ and $X \in \mathcal{P}(X)$.

Step 1 $\emptyset \subseteq X$ [Exercise 2.1.27], so $\emptyset \in \mathcal{P}(X)$ [Definition 2.1.29: $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$].

Step 2 $X \subseteq X$ [Example 2.1.8: let $a \in X$; then $a \in X$], so $X \in \mathcal{P}(X)$ [Definition 2.1.29].

Conclusion. Every power set contains $\emptyset$ and X ; in particular $\mathcal{P}(X)$ is never empty. □

Exercise 2.1.35

essential

Let X and Y be sets. Prove that if $X \subseteq Y$, then $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Given. $X \subseteq Y$. Transitivity of $\subseteq$ (Proposition 2.1.12).

To show. $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$ (shape: a subset statement about sets of sets: “let $U \in \mathcal{P}(X)$... so $U \in \mathcal{P}(Y)$ ”).

Step 1 Suppose $X \subseteq Y$ [Strategy 1.1.28]. Let $U \in \mathcal{P}(X)$ [Strategy 2.1.7].

Step 2 Then $U \subseteq X$ [Definition 2.1.29].

Step 3 From $U \subseteq X$ and $X \subseteq Y$: $U \subseteq Y$ [Proposition 2.1.12; or directly: let $a \in U$, then $a \in X$, then $a \in Y$].

Step 4 Hence $U \in \mathcal{P}(Y)$ [Definition 2.1.29]. Since U was arbitrary, $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Conclusion. If $X \subseteq Y$ then $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$. □

Compare with your work.

□ The element of $\mathcal{P}(X)$ is a *set* U ; the proof unfolds $\in \mathcal{P}$ into $\subseteq$ twice (Steps 2 and 4).

Exercise 2.1.36**recommended**

Determine whether each is true or false: (a) $\mathcal{P}(\emptyset) \in \{\emptyset\}$; (b) $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$; (c) $\mathcal{P}(\mathcal{P}(\emptyset)) \in \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$. Then repeat with $\in$ replaced by $\subseteq$.

Plan. Compute the power sets first, then test each membership/containment against list notation.

Step 1 $\mathcal{P}(\emptyset) = \{\emptyset\}$ [the only subset of $\emptyset$ is $\emptyset$]. $\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$ [subsets of a one-element set: $\emptyset$ and the set itself].

Step 2 (a) **False.** Is $\{\emptyset\}$ an element of $\{\emptyset\}$? The only element of $\{\emptyset\}$ is $\emptyset$, and $\{\emptyset\} \neq \emptyset$ (one has an element, the other none) [Axiom 2.1.14].

Step 3 (b) **True.** $\mathcal{P}(\emptyset) \subseteq \mathcal{P}(\emptyset)$ [Example 2.1.8], so $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$ [Definition 2.1.29].

Step 4 (c) **True.** $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$ is literally the second listed element of $\{\emptyset, \{\emptyset, \{\emptyset\}\}\}$.

Step 5 (a') $\{\emptyset\} \subseteq \{\emptyset\}$: **true** [Example 2.1.8].

Step 6 (b') $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$: **true**; the only element $\emptyset$ of the left set is listed on the right.

Step 7 (c') $\{\emptyset, \{\emptyset\}\} \subseteq \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$: **false**; $\{\emptyset\}$ is an element of the left set, but the right set's elements are $\emptyset$ and $\{\emptyset, \{\emptyset\}\}$, and $\{\emptyset\}$ equals neither ($\{\emptyset\} \neq \emptyset$ as it has an element; $\{\emptyset\} \neq \{\emptyset, \{\emptyset\}\}$ since $\{\emptyset\}$ is an element of the latter but not of the former).

Conclusion. (a) F, (b) T, (c) T; (a') T, (b') T, (c') F. □

Compare with your work.

- Always compute $\mathcal{P}(\emptyset) = \{\emptyset\}$ and $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$ first; then every question is a comparison of lists. “Different” must be justified by an element in one set and not the other.

B SET OPERATIONS**What you unfold in this section**

$x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$ (Definition 2.2.1); $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$ (Definition 2.2.21); $x \in X \setminus A \Leftrightarrow x \in X \wedge x \notin A$ (Definition 2.2.38); $x \in \bigcap_{i \in I} A_i \Leftrightarrow \forall i \in I, x \in A_i$ (Definition 2.2.16); $x \in \bigcup_{i \in I} A_i \Leftrightarrow \exists i \in I, x \in A_i$ (Definition 2.2.29). Every identity is proved by double containment or by a chain of “iff”s on membership, each justified by one of these and a logical law.

Exercise 2.2.4**recommended**

Prove that $[0, \infty) \cap \mathbb{Z} = \mathbb{N}$.

Given. $x \in [0, \infty)$ iff $x \in \mathbb{R}$ and $x \geq 0$; $\mathbb{N}$ = the non-negative integers (Definitions 0.6, 0.8: natural numbers are the integers reached by steps to the right from 0).

Step 1 ($\subseteq$) Let $x \in [0, \infty) \cap \mathbb{Z}$. Then $x \in [0, \infty)$ and $x \in \mathbb{Z}$ [Definition 2.2.1]: x is an integer with $x \geq 0$. An integer ≥ 0 is reached from 0 by unit steps to the right, so $x \in \mathbb{N}$ [Definitions 0.6, 0.8].

Step 2 ($\supseteq$) Let $n \in \mathbb{N}$. Then n is an integer [$\mathbb{N} \subseteq \mathbb{Z}$, Definition 0.8] and $n \geq 0$ [Definition 0.6], so $n \in [0, \infty)$. Hence $n \in [0, \infty) \cap \mathbb{Z}$ [Definition 2.2.1].

Conclusion. $[0, \infty) \cap \mathbb{Z} = \mathbb{N}$ [Axiom 2.1.14]. □

Exercise 2.2.5**essential**

Write down the elements of $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\}$.

Step 1 An element of the intersection must be in both lists [Definition 2.2.1]. Go through the first list: 0 (not in the second), 1 (yes), 4 (yes), 7 (no).

Answer. $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\} = \{1, 4\}$. □

Exercise 2.2.6**recommended**

Express $[-2, 5) \cap [4, 7)$ as a single interval.

Step 1 $x \in [-2, 5) \cap [4, 7)$ iff $(-2 \leq x < 5 \text{ and } 4 \leq x < 7)$ [Definitions 2.2.1, 0.28].

Step 2 Both lower bounds must hold: $x \geq -2$ and $x \geq 4$, i.e. $x \geq 4$. Both upper bounds: $x < 5$ and $x < 7$, i.e. $x < 5$. So the condition is $4 \leq x < 5$.

Answer. $[-2, 5) \cap [4, 7) = [4, 5)$ [Axiom 2.1.14: same membership condition]. □

Compare with your work.

□ Include the one-line justification “ x is in the set iff ...”; the bracket at 5 is round because $5 \notin [-2, 5)$.

Exercise 2.2.9**recommended**

Let X be a set. Prove that $X \cap \emptyset = \emptyset$ and $X \cap X = X$.

Step 1 $X \cap \emptyset = \emptyset$. Suppose, towards a contradiction, that $a \in X \cap \emptyset$ [Strategy 2.1.20]. Then $a \in \emptyset$ [Definition 2.2.1, $\wedge E_2$], contradicting that $\emptyset$ has no elements. So $X \cap \emptyset$ is empty, and any two empty sets are equal [Theorem 2.1.25]: $X \cap \emptyset = \emptyset$.

Step 2 $X \cap X = X$. Let a be arbitrary. $a \in X \cap X \Leftrightarrow (a \in X \wedge a \in X) \Leftrightarrow a \in X$ [Definition 2.2.1; $p \wedge p \equiv p$]. So the sets are equal [Axiom 2.1.14].

Conclusion. Both identities hold. □

Compare with your work.

□ For the first, the clean route is “assume an element, contradiction, hence empty, hence $= \emptyset$ by Theorem 2.1.25”.

Exercise 2.2.24**essential**

Write down the elements of $\{0, 1, 4, 7\} \cup \{1, 2, 3, 4, 5\}$.

Step 1 An element of the union is in at least one of the lists [Definition 2.2.21]. Merge the two lists and write each element once [repetition does not matter, Exercise 2.1.18].

Answer. $\{0, 1, 2, 3, 4, 5, 7\}$. □

Exercise 2.2.25**recommended**

Express $[-2, 5) \cup [4, 7)$ as a single interval.

Step 1 x is in the union iff $-2 \leq x < 5$ or $4 \leq x < 7$ [Definitions 2.2.21, 0.28].

Step 2 If $-2 \leq x < 5$ then $-2 \leq x < 7$; if $4 \leq x < 7$ then $-2 \leq x < 7$. Conversely if $-2 \leq x < 7$:

either $x < 5$ (then $x \in [-2, 5)$) or $x \geq 5$ (then $4 \leq x < 7$, so $x \in [4, 7)$). So the condition is $-2 \leq x < 7$.

Answer. $[-2, 5) \cup [4, 7) = [-2, 7)$. □

Compare with your work.

- The “conversely” half needs the overlap: the two ranges together cover $[-2, 7)$ without a gap because $4 < 5$.

Exercise 2.2.26

essential

Let X and Y be sets. Prove that $X \subseteq Y$ if and only if $X \cup Y = Y$.

To show. Two implications; the first contains a set equality (double containment).

Step 1 ($\Rightarrow$) Suppose $X \subseteq Y$. $X \cup Y \subseteq Y$: let $a \in X \cup Y$; then $a \in X$ or $a \in Y$ [Definition 2.2.21]. Case $a \in X$: then $a \in Y$ [$X \subseteq Y$]. Case $a \in Y$: done. In both cases $a \in Y$ [Strategy 1.1.19].

Step 2 $Y \subseteq X \cup Y$: let $a \in Y$; then $a \in X \vee a \in Y$ [$\vee I_2$], so $a \in X \cup Y$. Hence $X \cup Y = Y$ [Axiom 2.1.14].

Step 3 ($\Leftarrow$) Suppose $X \cup Y = Y$. Let $a \in X$ [Strategy 2.1.7]. Then $a \in X \cup Y$ [$\vee I_1$], and $X \cup Y = Y$, so $a \in Y$. Hence $X \subseteq Y$.

Conclusion. $X \subseteq Y \Leftrightarrow X \cup Y = Y$. □

Compare with your work.

- Three containments in total; the case split in Step 1 is where the hypothesis $X \subseteq Y$ is used.

Exercise 2.2.28

essential

Let X , Y and Z be sets. Prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.

Plan. Double containment. ($\subseteq$) cases on the union. ($\supseteq$) LEM on “ $x \in X$ ” (Strategy 1.1.61): if $x \notin X$, both memberships $x \in X \cup Y$, $x \in X \cup Z$ force $x \in Y$ and $x \in Z$.

Step 1 ($\subseteq$) Let $x \in X \cup (Y \cap Z)$. Then $x \in X$ or $x \in Y \cap Z$ [Definition 2.2.21].

Step 2 Case $x \in X$: then $x \in X \cup Y$ and $x \in X \cup Z$ [$\vee I_1$ twice], so $x \in (X \cup Y) \cap (X \cup Z)$ [Definition 2.2.1].

Step 3 Case $x \in Y \cap Z$: then $x \in Y$ and $x \in Z$ [Definition 2.2.1], so $x \in X \cup Y$ and $x \in X \cup Z$ [$\vee I_2$ twice], so $x \in (X \cup Y) \cap (X \cup Z)$. Both cases done.

Step 4 ($\supseteq$) Let $x \in (X \cup Y) \cap (X \cup Z)$, so $x \in X \cup Y$ and $x \in X \cup Z$ [Definition 2.2.1]. By LEM, $x \in X$ or $x \notin X$ [Strategy 1.1.61].

Step 5 Case $x \in X$: then $x \in X \cup (Y \cap Z)$ [$\vee I_1$].

Step 6 Case $x \notin X$: from $x \in X \cup Y$, i.e. $x \in X \vee x \in Y$, and $x \notin X$ we get $x \in Y$ [cases: the first disjunct contradicts $x \notin X$]. Likewise from $x \in X \cup Z$ we get $x \in Z$. So $x \in Y \cap Z$, hence $x \in X \cup (Y \cap Z)$ [$\vee I_2$].

Conclusion. $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$ [Axiom 2.1.14]. □

Compare with your work.

- The ($\supseteq$) direction is the one that needs an idea; the case split on $x \in X$ must be announced (“either $x \in X$ or not”).
- Alternatively prove the membership equivalence as the logical law $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ with a truth table (Strategy 1.3.11), then cite extensionality.

Exercise 2.2.35**essential**

Express $\bigcup_{n \geq 1} \left[0, 1 - \frac{1}{n}\right]$ as an interval.

Given. $x \in \bigcup_{n \geq 1} A_n$ iff $\exists n \geq 1, x \in A_n$ [Definition 2.2.29]. $\lceil y \rceil$ is the least integer $\geq y$ (Definition 0.10).

To show. Guess the interval, then prove equality by double containment. The sets $[0, 1 - \frac{1}{n}]$ are $[0, 0], [0, \frac{1}{2}], [0, \frac{2}{3}], \dots$, growing towards 1 but never containing 1: the union should be $[0, 1)$.

Plan. ($\subseteq$) an element of some $[0, 1 - \frac{1}{n}]$ is < 1 . ($\supseteq$) given $x \in [0, 1)$, build an index N with $1 - \frac{1}{N} \geq x$, i.e. $N \geq \frac{1}{1-x}$ (the technique of Example 2.2.19).

Step 1 Claim: $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}] = [0, 1)$.

Step 2 ($\subseteq$) Let x be in the union. Take $n \geq 1$ with $x \in [0, 1 - \frac{1}{n}]$ [Strategy 1.2.30]. Then $0 \leq x \leq 1 - \frac{1}{n} < 1$ [$\frac{1}{n} > 0$]. So $x \in [0, 1)$.

Step 3 ($\supseteq$) Let $x \in [0, 1)$, so $0 \leq x < 1$, hence $1 - x > 0$ and $\frac{1}{1-x} \geq 1$ [$0 < 1 - x \leq 1$].

Step 4 Define $N = \lceil \frac{1}{1-x} \rceil$. Then $N \in \mathbb{Z}$ and $N \geq \frac{1}{1-x} \geq 1$ [Definition 0.10], so $N \geq 1$ is a valid index.

Step 5 From $N \geq \frac{1}{1-x}$ and $N, 1 - x > 0$: $\frac{1}{N} \leq 1 - x$, i.e. $x \leq 1 - \frac{1}{N}$ [invert both sides of an inequality between positive numbers].

Step 6 So $0 \leq x \leq 1 - \frac{1}{N}$: $x \in [0, 1 - \frac{1}{N}]$, hence $x \in \bigcup_{n \geq 1} [0, 1 - \frac{1}{n}]$ [$\exists$ I with $n = N$].

Conclusion. $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}] = [0, 1)$ [Axiom 2.1.14]. □

Compare with your work.

- The ($\supseteq$) direction must *construct* N from x and check $N \geq 1$ and $x \leq 1 - \frac{1}{N}$. “Take n large enough” is not a proof.
- 1 is not in the union: $1 \leq 1 - \frac{1}{n}$ fails for every n . That is why the answer is $[0, 1)$ and not $[0, 1]$.

Exercise 2.2.36**essential**

Prove that $\bigcap_{n \in \mathbb{N}} [n] = \emptyset$ and $\bigcup_{n \in \mathbb{N}} [n] = \mathbb{N}$.

Given. $[n] = \{k \in \mathbb{N} \mid k < n\}$ (Definition 0.25); $[0] = \emptyset$ (Example 2.1.22). Definitions 2.2.16 and 2.2.29 for $\cap, \cup$ over the index set $\mathbb{N}$.

Step 1 Intersection. Suppose $x \in \bigcap_{n \in \mathbb{N}} [n]$ [Strategy 2.1.20]. Then $x \in [n]$ for every $n \in \mathbb{N}$ [Definition 2.2.16]; in particular $x \in [0]$ [Strategy 1.2.19 with $n = 0$].

Step 2 But $[0] = \{k \in \mathbb{N} \mid k < 0\}$ has no elements [no natural number is < 0]: contradiction. So the intersection is empty, hence equals $\emptyset$ [Theorem 2.1.25].

Step 3 Union, ($\subseteq$). Let $x \in \bigcup_{n \in \mathbb{N}} [n]$; take n with $x \in [n]$ [Definition 2.2.29, Strategy 1.2.30]. Then $x \in \mathbb{N}$ [Definition 0.25: elements of $[n]$ are natural numbers].

Step 4 Union, ($\supseteq$). Let $k \in \mathbb{N}$. Then $k + 1 \in \mathbb{N}$ and $k < k + 1$, so $k \in [k + 1]$ [Definition 0.25]. Hence $k \in \bigcup_{n \in \mathbb{N}} [n]$ [$\exists$ I with $n = k + 1$].

Conclusion. $\bigcap_{n \in \mathbb{N}} [n] = \emptyset$ and $\bigcup_{n \in \mathbb{N}} [n] = \mathbb{N}$. □

Compare with your work.

- For the intersection a single index ($n = 0$) kills every candidate. For the union each k needs its own index ($n = k + 1$; $n = k$ would be wrong since $k \notin [k]$).

Exercise 2.2.37

essential

Find a family $\{X_n \mid n \in \mathbb{N}\}$ of sets such that (i) $\bigcup_{n \in \mathbb{N}} X_n = \mathbb{N}$; (ii) $\bigcap_{n \in \mathbb{N}} X_n = \emptyset$; (iii) $X_i \cap X_j \neq \emptyset$ for all $i, j \in \mathbb{N}$.

To show. An example with three verifications. Conditions (ii) and (iii) together say: no number is in all the sets, yet any two sets overlap. So each X_n should omit only a little.

Plan. Let X_n be $\mathbb{N}$ with the single number n removed.

Step 1 Define $X_n = \{k \in \mathbb{N} \mid k \neq n\} = \mathbb{N} \setminus \{n\}$ for $n \in \mathbb{N}$.

Step 2 (i) ($\subseteq$) each $X_n \subseteq \mathbb{N}$, so the union is $\subseteq \mathbb{N}$. ($\supseteq$) Let $k \in \mathbb{N}$; then $k \neq k+1$, so $k \in X_{k+1}$, hence k is in the union [$\exists$].

Step 3 (ii) Suppose $k \in \bigcap_{n \in \mathbb{N}} X_n$. Then $k \in X_n$ for all n , in particular $k \in X_k$ [Strategy 1.2.19], i.e. $k \neq k$: contradiction. So the intersection is empty [Strategy 2.1.20, Theorem 2.1.25].

Step 4 (iii) Let $i, j \in \mathbb{N}$. Define $m = i + j + 1$. Then $m \in \mathbb{N}$, $m \neq i$ [$m \geq i + 1 > i$] and $m \neq j$ [$m > j$]. So $m \in X_i$ and $m \in X_j$, i.e. $m \in X_i \cap X_j$, which is therefore non-empty [Strategy 2.1.20].

Conclusion. The family $X_n = \mathbb{N} \setminus \{n\}$ satisfies (i)–(iii). □

Compare with your work.

- Any correct family is acceptable, but all three conditions must be verified, (iii) for *arbitrary* i, j (including $i = j$).

Exercise 2.2.40

essential

Write down the elements of $\{0, 1, 4, 7\} \setminus \{1, 2, 3, 4, 5\}$.

Step 1 Keep the elements of the first set that are *not* in the second [Definition 2.2.38]: 0 (keep), 1 (remove), 4 (remove), 7 (keep).

Answer. $\{0, 7\}$. □

Exercise 2.2.41

recommended

Express $[-2, 5) \setminus [4, 7)$ and $[4, 7) \setminus [-2, 5)$ as intervals.

Step 1 $x \in [-2, 5) \setminus [4, 7)$ iff $-2 \leq x < 5$ and $\neg(4 \leq x < 7)$ [Definition 2.2.38]. Within $-2 \leq x < 5$, the condition $4 \leq x < 7$ reduces to $x \geq 4$, so its negation is $x < 4$. Hence the condition is $-2 \leq x < 4$: the set is $[-2, 4)$.

Step 2 $x \in [4, 7) \setminus [-2, 5)$ iff $4 \leq x < 7$ and $\neg(-2 \leq x < 5)$. Within $4 \leq x < 7$, the condition $-2 \leq x < 5$ reduces to $x < 5$, so its negation is $x \geq 5$. Hence $5 \leq x < 7$: the set is $[5, 7)$.

Answer. $[-2, 5) \setminus [4, 7) = [-2, 4)$ and $[4, 7) \setminus [-2, 5) = [5, 7)$. □

Compare with your work.

- Bracket types: $4 \notin [-2, 4)$ because $4 \in [4, 7)$ is removed; $5 \in [5, 7)$ because $5 \notin [-2, 5)$ is not removed.

Exercise 2.2.45

recommended

Complete the proof of Theorem 2.2.44 (de Morgan's laws for sets): (b) $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$; (c) $X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i)$; (d) $X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$.

Given. The membership unfoldings and de Morgan's laws for $\wedge, \vee$ (Theorem 1.3.24) and for $\forall, \exists$ (Theorem

1.3.29). Part (a) is proved in the book by double containment; we follow the same pattern for (b) and use membership chains for (c), (d).

Step 1 (b), ($\subseteq$). Let $x \in X \setminus (A \cap B)$: $x \in X$ and $x \notin A \cap B$, i.e. $\neg(x \in A \wedge x \in B)$ [Definitions 2.2.38, 2.2.1]. By de Morgan, $x \notin A$ or $x \notin B$ [Theorem 1.3.24(a)]. Case $x \notin A$: $x \in X \setminus A$. Case $x \notin B$: $x \in X \setminus B$. Either way $x \in (X \setminus A) \cup (X \setminus B)$ [Definition 2.2.21].

Step 2 (b), ($\supseteq$). Let $x \in (X \setminus A) \cup (X \setminus B)$. Case $x \in X \setminus A$: $x \in X$ and $x \notin A$; then $x \notin A \cap B$ (an element of $A \cap B$ would be in A), so $x \in X \setminus (A \cap B)$. Case $x \in X \setminus B$: identical with B . Hence (b) [Axiom 2.1.14].

Step 3 (c). For any x :

$$\begin{aligned} x \in X \setminus \bigcup_{i \in I} A_i &\Leftrightarrow x \in X \wedge \neg(\exists i \in I, x \in A_i) \quad [\text{Definitions 2.2.38, 2.2.29}] \\ &\Leftrightarrow x \in X \wedge \forall i \in I, x \notin A_i \quad [\text{Theorem 1.3.29(b)}] \\ &\Leftrightarrow \forall i \in I, (x \in X \wedge x \notin A_i) \quad [x \in X \text{ does not depend on } i; \text{ see note}] \\ &\Leftrightarrow \forall i \in I, x \in X \setminus A_i \quad [\text{Definition 2.2.38}] \\ &\Leftrightarrow x \in \bigcap_{i \in I} (X \setminus A_i) \quad [\text{Definition 2.2.16}]. \end{aligned}$$

Note. The third step needs $I \neq \emptyset$ for the direction $\Leftarrow$ (if I is empty the right side is vacuously true while $x \in X$ may fail); for $I = \emptyset$ the book's conventions on empty intersections apply. Hence (c) by extensionality.

Step 4 (d). For any x :

$$\begin{aligned} x \in X \setminus \bigcap_{i \in I} A_i &\Leftrightarrow x \in X \wedge \neg(\forall i \in I, x \in A_i) \quad [\text{Definitions 2.2.38, 2.2.16}] \\ &\Leftrightarrow x \in X \wedge \exists i \in I, x \notin A_i \quad [\text{Theorem 1.3.29(a)}] \\ &\Leftrightarrow \exists i \in I, (x \in X \wedge x \notin A_i) \quad [x \in X \text{ independent of } i] \\ &\Leftrightarrow \exists i \in I, x \in X \setminus A_i \Leftrightarrow x \in \bigcup_{i \in I} (X \setminus A_i) \quad [\text{Definitions 2.2.38, 2.2.29}]. \end{aligned}$$

Conclusion. (b), (c), (d) hold. □

Compare with your work.

- Each line of a membership chain must cite the definition or logical law used; the de Morgan step is the heart of each part.
- For (b) two directions with cases; for (c)/(d) the quantifier version of de Morgan (Theorem 1.3.29), which cannot be replaced by a truth table.

C PRODUCTS

What you unfold here

Ordered lists (Definition 2.2.46): $(x_0, \dots, x_{k-1}) = (y_0, \dots, y_{\ell-1})$ iff $k = \ell$ and $x_i = y_i$ for all $i \in [k]$. Cartesian product (Definition 2.2.49): $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$, so $(x, y) \in X \times Y$ iff $x \in X$ and $y \in Y$.

Exercise 2.2.51

essential

Write down the elements of the set $\{1, 2\} \times \{3, 4, 5\}$.

Step 1 The elements are the ordered pairs (x, y) with $x \in \{1, 2\}$ and $y \in \{3, 4, 5\}$ [Definition 2.2.49]. For $x = 1$: $(1, 3), (1, 4), (1, 5)$; for $x = 2$: $(2, 3), (2, 4), (2, 5)$.

Answer. $\{1, 2\} \times \{3, 4, 5\} = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$, six pairs $(2 \cdot 3)$. □

Compare with your work.

- The elements are pairs, with the element of the *first* set first: $(3, 1)$ is not in this product.

Exercise 2.2.53

recommended

Let X and Y be sets. Under what conditions is it true that $X \times Y = Y \times X$?

To show. A condition, with a proof of both directions. Guess: equality holds when $X = Y$, and also when one of the sets is empty (both products are then empty).

Step 1 Claim. $X \times Y = Y \times X$ if and only if $X = Y$ or $X = \emptyset$ or $Y = \emptyset$.

Step 2 ($\Leftarrow$) If $X = Y$, both sides are $X \times X$. If $X = \emptyset$: a pair $(x, y) \in X \times Y$ would have $x \in \emptyset$, impossible, so $X \times Y = \emptyset$; likewise $Y \times X = \emptyset$ [Exercise 2.2.52]; so both sides are $\emptyset$. The case $Y = \emptyset$ is symmetric.

Step 3 ($\Rightarrow$) Suppose $X \times Y = Y \times X$. If $X = \emptyset$ or $Y = \emptyset$ the disjunction holds, so assume $X \neq \emptyset$ and $Y \neq \emptyset$ [LEM, Strategy 1.1.61]; we prove $X = Y$ by double containment.

Step 4 $X \subseteq Y$: let $x \in X$. Since $Y \neq \emptyset$, fix $y \in Y$ [Strategy 1.2.30]. Then $(x, y) \in X \times Y = Y \times X$ [Definition 2.2.49], so (x, y) is a pair whose first term is in Y : $x \in Y$.

Step 5 $Y \subseteq X$: let $y \in Y$; fix $x \in X$; then $(y, x) \in Y \times X = X \times Y$, so $y \in X$.

Step 6 Hence $X = Y$ [Axiom 2.1.14].

Conclusion. $X \times Y = Y \times X$ exactly when $X = Y$ or one of X, Y is empty. □

Compare with your work.

- “ $X = Y$ ” alone is not the full answer: $\emptyset \times \{1\} = \{1\} \times \emptyset = \emptyset$ with $\emptyset \neq \{1\}$.
 □ In ($\Rightarrow$) the non-emptiness of the *other* set is what supplies the partner y for the pair.

D CHAPTER 2 EXERCISES

How to answer the chapter exercises

Prove: unfold membership, then “let $a \in \dots$ ” for $\subseteq$, double containment for $=$, “assume $a \in \dots$, contradiction” for $\neq$. **True/false:** proof or counterexample. **Always/sometimes/never:** proof of the conclusion / one example and one counterexample / proof of the negated conclusion. The tiny sets $\emptyset, \{0\}, \{1\}, \{0, 1\}, \{\emptyset\}$ settle nearly every “sometimes”.

Exercise 2.E.4

recommended

Let X be a set and let $U, V \subseteq X$. Prove that U and V are disjoint if and only if $U \subseteq X \setminus V$.

Given. $U \subseteq X, V \subseteq X$. Disjoint: $U \cap V$ is empty (Definition 2.2.10). $a \in X \setminus V$ iff $a \in X$ and $a \notin V$ (Definition 2.2.38).

Step 1 ($\Rightarrow$) Suppose $U \cap V = \emptyset$. Let $a \in U$ [Strategy 2.1.7]. Then $a \in X$ [$U \subseteq X$].

Step 2 If $a \in V$, then $a \in U \cap V$ [Definition 2.2.1], contradicting $U \cap V = \emptyset$; so $a \notin V$ [$\neg I$].

Step 3 Hence $a \in X \setminus V$ [Definition 2.2.38]. So $U \subseteq X \setminus V$.

Step 4 ($\Leftarrow$) Suppose $U \subseteq X \setminus V$. To show $U \cap V$ is empty, suppose $a \in U \cap V$ [Strategy 2.1.20]. Then

$a \in U$ and $a \in V$.

Step 5 From $a \in U$: $a \in X \setminus V$ [hypothesis], so $a \notin V$. But $a \in V$: contradiction. Hence $U \cap V$ is empty: U, V are disjoint.

Conclusion. $U \cap V = \emptyset \Leftrightarrow U \subseteq X \setminus V$. □

Compare with your work.

- The hypothesis $U \subseteq X$ is needed exactly once (Step 1), to get $a \in X$; say where.

Exercise 2.E.5

recommended

For each of the following statements, determine whether it is true for all sets A and X , and prove your claim: (a) if $X \setminus A = \emptyset$ then $X = A$; (b) if $X \setminus A = X$ then $A = \emptyset$; (c) if $X \setminus A = A$ then $A = \emptyset$; (d) $X \setminus (X \setminus A) = A$.

Plan. Test each with tiny sets before trying to prove it. $X \setminus A = \emptyset$ only says $X \subseteq A$; $X \setminus A = X$ only says $X \cap A = \emptyset$; $X \setminus (X \setminus A) = X \cap A$ (Exercise 2.2.43).

Step 1 (a) *False.* $X = \emptyset, A = \{0\}$: $X \setminus A = \emptyset$ but $X \neq A$ ($0 \in A, 0 \notin X$).

Step 2 (b) *False.* $X = \emptyset, A = \{0\}$: $X \setminus A = \emptyset = X$ but $A \neq \emptyset$.

Step 3 (c) *True.* Suppose $X \setminus A = A$. Suppose, towards a contradiction, that $a \in A$ [Strategy 2.1.20]. Then $a \in X \setminus A$ [$A = X \setminus A$], so $a \notin A$ [Definition 2.2.38]. Contradiction with $a \in A$. Hence A is empty.

Step 4 (d) *False.* $X = \emptyset, A = \{0\}$: $X \setminus A = \emptyset$, so $X \setminus (X \setminus A) = \emptyset \setminus \emptyset = \emptyset \neq A$.

Conclusion. Only (c) holds for all sets. □

Compare with your work.

- One counterexample per false statement, with the two sides computed explicitly.
- For (c) the proof is three lines and uses only the definition of $\setminus$; no cases needed.

Exercise 2.E.6

essential

For each statement, determine whether it is true for all sets X, Y , false for all sets X, Y , or true for some and false for others: (a) $\mathcal{P}(X \cup Y) = \mathcal{P}(X) \cup \mathcal{P}(Y)$; (b) $\mathcal{P}(X \cap Y) = \mathcal{P}(X) \cap \mathcal{P}(Y)$; (c) $\mathcal{P}(X \times Y) = \mathcal{P}(X) \times \mathcal{P}(Y)$; (d) $\mathcal{P}(X \setminus Y) = \mathcal{P}(X) \setminus \mathcal{P}(Y)$.

Given. $U \in \mathcal{P}(Z) \Leftrightarrow U \subseteq Z$; $\emptyset \in \mathcal{P}(Z)$ for every Z (Exercise 2.1.32).

Step 1 (a) *Sometimes.* True for $X = Y = \emptyset$: both sides are $\mathcal{P}(\emptyset) = \{\emptyset\}$. False for $X = \{0\}, Y = \{1\}$: $\{0, 1\} \subseteq X \cup Y$ so $\{0, 1\} \in \mathcal{P}(X \cup Y)$; but $\{0, 1\} \not\subseteq X$ and $\{0, 1\} \not\subseteq Y$, so $\{0, 1\} \notin \mathcal{P}(X) \cup \mathcal{P}(Y)$.

Step 2 (b) *Always.* Let U be arbitrary. $U \in \mathcal{P}(X \cap Y) \Leftrightarrow U \subseteq X \cap Y$ [Definition 2.1.29]. Now $U \subseteq X \cap Y$ iff $(U \subseteq X \text{ and } U \subseteq Y)$: if $U \subseteq X \cap Y$ and $a \in U$ then $a \in X \cap Y$, so $a \in X$ and $a \in Y$; conversely if $U \subseteq X, U \subseteq Y$ and $a \in U$ then $a \in X$ and $a \in Y$, so $a \in X \cap Y$. Finally $(U \subseteq X \wedge U \subseteq Y) \Leftrightarrow U \in \mathcal{P}(X) \wedge U \in \mathcal{P}(Y) \Leftrightarrow U \in \mathcal{P}(X) \cap \mathcal{P}(Y)$. By extensionality the sets are equal.

Step 3 (c) *Never.* $\emptyset \in \mathcal{P}(X \times Y)$ [Exercise 2.1.32]. The elements of $\mathcal{P}(X) \times \mathcal{P}(Y)$ are ordered pairs (U, V) [Definition 2.2.49], and $\emptyset$ is not an ordered pair, so $\emptyset \notin \mathcal{P}(X) \times \mathcal{P}(Y)$. The two sets differ in the element $\emptyset$, for every X, Y [Axiom 2.1.14].

Step 4 (d) *Never.* $\emptyset \in \mathcal{P}(X \setminus Y)$. But $\emptyset \in \mathcal{P}(Y)$, so $\emptyset \notin \mathcal{P}(X) \setminus \mathcal{P}(Y)$ [Definition 2.2.38]. So the sets differ for every X, Y .

Conclusion. (a) sometimes, (b) always, (c) never, (d) never. □

Compare with your work.

- “Never” is proved by a *uniform* witness ($\emptyset$) that lies in one side and not the other for all X, Y .
- In (a) the inclusion $\supseteq$ always holds; the failure is in $\subseteq$, shown by a set that is a subset of the union but of neither part.

Exercise 2.E.16

recommended

For each of the following subsets of $\mathbb{R}$, determine (with proof) whether it is open: (a) $\emptyset$; (b) $(0, 1)$; (c) $(0, 1]$; (d) $\mathbb{Z}$; (e) $\mathbb{R} \setminus \mathbb{Z}$; (f) $\mathbb{Q}$.

Given. Definition 2.E.15: $U \subseteq \mathbb{R}$ is **open** if $\forall a \in U, \exists \delta > 0, (a - \delta, a + \delta) \subseteq U$. Its negation (maximally negated): $\exists a \in U, \forall \delta > 0, \exists x \in (a - \delta, a + \delta), x \notin U$.

Plan. Open: let $a \in U$; define δ in terms of a ; check $\delta > 0$; show $(a - \delta, a + \delta) \subseteq U$. **Not open:** pick a specific $a \in U$; let $\delta > 0$ be arbitrary; produce x within δ of a with $x \notin U$.

Step 1 (a) *Open.* “For all $a \in \emptyset \dots$ ” is vacuously true [Exercise 2.1.24].

Step 2 (b) *Open.* Let $a \in (0, 1)$. Define $\delta = \min(a, 1 - a)$; then $\delta > 0$ [$a > 0$ and $1 - a > 0$]. Let $x \in (a - \delta, a + \delta)$. Then $x > a - \delta \geq a - a = 0$ [$\delta \leq a$] and $x < a + \delta \leq a + (1 - a) = 1$ [$\delta \leq 1 - a$]. So $x \in (0, 1)$; hence $(a - \delta, a + \delta) \subseteq (0, 1)$.

Step 3 (c) *Not open.* Take $a = 1 \in (0, 1]$. Let $\delta > 0$. Define $x = 1 + \frac{\delta}{2}$. Then $1 - \delta < x < 1 + \delta$ [$0 < \frac{\delta}{2} < \delta$], so $x \in (a - \delta, a + \delta)$, but $x > 1$ so $x \notin (0, 1]$. Thus no δ works for $a = 1$.

Step 4 (d) *Not open.* Take $a = 0 \in \mathbb{Z}$. Let $\delta > 0$. Define $x = \frac{1}{2} \min(\delta, 1)$. Then $0 < x < \delta$, so $x \in (-\delta, \delta)$; and $0 < x \leq \frac{1}{2}$, so x is strictly between the consecutive integers 0 and 1 and is not an integer [page 6].

Step 5 (e) *Open.* Let $a \in \mathbb{R} \setminus \mathbb{Z}$. Let $n = \lfloor a \rfloor$, so $n \leq a < n + 1$ [Definition 0.10]; since $a \notin \mathbb{Z}$, $a \neq n$, so $n < a < n + 1$. Define $\delta = \min(a - n, n + 1 - a) > 0$. If $x \in (a - \delta, a + \delta)$ then $n \leq a - \delta < x < a + \delta \leq n + 1$, so x lies strictly between n and $n + 1$: $x \notin \mathbb{Z}$, i.e. $x \in \mathbb{R} \setminus \mathbb{Z}$.

Step 6 (f) *Not open.* Take $a = 0 \in \mathbb{Q}$. Let $\delta > 0$. Choose $n \in \mathbb{N}$ with $n > \frac{\sqrt{2}}{\delta}$ [$\mathbb{N}$ is unbounded in $\mathbb{R}$] and define $x = \frac{\sqrt{2}}{n}$. Then $0 < x < \delta$, so $x \in (-\delta, \delta)$. And $x \notin \mathbb{Q}$: if $\frac{\sqrt{2}}{n} = r \in \mathbb{Q}$ then $\sqrt{2} = nr \in \mathbb{Q}$ [Theorem 0.16], contradicting Assumption 0.49.

Conclusion. Open: (a), (b), (e). Not open: (c), (d), (f). □

Compare with your work.

- For “open”, δ must depend on a and be checked positive; for “not open”, a is specific and δ is arbitrary; the x depends on δ .
- In (f) the irrational point near 0 must be shown irrational (via Assumption 0.49), not just asserted.

Exercise 2.E.18

bonus

(a) Let $n \geq 1$ and let $U_1, \dots, U_n$ be open subsets of $\mathbb{R}$. Prove that $U_1 \cap \dots \cap U_n$ is open. (b) Prove that $(0, 1 + \frac{1}{n})$ is open for all $n \geq 1$, but that $\bigcap_{n \geq 1} (0, 1 + \frac{1}{n})$ is not open.

Step 1 (a) Let $a \in U_1 \cap \dots \cap U_n$. For each i , $a \in U_i$ and U_i is open, so there is $\delta_i > 0$ with $(a - \delta_i, a + \delta_i) \subseteq U_i$ [Definition 2.E.15, Strategy 1.2.30]. Define $\delta = \min(\delta_1, \dots, \delta_n)$; $\delta > 0$ as the minimum of finitely many positive numbers.

Step 2 Let $x \in (a - \delta, a + \delta)$ and let $i \in \{1, \dots, n\}$. Since $\delta \leq \delta_i$: $a - \delta_i \leq a - \delta < x < a + \delta \leq a + \delta_i$, so $x \in U_i$. As i was arbitrary, $x \in U_1 \cap \dots \cap U_n$. Hence $(a - \delta, a + \delta) \subseteq U_1 \cap \dots \cap U_n$: open.

Step 3 (b), each interval open. Fix $n \geq 1$, let $a \in (0, 1 + \frac{1}{n})$ and define $\delta = \min(a, 1 + \frac{1}{n} - a) > 0$. If $a - \delta < x < a + \delta$ then $x > 0$ and $x < 1 + \frac{1}{n}$ [as in 2.E.16(b)], so the interval is open.

Step 4 (b), the intersection. Claim $\bigcap_{n \geq 1} (0, 1 + \frac{1}{n}) = (0, 1]$. ($\supseteq$) if $0 < x \leq 1$ then $0 < x < 1 + \frac{1}{n}$ for all $n \geq 1$. ($\subseteq$) let x be in every $(0, 1 + \frac{1}{n})$; then $x > 0$; if $x > 1$, choose $n \geq 1$ with $n > \frac{1}{x-1}$, so $\frac{1}{n} < x - 1$, i.e. $1 + \frac{1}{n} < x$, contradicting $x \in (0, 1 + \frac{1}{n})$; so $x \leq 1$.

Step 5 $(0, 1]$ is not open [Exercise 2.E.16(c)]. So an intersection of infinitely many open sets need not be open: the proof of (a) fails because infinitely many δ_n ($\delta_n \rightarrow 0$ here) need not have a positive minimum.

Conclusion. (a) finite intersections of open sets are open; (b) infinite ones need not be. $\square$

Exercise 2.E.28

recommended

Let a, b, c be arbitrary objects. Then $\{a, b\} = \{a, c\}$. (Always, sometimes or never?)

Step 1 Sometimes. Holds: $a = b = c = 0$ (or any $b = c$): both sets are $\{0\}$.

Step 2 Fails: $a = 0, b = 1, c = 2$: $1 \in \{0, 1\}$ but $1 \notin \{0, 2\}$ [list notation], so the sets are different [Axiom 2.1.14].

Conclusion. Sometimes (exactly when $b = c$, or $a = b = c$ in disguise, or $b = a = c$). $\square$

Exercise 2.E.32

recommended

Let X and Y be sets with $X \cap Y = \emptyset$. Then $\forall a, (a \in X \Rightarrow a \notin Y)$.

Step 1 Always. Let a be arbitrary and suppose $a \in X$ [Strategies 1.2.10, 1.1.28]. Goal: $a \notin Y$, a negation.

Step 2 Suppose $a \in Y$ [Strategy 1.1.51]. Then $a \in X \cap Y$ [Definition 2.2.1], but $X \cap Y = \emptyset$ has no elements: contradiction. Hence $a \notin Y$.

Conclusion. Always: this is just what “ $X \cap Y$ is empty” means, unfolded. $\square$

Exercise 2.E.33

essential

Let X and Y be sets with $X \neq Y$. Then $\exists a, (a \in X \wedge a \notin Y)$.

Given. $X \neq Y$ means, by Axiom 2.1.14, $\neg \forall a, (a \in X \Leftrightarrow a \in Y)$, i.e. some a lies in exactly one of X, Y : $a \in X \setminus Y$ or $a \in Y \setminus X$. The hypothesis does not say which.

Step 1 Sometimes. Holds: $X = \{0\}, Y = \emptyset$. Then $X \neq Y$, and $a = 0$ satisfies $a \in X, a \notin Y$.

Step 2 Fails: $X = \emptyset, Y = \{0\}$. Then $X \neq Y$, but no a satisfies $a \in X$ (there is nothing in X), so the conclusion is false.

Conclusion. Sometimes. (The conclusion holds exactly when $X \not\subseteq Y$.) $\square$

Compare with your work.

- $\square$ The trap is to read $X \neq Y$ as “ X has something Y lacks”; it may equally be Y that has the extra element. Both examples must be given.

Exercise 2.E.38**essential**

Let X and Y be sets such that $\mathcal{P}(X) = \mathcal{P}(Y)$. Then $X = Y$.

Step 1 *Always.* $X \subseteq X$ [Example 2.1.8], so $X \in \mathcal{P}(X)$ [Definition 2.1.29]. Since $\mathcal{P}(X) = \mathcal{P}(Y)$, $X \in \mathcal{P}(Y)$, i.e. $X \subseteq Y$.

Step 2 Symmetrically $Y \in \mathcal{P}(Y) = \mathcal{P}(X)$, so $Y \subseteq X$.

Step 3 Hence $X = Y$ [Axiom 2.1.14, double containment].

Conclusion. Always. □

Compare with your work.

□ The whole proof is: “ X is an element of its own power set”. Two lines plus extensionality.

Exercise 2.E.41**recommended**

Let X and Y be sets. Then $X \cup Y = X \cap Y$.

Step 1 *Sometimes. Holds:* $X = Y = \{0\}$: $X \cup Y = \{0\} = X \cap Y$ (in general $X \cup X = X = X \cap X$, Exercises 2.2.26 and 2.2.9).

Step 2 *Fails:* $X = \{0\}$, $Y = \emptyset$: $X \cup Y = \{0\}$ but $X \cap Y = \emptyset$ [Exercise 2.2.9], and $\{0\} \neq \emptyset$.

Step 3 *Extra (when exactly):* $X \cup Y = X \cap Y$ iff $X = Y$. ($\Leftarrow$) above. ($\Rightarrow$) if $a \in X$ then $a \in X \cup Y = X \cap Y$, so $a \in Y$: $X \subseteq Y$; symmetrically $Y \subseteq X$.

Conclusion. Sometimes. □

Chapter 3

Functions

Contents

- A Functions (§3.1)
- B Injections, surjections (§3.2)
- C Chapter 3 exercises (3.E)

Planning

★ 3.1: 11, 15abc, 36 · 3.2: 5, 8, 12, 14, 20, 40, 47 · 3.E: 13, 17, 19, 24, 25, 34, 41, 48
 ★ 3.1: 8, 16, 19, 33, 40, 41 · 3.2: 6, 16, 19, 25, 39ab, 45, 46 · 3.E: 2, 35, 37, 38, 47; later 25e, 26b
 Bonus: 3.1.21. Tested in **Quiz 2** (7 Oct).

A FUNCTIONS

What you unfold in this section

Function (Definition 3.1.1): $f : X \rightarrow Y$ specifies for each $x \in X$ exactly one $f(x) \in Y$: $\forall x \in X, \exists! y \in Y, y = f(x)$. **Well-defined** (page 101): totality (every x gets a value), existence (the value exists and is in Y), uniqueness (only one value). **Graph** (Definition 3.1.6, Theorem 3.1.9): $G \subseteq X \times Y$ is the graph of a function $X \rightarrow Y$ iff $\forall x \in X, \exists! y \in Y, (x, y) \in G$. **Composite** (Definition 3.1.17): $(g \circ f)(x) = g(f(x))$; f first. **Image, preimage** (Definitions 3.1.31, 3.1.37): $f[U] = \{f(x) \mid x \in U\}$, so $y \in f[U] \Leftrightarrow \exists x \in U, y = f(x)$; $f^{-1}[V] = \{x \in X \mid f(x) \in V\}$, so $x \in f^{-1}[V] \Leftrightarrow f(x) \in V$.

Exercise 3.1.8

recommended

Find a function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ whose graph is $\{\dots, (-2, -5), (-1, -2), (0, 1), (1, 4), (2, 7), (3, 10), \dots\}$.

Given. $\text{Gr}(f) = \{(n, f(n)) \mid n \in \mathbb{Z}\}$ (Definition 3.1.6): the pair (n, y) is in the graph iff $y = f(n)$.

Plan. Read off a pattern: second coordinates increase by 3 per unit step; $(0, 1)$ gives the constant.

Step 1 Guess $f(n) = 3n + 1$. Check: $f(-2) = -5$, $f(-1) = -2$, $f(0) = 1$, $f(1) = 4$, $f(2) = 7$, $f(3) = 10$: all listed pairs are $(n, 3n + 1)$.

Step 2 Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(n) = 3n + 1$. It is well-defined (a formula producing an integer for each integer). Its graph is $\{(n, 3n + 1) \mid n \in \mathbb{Z}\}$ [Definition 3.1.6], which is the given set (the “...” are understood to continue the pattern).

Answer. $f(n) = 3n + 1$. □

Exercise 3.1.11

essential

For each of the following, determine whether G is the graph of a function from X to Y : (a) $X = Y = \mathbb{R}$, $G = \{(a, a^2) \mid a \in \mathbb{R}\}$; (b) $X = Y = \mathbb{R}$, $G = \{(a^2, a) \mid a \in \mathbb{R}\}$; (c) $X = Y = [0, \infty)$, $G = \{(a^2, a) \mid a \in [0, \infty)\}$; (d) $X = Y = \mathbb{Q}$, $G = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid xy = 1\}$; (e) $X = Y = \mathbb{Q}$, $G = \{(a, a) \mid a \in \mathbb{Z}\}$.

Given. Theorem 3.1.9: G is a graph of a function $X \rightarrow Y$ iff for every $x \in X$ there is *exactly one* $y \in Y$ with $(x, y) \in G$.

Plan. For each $x \in X$ count the y with $(x, y) \in G$. “Yes” needs the count to be 1 for all x ; “no” needs one x with count 0 or count ≥ 2 .

Step 1 (a) *Yes.* Let $x \in \mathbb{R}$. $(x, y) \in G$ iff $y = x^2$ [the pair (a, a^2) with $a = x$]. Exactly one such y , and $x^2 \in \mathbb{R}$. So $G = \text{Gr}(x \mapsto x^2)$.

Step 2 (b) *No.* For $x = -1 \in \mathbb{R}$: $(-1, y) \in G$ would need $-1 = a^2$ for some real a , impossible. Count 0. (Also $x = 1$: $(1, 1)$ and $(1, -1)$ are both in G , count 2.) Either failure suffices.

Step 3 (c) *Yes.* Let $x \in [0, \infty)$. $(x, y) \in G$ iff $y \in [0, \infty)$ and $y^2 = x$. There is exactly one such y : $y = \sqrt{x}$ [for $x > 0$ by Example 1.2.36; for $x = 0$, $y^2 = 0$ forces $y = 0$]. So $G = \text{Gr}(x \mapsto \sqrt{x})$.

Step 4 (d) *No.* For $x = 0 \in \mathbb{Q}$: $(0, y) \in G$ needs $0 \cdot y = 1$, impossible. Count 0.

Step 5 (e) *No.* For $x = \frac{1}{2} \in \mathbb{Q}$: every element of G has an *integer* first coordinate, so no $(\frac{1}{2}, y) \in G$. Count 0 (totality fails).

Conclusion. (a) yes, (b) no, (c) yes, (d) no, (e) no. □

Compare with your work.

- ☐ Always quote the “exactly one” criterion; “no” needs a *specific* x with the count stated.
- ☐ (b) versus (c): same formula, different domain and codomain. Domain and codomain are part of the function.

Exercise 3.1.15

essential (a)–(c)

Which of the following specifications of functions are well-defined? (a) $g : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $(x+1)g(x) = 1$ for all $x \in \mathbb{Q}$; (b) $h : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $(x+1)h(x) = 1$ for all $x \in \mathbb{N}$; (c) $k : \mathbb{N} \rightarrow \mathbb{N}$ defined by $(x+1)k(x) = 1$ for all $x \in \mathbb{N}$; (d) $\ell : \mathbb{N} \rightarrow \mathbb{N}$ defined by $\ell(x) = \ell(x)$ for all $x \in \mathbb{N}$.

Given. The three checks: totality, existence (a value *in the codomain*), uniqueness.

Plan. For each x in the domain, solve the defining equation and count its solutions *in the codomain*.

Step 1 (a) *Not well-defined.* For $x = -1 \in \mathbb{Q}$ the condition is $0 \cdot g(-1) = 1$, i.e. $0 = 1$: no rational number $g(-1)$ satisfies it. Existence fails at $x = -1$.

Step 2 (b) *Well-defined.* Let $x \in \mathbb{N}$. Then $x+1 \geq 1$, so $x+1 \neq 0$ and the equation $(x+1)h(x) = 1$ has exactly one solution, $h(x) = \frac{1}{x+1}$ [divide by the non-zero $x+1$], which is rational [Definition 0.15: $1, x+1 \in \mathbb{Z}$]. Totality, existence and uniqueness hold.

Step 3 (c) *Not well-defined.* As in (b) the only candidate is $k(x) = \frac{1}{x+1}$; for $x = 1$ this is $\frac{1}{2}$, which is not a natural number. Existence of a value in the codomain $\mathbb{N}$ fails at $x = 1$.

Step 4 (d) *Not well-defined.* The equation $\ell(x) = \ell(x)$ holds for every choice of $\ell(x) \in \mathbb{N}$; it does not single out one value. Uniqueness fails (at every x).

Conclusion. Only (b) is well-defined. □

Compare with your work.

- ☐ Name the failing check and the failing x . For (c), “the value is in $\mathbb{Q}$, not $\mathbb{N}$ ” is the whole point: a value outside the codomain is no value.

Exercise 3.1.16

recommended

Find a condition on sets X and Y such that the specification of $i : X \cup Y \rightarrow \{0, 1\}$ given by $i(z) = 0$ if $z \in X$ and $i(z) = 1$ if $z \in Y$ is well-defined.

Given. A specification by cases is well-defined iff the cases are exhaustive (totality) and compatible (uniqueness: overlapping cases give the same value), page 101.

Step 1 *Exhaustive*: every $z \in X \cup Y$ satisfies $z \in X$ or $z \in Y$ [Definition 2.2.21], so each z gets at least one value. Totality and existence hold always.

Step 2 *Compatible*: if $z \in X$ and $z \in Y$, the cases prescribe $i(z) = 0$ and $i(z) = 1$, two different values: uniqueness fails. If no z lies in both, every z falls under exactly one case and uniqueness holds.

Step 3 Hence: well-defined iff no z is in $X \cap Y$, i.e. iff $X \cap Y = \emptyset$.

Answer. i is well-defined if and only if X and Y are disjoint. (Then $i = \chi_Y$ on $X \cup Y$, Definition 3.1.23.) $\square$

Compare with your work.

- ☐ Both “if” and “only if” should be argued: disjoint $\Rightarrow$ well-defined, and an element of $X \cap Y \Rightarrow$ two values.

Exercise 3.1.19

recommended

With $f, g, h, k : \mathbb{Q} \rightarrow \mathbb{Q}$ as in Example 3.1.18 ($f(x) = 2x$, $g(x) = x + 5$, $h(x) = x^2$, $k(x) = \frac{x}{6}$), compute equations defining (a) $f \circ g$; (b) $g \circ f$; (c) $((f \circ g) \circ h) \circ k$; (d) $f \circ (g \circ (h \circ k))$; (e) $(g \circ g) \circ (g \circ g)$.

Given. $(u \circ v)(x) = u(v(x))$: apply the *right* function first (Definition 3.1.17).

Step 1 (a) $(f \circ g)(x) = f(g(x)) = f(x + 5) = 2(x + 5) = 2x + 10$.

Step 2 (b) $(g \circ f)(x) = g(f(x)) = g(2x) = 2x + 5$. (So $f \circ g \neq g \circ f$.)

Step 3 (c) Apply k , then h , then g , then f :

$$x \xrightarrow{k} \frac{x}{6} \xrightarrow{h} \frac{x^2}{36} \xrightarrow{g} \frac{x^2}{36} + 5 \xrightarrow{f} \frac{x^2}{18} + 10.$$

$$\text{So } (((f \circ g) \circ h) \circ k)(x) = \frac{x^2}{18} + 10.$$

Step 4 (d) $(f \circ (g \circ (h \circ k)))(x) = f(g(h(k(x))))$: the same chain, so again $\frac{x^2}{18} + 10$ [associativity, Exercise 3.1.21].

Step 5 (e) $(g \circ g)(x) = g(x + 5) = x + 10$; then $((g \circ g) \circ (g \circ g))(x) = (g \circ g)(x + 10) = x + 20$.

Answer. (a) $2x + 10$; (b) $2x + 5$; (c), (d) $\frac{x^2}{18} + 10$; (e) $x + 20$. $\square$

Compare with your work.

- ☐ The common error is to apply the left function first; (a) and (b) are there to catch it.

Exercise 3.1.21**bonus**

Prove that composition is associative: for $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $h : Z \rightarrow W$, $h \circ (g \circ f) = (h \circ g) \circ f$.

Given. Strategy 3.1.5: two functions with the same domain and codomain are equal iff they agree at every element.

Step 1 Both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ have domain X and codomain W [Definition 3.1.17 twice].

Step 2 Let $x \in X$. Then

$$(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))) = (h \circ g)(f(x)) = ((h \circ g) \circ f)(x),$$

each equality by Definition 3.1.17.

Step 3 Since x was arbitrary, the functions are equal [Strategy 3.1.5].

Conclusion. $h \circ (g \circ f) = (h \circ g) \circ f$, so brackets may be dropped: $h \circ g \circ f$. □

Exercise 3.1.33**recommended**

Describe the image $f[U]$ in each case: (a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 3n$, $U = \mathbb{N}$; (b) $f : X \rightarrow X \times X$, $f(x) = (x, x)$, $U = X$; (c) $f : \{a, b, c\} \rightarrow \{1, 2, 3\}$ with $f(a) = 1$, $f(b) = 3$, $f(c) = 1$, $U = \{a, b, c\}$.

Given. $f[U] = \{f(x) \mid x \in U\}$ (Definition 3.1.31).

Step 1 (a) $f[\mathbb{N}] = \{3n \mid n \in \mathbb{N}\} = \{0, 3, 6, 9, \dots\}$: the natural multiples of 3.

Step 2 (b) $f[X] = \{(x, x) \mid x \in X\}$: the “diagonal” of $X \times X$.

Step 3 (c) $f[\{a, b, c\}] = \{f(a), f(b), f(c)\} = \{1, 3, 1\} = \{1, 3\}$ [repetition does not matter].

Answer. As above. □

Exercise 3.1.36**essential**

Example 3.1.35 showed that $f[U \cap V] \subseteq f[U] \cap f[V]$. Determine which of the following are true for all sets U, V, X, Y and functions $f : X \rightarrow Y$, with a proof or a counterexample: (a) $f[U] \cap f[V] \subseteq f[U \cap V]$; (b) $f[U \cup V] \subseteq f[U] \cup f[V]$; (c) $f[U] \cup f[V] \subseteq f[U \cup V]$.

Given. $y \in f[U]$ iff there is $x \in U$ with $y = f(x)$; from $y \in f[U]$ you may fix such an x , but it is not unique.

Plan. (a) fails when two different inputs, one in U and one in V , share a value. (b), (c): “let $y \in \dots$, fix x , case split”.

Step 1 (a) *False.* $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{1\}$, $V = \{-1\}$. Then $f[U] = \{1\}$ and $f[V] = \{1\}$, so $f[U] \cap f[V] = \{1\}$. But $U \cap V = \emptyset$ and $f[\emptyset] = \emptyset$ [no $x \in \emptyset$ to produce a value]. So $1 \in f[U] \cap f[V]$ but $1 \notin f[U \cap V]$.

Step 2 (b) *True.* Let $y \in f[U \cup V]$. Fix $x \in U \cup V$ with $y = f(x)$ [Definition 3.1.31, Strategy 1.2.30]. Then $x \in U$ or $x \in V$. Case $x \in U$: $y = f(x) \in f[U]$. Case $x \in V$: $y \in f[V]$. Either way $y \in f[U] \cup f[V]$ [Definition 2.2.21].

Step 3 (c) *True.* Let $y \in f[U] \cup f[V]$. Case $y \in f[U]$: fix $x \in U$ with $y = f(x)$; then $x \in U \cup V$, so $y \in f[U \cup V]$. Case $y \in f[V]$: identical. So $y \in f[U \cup V]$.

Conclusion. (a) false; (b), (c) true, so $f[U \cup V] = f[U] \cup f[V]$, while for intersections only $f[U \cap V] \subseteq f[U] \cap f[V]$ holds. □

Compare with your work.

□ In (a) the witnesses for $y \in f[U]$ and $y \in f[V]$ are *different* elements; that is exactly why the inclusion

fails, and your counterexample should make this visible.

Exercise 3.1.40

recommended

Let $f : X \rightarrow Y$ and $A, B \subseteq Y$. Prove that $f^{-1}[A \cap B] = f^{-1}[A] \cap f^{-1}[B]$, $f^{-1}[A \cup B] = f^{-1}[A] \cup f^{-1}[B]$ and $f^{-1}[Y \setminus A] = X \setminus f^{-1}[A]$.

Given. $x \in f^{-1}[V] \Leftrightarrow x \in X \wedge f(x) \in V$ (Definition 3.1.37). Unlike images, no choice is involved, so membership chains work.

Step 1 Let $x \in X$ be arbitrary.

$$\begin{aligned} x \in f^{-1}[A \cap B] &\Leftrightarrow f(x) \in A \cap B \Leftrightarrow f(x) \in A \wedge f(x) \in B \quad [\text{Definitions 3.1.37, 2.2.1}] \\ &\Leftrightarrow x \in f^{-1}[A] \wedge x \in f^{-1}[B] \Leftrightarrow x \in f^{-1}[A] \cap f^{-1}[B]. \end{aligned}$$

Step 2 Likewise

$$x \in f^{-1}[A \cup B] \Leftrightarrow f(x) \in A \vee f(x) \in B \Leftrightarrow x \in f^{-1}[A] \vee x \in f^{-1}[B] \Leftrightarrow x \in f^{-1}[A] \cup f^{-1}[B].$$

Step 3 And

$$x \in f^{-1}[Y \setminus A] \Leftrightarrow f(x) \in Y \wedge f(x) \notin A \Leftrightarrow f(x) \notin A \Leftrightarrow x \notin f^{-1}[A] \Leftrightarrow x \in X \setminus f^{-1}[A],$$

where $f(x) \in Y$ holds automatically (codomain) and $x \in X$ throughout.

Step 4 Each chain shows that the two sets have the same elements; they are equal by extensionality [Axiom 2.1.14]. (All three sets are subsets of X , so restricting to $x \in X$ loses nothing.)

Conclusion. Preimages commute with $\cap$, $\cup$ and complements. □

Compare with your work.

- Compare with 3.1.36: for images only inclusions hold, because of the existential quantifier in $f[U]$; for preimages everything is an “iff”.

Exercise 3.1.41

recommended

Let $f : X \rightarrow Y$. Prove that $f^{-1}[\emptyset] = \emptyset$ and $f^{-1}[Y] = X$.

Step 1 $f^{-1}[\emptyset] = \emptyset$. Suppose $x \in f^{-1}[\emptyset]$ [Strategy 2.1.20]. Then $f(x) \in \emptyset$ [Definition 3.1.37], impossible. So $f^{-1}[\emptyset]$ is empty [Theorem 2.1.25].

Step 2 $f^{-1}[Y] = X$. ($\subseteq$) $f^{-1}[Y] \subseteq X$ by definition (it is $\{x \in X \mid \dots\}$). ($\supseteq$) Let $x \in X$. Then $f(x) \in Y$ [Y is the codomain], so $x \in f^{-1}[Y]$.

Conclusion. Both hold. □

B INJECTIONS, SURJECTIONS

What you unfold in this section

Injective (Definition 3.2.1): $\forall a, b \in X, f(a) = f(b) \Rightarrow a = b$; proof: “let $a, b \in X$ with $f(a) = f(b) \dots$ so $a = b$ ” (Strategy 3.2.2). Not injective: two specific $a \neq b$ with $f(a) = f(b)$. **Surjective** (Definition 3.2.9): $\forall y \in Y, \exists x \in X, f(x) = y$; proof: “let $y \in Y$; define $x = \dots$; check $x \in X$ and $f(x) = y$ ” (Strategy

3.2.10). Not surjective: a specific y with no preimage. **Bijjective**: both. **Inverses** (Definitions 3.2.23, 3.2.29, 3.2.37): g is a left inverse of f if $g \circ f = \text{id}_X$, a right inverse if $f \circ g = \text{id}_Y$, an inverse if both. Left inverse $\Rightarrow$ injective (3.2.25); right inverse $\Rightarrow$ surjective (3.2.31); inverse $\Leftrightarrow$ bijective (3.2.40, Strategy 3.2.41).

Exercise 3.2.5**essential**

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Prove that if $g \circ f$ is injective then f is injective.

Given. $g \circ f$ injective: $\forall a, b \in X, g(f(a)) = g(f(b)) \Rightarrow a = b$.

To show. f injective: let $a, b \in X$ with $f(a) = f(b)$; show $a = b$.

Step 1 Suppose $g \circ f$ is injective [Strategy 1.1.28]. Let $a, b \in X$ and suppose $f(a) = f(b)$ [Strategy 3.2.2].

Step 2 Apply g to both sides of $f(a) = f(b)$: $g(f(a)) = g(f(b))$ [a function gives equal values on equal inputs], i.e. $(g \circ f)(a) = (g \circ f)(b)$ [Definition 3.1.17].

Step 3 Since $g \circ f$ is injective, $a = b$ [Definition 3.2.1 applied to $g \circ f$].

Conclusion. f is injective. □

Compare with your work.

□ Three lines; the only move is “apply g ”. Note that nothing can be said about g (see 3.E.35).

Exercise 3.2.6**recommended**

Write out what it means to say a function $f : X \rightarrow Y$ is *not* injective; describe how to prove it; give an example and prove that it is not injective.

Step 1 Negate Definition 3.2.1: $\neg \forall a, b \in X, (f(a) = f(b) \Rightarrow a = b) \equiv \exists a, b \in X, \neg(f(a) = f(b) \Rightarrow a = b)$ [Theorem 1.3.29] $\equiv \exists a, b \in X, (f(a) = f(b) \wedge a \neq b)$ [Exercise 1.3.27].

Step 2 *How to prove it:* exhibit two specific elements $a, b \in X$, check $a \neq b$, and compute $f(a) = f(b)$ [Strategy 1.2.24].

Step 3 *Example:* $g : \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = n^2$. Take $a = 1, b = -1$: $a \neq b$, and $g(1) = 1 = g(-1)$. So g is not injective.

Conclusion. As above. □

Exercise 3.2.8**essential**

Let $a, b \in \mathbb{R}$ with $b \neq 0$, and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) = a + bt$. Prove that f is injective.

Step 1 Let $s, t \in \mathbb{R}$ and suppose $f(s) = f(t)$, i.e. $a + bs = a + bt$ [Strategy 3.2.2].

Step 2 Subtract a : $bs = bt$. Since $b \neq 0$, divide by b : $s = t$ [algebra; division needs $b \neq 0$].

Conclusion. f is injective. □

Compare with your work.

□ Say where $b \neq 0$ is used; for $b = 0$ the function is constant and not injective.

Exercise 3.2.12**essential**

For each pair (X, Y) , determine whether the function $f : X \rightarrow Y$ defined by $f(x) = 2x + 1$ is surjective: (a) $X = \mathbb{Z}, Y = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$; (b) $X = Y = \mathbb{Z}$; (c) $X = Y = \mathbb{Q}$; (d) $X = Y = \mathbb{R}$.

Plan. Solve $2x + 1 = y$ on scrap paper: $x = \frac{y-1}{2}$. Then check whether that x lies in the domain for every y in the codomain.

Step 1 (a) *Surjective.* Let $y \in Y$, so y is odd. Then $y = 2k + 1$ for some $k \in \mathbb{Z}$ [Proposition 1.1.58]. Define $x = k \in \mathbb{Z}$; then $f(x) = 2k + 1 = y$ [Strategy 3.2.10].

Step 2 (b) *Not surjective.* Take $y = 0 \in \mathbb{Z}$. If $f(x) = 0$ then $2x = -1$, which has no integer solution (-1 is odd, not $2 \cdot$ integer; or $x = -\frac{1}{2} \notin \mathbb{Z}$). So nothing maps to 0.

Step 3 (c) *Surjective.* Let $y \in \mathbb{Q}$. Define $x = \frac{y-1}{2}$. Then $x \in \mathbb{Q}$ [Theorem 0.16] and $f(x) = 2 \cdot \frac{y-1}{2} + 1 = y$.

Step 4 (d) *Surjective.* Let $y \in \mathbb{R}$; $x = \frac{y-1}{2} \in \mathbb{R}$ and $f(x) = y$.

Conclusion. (a) yes, (b) no, (c) yes, (d) yes. □

Compare with your work.

- A surjectivity proof must *check that x is in the domain* (in (b) it is not) and *compute $f(x) = y$* .
- In (a) the step from “odd” to “ $= 2k + 1$ ” quotes Proposition 1.1.58.

Exercise 3.2.14**essential**

Let $f : X \rightarrow Y$. Prove that f is surjective if and only if $Y = f[X]$.

Given. $f[X] = \{f(x) \mid x \in X\} \subseteq Y$ always (its elements are values, which lie in Y). So $Y = f[X]$ iff $Y \subseteq f[X]$.

Step 1 ($\Rightarrow$) Suppose f is surjective. $f[X] \subseteq Y$ holds always. For $Y \subseteq f[X]$: let $y \in Y$; by surjectivity there is $x \in X$ with $f(x) = y$ [Definition 3.2.9]; so $y \in f[X]$ [Definition 3.1.31]. Hence $Y = f[X]$ [Axiom 2.1.14].

Step 2 ($\Leftarrow$) Suppose $Y = f[X]$. Let $y \in Y$. Then $y \in f[X]$, so $y = f(x)$ for some $x \in X$ [Definition 3.1.31]. This is exactly $\exists x \in X, f(x) = y$; as y was arbitrary, f is surjective.

Conclusion. f surjective $\Leftrightarrow f[X] = Y$. □

Compare with your work.

- The two definitions say the same thing in different words; the proof is the unfolding, in both directions, plus the remark $f[X] \subseteq Y$.

Exercise 3.2.16**recommended**

Let $f : X \rightarrow \mathcal{P}(X)$. By considering the set $B = \{x \in X \mid x \notin f(x)\}$, prove that f is not surjective.

Given. $f(x) \subseteq X$ for every x (values lie in $\mathcal{P}(X)$). $B \subseteq X$, so $B \in \mathcal{P}(X)$. “Not surjective”: $\exists V \in \mathcal{P}(X), \forall a \in X, f(a) \neq V$.

Plan. Show B is not a value: if $f(a) = B$, ask whether $a \in B$; both answers contradict.

Step 1 $B \subseteq X$ [set-builder with $x \in X$], so $B \in \mathcal{P}(X)$: B is an element of the codomain.

Step 2 Suppose, towards a contradiction, that $a \in X$ and $f(a) = B$ [Strategy 1.1.51].

Step 3 By LEM, $a \in B$ or $a \notin B$ [Strategy 1.1.61].

Step 4 *Case $a \in B$:* by definition of B , $a \notin f(a)$. But $f(a) = B$, so $a \notin B$: contradiction.

Step 5 Case $a \notin B$: then $a \notin f(a)$ (as $f(a) = B$), so a satisfies the defining condition of B , i.e. $a \in B$: contradiction.

Step 6 Both cases are contradictory, so no $a \in X$ has $f(a) = B$ $[\neg I]$.

Conclusion. B is not in the image of f ; f is not surjective. (Cantor's theorem: no set surjects onto its power set.) $\square$

Compare with your work.

$\square$ Both cases must be written; the contradiction in each must name what clashes ($a \in B$ versus $a \notin B$).

Exercise 3.2.19

recommended

Let X be a set. Prove that $\text{id}_X : X \rightarrow X$ is a bijection.

Step 1 *Injective*: let $a, b \in X$ with $\text{id}_X(a) = \text{id}_X(b)$. Since $\text{id}_X(a) = a$ and $\text{id}_X(b) = b$ [Definition 3.1.12], $a = b$.

Step 2 *Surjective*: let $x \in X$. Define the preimage to be x itself: $x \in X$ and $\text{id}_X(x) = x$.

Conclusion. id_X is injective and surjective, hence bijective [Definition 3.2.17]. $\square$

Exercise 3.2.20

essential

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f$ is a bijection.

Given. f, g injective and surjective. Proposition 3.2.4: a composite of injections is injective.

Step 1 *Injective*: f and g are injective, so $g \circ f$ is injective [Proposition 3.2.4]. (Directly: if $g(f(a)) = g(f(b))$ then $f(a) = f(b)$ by injectivity of g , then $a = b$ by injectivity of f .)

Step 2 *Surjective*: let $z \in Z$. Since g is surjective, fix $y \in Y$ with $g(y) = z$ [Definition 3.2.9, Strategy 1.2.30]. Since f is surjective, fix $x \in X$ with $f(x) = y$.

Step 3 Then $(g \circ f)(x) = g(f(x)) = g(y) = z$ [Definition 3.1.17]. So z has a preimage; $g \circ f$ is surjective.

Conclusion. $g \circ f$ is a bijection [Definition 3.2.17]. $\square$

Compare with your work.

$\square$ For surjectivity work *backwards*: first the preimage under g , then under f .

Exercise 3.2.25

recommended

Let $f : X \rightarrow Y$. Prove that if f has a left inverse then f is injective.

Given. A left inverse: $g : Y \rightarrow X$ with $g \circ f = \text{id}_X$, i.e. $g(f(x)) = x$ for all $x \in X$ (Definition 3.2.23).

Step 1 Suppose $g : Y \rightarrow X$ is a left inverse of f . Let $a, b \in X$ with $f(a) = f(b)$ [Strategy 3.2.2].

Step 2 Apply g : $g(f(a)) = g(f(b))$. But $g(f(a)) = a$ and $g(f(b)) = b$ [$g \circ f = \text{id}_X$]. So $a = b$.

Conclusion. f is injective. $\square$

Compare with your work.

$\square$ This is Exercise 3.2.5 with $g \circ f = \text{id}_X$ (which is injective by 3.2.19) as the composite.

Exercise 3.2.39

recommended (a), (b)

Find the inverse of each function and prove that it is an inverse: (a) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{2x+1}{3}$; (b) $g : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N})$, $g(X) = \mathbb{N} \setminus X$.

Given. Strategy 3.2.41: h is an inverse of f iff $h(f(x)) = x$ for all x and $f(h(y)) = y$ for all y .

Step 1 (a) *Find:* solve $y = \frac{2x+1}{3}$ for x : $3y = 2x + 1$, $x = \frac{3y-1}{2}$. Define $h : \mathbb{R} \rightarrow \mathbb{R}$, $h(y) = \frac{3y-1}{2}$ (well-defined: a formula on $\mathbb{R}$).

Step 2 *Left:* for $x \in \mathbb{R}$, $h(f(x)) = \frac{3 \cdot \frac{2x+1}{3} - 1}{2} = \frac{2x+1-1}{2} = x$.

Step 3 *Right:* for $y \in \mathbb{R}$, $f(h(y)) = \frac{2 \cdot \frac{3y-1}{2} + 1}{3} = \frac{3y-1+1}{3} = y$.

Step 4 So h is an inverse of f [Definition 3.2.37]; $f^{-1}(y) = \frac{3y-1}{2}$.

Step 5 (b) *Find:* removing X from $\mathbb{N}$ twice gives X back, so guess $g^{-1} = g$. Check: for $X \in \mathcal{P}(\mathbb{N})$, $g(g(X)) = \mathbb{N} \setminus (\mathbb{N} \setminus X) = \mathbb{N} \cap X$ [Exercise 2.2.43] $= X$ [$X \subseteq \mathbb{N}$, Proposition 2.2.8].

Step 6 This single computation shows $g \circ g = \text{id}_{\mathcal{P}(\mathbb{N})}$, which is at once the left-inverse and the right-inverse condition for g with itself. So g is its own inverse.

Conclusion. (a) $f^{-1}(y) = \frac{3y-1}{2}$; (b) $g^{-1} = g$. □

Compare with your work.

- ☐ Both compositions must be computed in (a); checking only $h \circ f = \text{id}$ proves only that f is injective.
- ☐ In (b) the key step $\mathbb{N} \setminus (\mathbb{N} \setminus X) = X$ needs $X \subseteq \mathbb{N}$ and the cited exercise.

Exercise 3.2.40

essential

Let $f : X \rightarrow Y$. Prove that f is bijective if and only if f has an inverse.

Given. Definition 3.1.1 (a proposition $\forall y \in Y, \exists! x \in X, p(x, y)$ defines a function $Y \rightarrow X$); Exercise 3.2.25 (left inverse $\Rightarrow$ injective).

Step 1 ($\Rightarrow$) Suppose f is bijective. Let $y \in Y$. There is $x \in X$ with $f(x) = y$ [surjective], and it is unique: if $f(x) = y = f(x')$ then $x = x'$ [injective]. So $\forall y \in Y, \exists! x \in X, f(x) = y$.

Step 2 Hence we may define $g : Y \rightarrow X$ by: $g(y)$ is the unique $x \in X$ with $f(x) = y$ [Definition 3.1.1; well-defined by Step 1]. By construction $f(g(y)) = y$ for all $y \in Y$: g is a right inverse.

Step 3 For $x \in X$: the unique element mapping to $f(x)$ is x itself, so $g(f(x)) = x$: g is a left inverse. Hence g is an inverse of f [Definition 3.2.37].

Step 4 ($\Leftarrow$) Suppose $g : Y \rightarrow X$ is an inverse of f . *Injective:* g is a left inverse, so f is injective [Exercise 3.2.25]. *Surjective:* let $y \in Y$; define $x = g(y) \in X$; then $f(x) = f(g(y)) = y$ [$f \circ g = \text{id}_Y$]. So f is surjective.

Conclusion. f bijective $\Leftrightarrow f$ has an inverse. □

Compare with your work.

- ☐ In ($\Rightarrow$) the inverse must be *defined* (via unique existence), then both identities checked. In ($\Leftarrow$) each identity gives one property.

Exercise 3.2.45

recommended

Let $f : X \rightarrow Y$ be a bijection. Prove that $f^{-1} : Y \rightarrow X$ is a bijection.

Given. f^{-1} is the (unique, Proposition 3.2.42) inverse of f : $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

Plan. Read the two identities as saying that f is an inverse of f^{-1} , then apply Exercise 3.2.40 to f^{-1} .

Step 1 By Exercise 3.2.40, f has an inverse $f^{-1} : Y \rightarrow X$ with $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

Step 2 For the function $f^{-1} : Y \rightarrow X$, a left inverse is $h : X \rightarrow Y$ with $h \circ f^{-1} = \text{id}_Y$; the identity $f \circ f^{-1} = \text{id}_Y$ says $h = f$ works. A right inverse is h with $f^{-1} \circ h = \text{id}_X$; the identity $f^{-1} \circ f = \text{id}_X$ says $h = f$ works.

Step 3 So f is an inverse of f^{-1} [Definition 3.2.37]. By Exercise 3.2.40 (direction $\Leftarrow$), f^{-1} is a bijection.

Conclusion. f^{-1} is a bijection, and $(f^{-1})^{-1} = f$. □

Compare with your work.

- No computation is needed: the two defining equations of an inverse are symmetric in f and f^{-1} . Alternatively prove injective/surjective directly from the identities.

Exercise 3.2.46

recommended

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f$ is a bijection and express its inverse in terms of f^{-1} and g^{-1} .

Plan. Guess $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (undo the last step first) and verify both identities with associativity.

Step 1 Candidate: $h = f^{-1} \circ g^{-1} : Z \rightarrow X$ (well-defined: $g^{-1} : Z \rightarrow Y$, $f^{-1} : Y \rightarrow X$).

Step 2 $h \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f$ [associativity, Exercise 3.1.21] $= f^{-1} \circ \text{id}_Y \circ f$ [$g^{-1} \circ g = \text{id}_Y$] $= f^{-1} \circ f = \text{id}_X$ [Example 3.1.20; $f^{-1} \circ f = \text{id}_X$].

Step 3 $(g \circ f) \circ h = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_Y \circ g^{-1} = g \circ g^{-1} = \text{id}_Z$.

Step 4 So h is an inverse of $g \circ f$; hence $g \circ f$ is a bijection [Strategy 3.2.41 / Exercise 3.2.40] with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ [uniqueness of inverses, Proposition 3.2.42].

Conclusion. $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (“socks and shoes”). □

Compare with your work.

- The order reverses; $g^{-1} \circ f^{-1}$ is not even composable in general.

Exercise 3.2.47

essential

Let $f : X \rightarrow A$ and $g : Y \rightarrow B$ be bijections. Prove that there is a bijection $X \times Y \rightarrow A \times B$, and describe its inverse.

Plan. Act coordinatewise: $(x, y) \mapsto (f(x), g(y))$, with inverse $(a, b) \mapsto (f^{-1}(a), g^{-1}(b))$; verify both identities (Strategy 3.2.41).

Step 1 Define $h : X \times Y \rightarrow A \times B$ by $h(x, y) = (f(x), g(y))$. Well-defined: for $(x, y) \in X \times Y$, $f(x) \in A$ and $g(y) \in B$, so $(f(x), g(y)) \in A \times B$ [Definition 2.2.49].

Step 2 Define $k : A \times B \rightarrow X \times Y$ by $k(a, b) = (f^{-1}(a), g^{-1}(b))$ [f^{-1}, g^{-1} exist by Exercise 3.2.40].

Step 3 For $(x, y) \in X \times Y$: $k(h(x, y)) = k(f(x), g(y)) = (f^{-1}(f(x)), g^{-1}(g(y))) = (x, y)$ [$f^{-1} \circ f = \text{id}_X$, $g^{-1} \circ g = \text{id}_Y$].

Step 4 For $(a, b) \in A \times B$: $h(k(a, b)) = h(f^{-1}(a), g^{-1}(b)) = (f(f^{-1}(a)), g(g^{-1}(b))) = (a, b)$.

Step 5 So k is an inverse of h ; h is a bijection with $h^{-1} = k$ [Strategy 3.2.41].

Conclusion. $h(x, y) = (f(x), g(y))$ is a bijection $X \times Y \rightarrow A \times B$ with inverse $(a, b) \mapsto (f^{-1}(a), g^{-1}(b))$. □

Compare with your work.

- Equality of pairs is coordinatewise (Definition 2.2.46), which is why checking each coordinate suffices.

C CHAPTER 3 EXERCISES**Exercise 3.E.2****recommended**

Let X be a set. Prove that $\forall a \in X, \exists! U \in \mathcal{P}(X), (a \in U \wedge \exists! x \in X, x \in U)$, and describe explicitly the function $X \rightarrow \mathcal{P}(X)$ that this suggests.

Given. The inner $\exists! x \in X, x \in U$ says “ U has exactly one element”. Strategy 1.2.35.

Step 1 Let $a \in X$ [Strategy 1.2.10].

Step 2 *Existence:* define $U = \{a\}$. Then $U \subseteq X$ (its only element is $a \in X$), so $U \in \mathcal{P}(X)$. $a \in U$. And U has exactly one element: $a \in U$, and any $x \in U$ equals a [list notation]. So U has the property.

Step 3 *Uniqueness:* let $U, V \in \mathcal{P}(X)$ both have the property. Let $u \in U$; since $a \in U$ and U has exactly one element, $u = a$. So $U \subseteq \{a\}$; with $a \in U$, $U = \{a\}$ [double containment]. The same for V : $V = \{a\}$. Hence $U = V$.

Step 4 The function suggested: $s : X \rightarrow \mathcal{P}(X), s(a) = \{a\}$ (the *singleton* map), well-defined because for each a the set $\{a\}$ is the unique U with the property [Definition 3.1.1].

Conclusion. As stated. □

Exercise 3.E.13**essential**

Let $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = a + 2b$, and $U = \{1\} \times \mathbb{Z}$. Find $f[U]$.

Step 1 Elements of U are the pairs $(1, b)$ with $b \in \mathbb{Z}$ [Definition 2.2.49]. So $f[U] = \{f(1, b) \mid b \in \mathbb{Z}\} = \{1 + 2b \mid b \in \mathbb{Z}\}$ [Definition 3.1.31].

Step 2 Claim: this is the set O of odd integers. $(\subseteq)$ $1 + 2b = 2b + 1$ is odd [Proposition 1.1.58]. $(\supseteq)$ if n is odd then $n = 2b + 1$ for some $b \in \mathbb{Z}$ [Proposition 1.1.58], and $n = f(1, b)$ with $(1, b) \in U$.

Answer. $f[U] = \{1 + 2b \mid b \in \mathbb{Z}\}$, the set of odd integers. □

Exercise 3.E.17**essential**

Let $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = a + 2b$, and $V = \{n \in \mathbb{Z} \mid n \text{ is odd}\}$. Find $f^{-1}[V]$.

Step 1 $f^{-1}[V] = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a + 2b \in V\} = \{(a, b) \mid a + 2b \text{ is odd}\}$ [Definition 3.1.37].

Step 2 Claim: $a + 2b$ is odd iff a is odd. If $a = 2k + 1$ then $a + 2b = 2(k + b) + 1$ is odd [Proposition 1.1.58]. If a is even, $a = 2k$, then $a + 2b = 2(k + b)$ is even, hence not odd [Definition 0.41]. By the division theorem these are the only cases.

Step 3 So $f^{-1}[V] = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \text{ is odd}\} = O \times \mathbb{Z}$, where O is the set of odd integers.

Answer. $f^{-1}[V] = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \text{ odd}\}; b$ is unrestricted. □

Compare with your work.

□ Both directions of “ $a + 2b$ odd iff a odd” are needed to describe the preimage exactly.

Exercise 3.E.19**essential**

Let $f : X \rightarrow Y$. Prove or find a counterexample: (a) $U \subseteq f^{-1}[f[U]]$ for all $U \subseteq X$; (b) $f^{-1}[f[U]] \subseteq U$ for all $U \subseteq X$; (c) $V \subseteq f[f^{-1}[V]]$ for all $V \subseteq Y$; (d) $f[f^{-1}[V]] \subseteq V$ for all $V \subseteq Y$.

Plan. (a), (d) are true by unfolding; (b) fails when f is not injective, (c) when f is not surjective. Use $f(x) = x^2$ on $\mathbb{R}$ for both counterexamples.

Step 1 (a) *True.* Let $x \in U$. Then $f(x) \in f[U]$ [Definition 3.1.31, witness x]. So $x \in f^{-1}[f[U]]$ [Definition 3.1.37].

Step 2 (b) *False.* $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{1\}$. $f[U] = \{1\}$; $f^{-1}[\{1\}] = \{x \in \mathbb{R} \mid x^2 = 1\} = \{-1, 1\}$; and $-1 \notin U$.

Step 3 (c) *False.* Same f , $V = \{-1\}$. $f^{-1}[V] = \{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$, so $f[f^{-1}[V]] = f[\emptyset] = \emptyset$, which does not contain -1 .

Step 4 (d) *True.* Let $y \in f[f^{-1}[V]]$. Fix $x \in f^{-1}[V]$ with $y = f(x)$ [Definition 3.1.31]. Then $f(x) \in V$ [Definition 3.1.37], i.e. $y \in V$.

Conclusion. (a), (d) hold always; (b), (c) fail in general. (Equality in (b) holds for injective f , in (c) for surjective f .) □

Compare with your work.

□ For (a) and (d) the proof is two lines of unfolding; for (b) and (c) compute both sides explicitly.

Exercise 3.E.24**essential**

(a) Prove that for all functions $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, if $g \circ f$ is bijective then f is injective and g is surjective. (b) Find functions f, g such that $g \circ f$ is bijective, f is not surjective and g is not injective.

Step 1 (a), f injective: $g \circ f$ is injective, so f is injective [Exercise 3.2.5].

Step 2 (a), g surjective: let $z \in Z$. Since $g \circ f$ is surjective, fix $x \in X$ with $g(f(x)) = z$. Define $y = f(x) \in Y$; then $g(y) = z$. So g is surjective.

Step 3 (b) Take $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$; $f(0) = 0$; $g(0) = g(1) = 0$. Then $g \circ f : \{0\} \rightarrow \{0\}$ sends $0 \mapsto 0$, so $g \circ f = \text{id}_{\{0\}}$, a bijection [Exercise 3.2.19]. But $1 \in Y$ is not a value of f (f not surjective), and $g(0) = g(1)$ with $0 \neq 1$ (g not injective).

Conclusion. (a) proved; (b) the example above. □

Compare with your work.

□ The pattern: the *first* function inherits injectivity, the *last* inherits surjectivity, and nothing more (part (b)).

Exercise 3.E.25**essential; (e) recommended later**

For each pair (U, V) of subsets of $\mathbb{R}$, determine whether “ $f(x) = x^2 - 4x + 7$ for all $x \in U$ ” defines a function $f : U \rightarrow V$, and if so whether f is injective and whether it is surjective. (a) $U = V = \mathbb{R}$; (b) $U = (1, 4)$, $V = [3, 7]$; (c) $U = [3, 4]$, $V = [4, 7]$; (d) $U = (3, 4]$, $V = [4, 7]$; (e) $U = [2, \infty)$, $V = [3, \infty)$; (f) $U = [2, \infty)$, $V = \mathbb{R}$.

Plan. Complete the square once: $f(x) = (x - 2)^2 + 3$. Facts: $f(x) \geq 3$, with $f(x) = 3$ only at $x = 2$;

$f(2+t) = f(2-t)$ (symmetric about 2); for $x \geq 2$ and $y \geq 3$, $f(x) = y$ iff $x = 2 + \sqrt{y-3}$. Then for each pair: well-defined means $f(x) \in V$ for all $x \in U$.

Step 1 $x^2 - 4x + 7 = (x-2)^2 + 3$ [$(x-2)^2 = x^2 - 4x + 4$]. Since $(x-2)^2 \geq 0$, $f(x) \geq 3$.

Step 2 (a) Defined (real values). *Not injective*: $f(0) = 7 = f(4)$, $0 \neq 4$. *Not surjective*: $0 \in V$ but $f(x) \geq 3 > 0$ for all x .

Step 3 (b) For $1 < x < 4$: $-1 < x-2 < 2$, so $0 \leq (x-2)^2 < 4$ and $3 \leq f(x) < 7$: values lie in V , defined. *Not injective*: $f(1.5) = f(2.5) = 3.25$. *Surjective*: let $y \in [3, 7)$; define $x = 2 + \sqrt{y-3}$. Then $0 \leq \sqrt{y-3} < 2$, so $2 \leq x < 4$, hence $x \in (1, 4)$; and $f(x) = (\sqrt{y-3})^2 + 3 = y$.

Step 4 (c) For $3 \leq x < 4$: $1 \leq x-2 < 2$, so $1 \leq (x-2)^2 < 4$ and $4 \leq f(x) < 7$: defined. *Injective*: let $a, b \in [3, 4)$ with $f(a) = f(b)$; then $(a-2)^2 = (b-2)^2$ with $a-2, b-2 > 0$, so $a-2 = b-2$ [positive square roots are unique, Example 1.2.36], $a = b$. *Surjective*: for $y \in [4, 7)$, $x = 2 + \sqrt{y-3}$ satisfies $1 \leq \sqrt{y-3} < 2$, so $x \in [3, 4)$, and $f(x) = y$. Bijective.

Step 5 (d) Not a function: $4 \in U$ but $f(4) = 7 \notin [4, 7)$ (existence of a value in the codomain fails).

Step 6 (e) For $x \geq 2$, $f(x) \geq 3$: defined. *Injective* as in (c) (now $a-2, b-2 \geq 0$, and $\sqrt{\cdot}$ is still unique on $[0, \infty)$). *Surjective*: for $y \geq 3$, $x = 2 + \sqrt{y-3} \geq 2$ and $f(x) = y$. Bijective.

Step 7 (f) Defined and injective as in (e). *Not surjective*: $0 \in \mathbb{R}$ is not a value since $f(x) \geq 3$.

Conclusion. (a) function, neither; (b) function, surjective only; (c) bijection; (d) not a function; (e) bijection; (f) function, injective only. $\square$

Compare with your work.

- ☐ “Defined” requires checking that every value lies in V ; (d) fails there. Then injectivity (symmetry about 2 is the enemy) and surjectivity (solve $f(x) = y$ and check $x \in U$).
- ☐ In (b) both $f(1.5) = f(2.5)$ and the two-sided range check are needed.

Exercise 3.E.26(b)

recommended (later)

Find (with proof) a bijection $\mathbb{R} \rightarrow (-1, 1)$.

Plan. Squash the line with $x \mapsto \frac{x}{1+|x|}$; invert with $y \mapsto \frac{y}{1-|y|}$; verify both identities (Strategy 3.2.41).

Step 1 Define $f: \mathbb{R} \rightarrow (-1, 1)$ by $f(x) = \frac{x}{1+|x|}$. Well-defined: $1+|x| > 0$, and $|f(x)| = \frac{|x|}{1+|x|} < 1$, so $f(x) \in (-1, 1)$.

Step 2 Define $g: (-1, 1) \rightarrow \mathbb{R}$ by $g(y) = \frac{y}{1-|y|}$. Well-defined: $|y| < 1$ so $1-|y| > 0$.

Step 3 For $x \in \mathbb{R}$: $|f(x)| = \frac{|x|}{1+|x|}$, so $1-|f(x)| = \frac{1}{1+|x|}$, and $g(f(x)) = \frac{x}{1+|x|} \cdot (1+|x|) = x$.

Step 4 For $y \in (-1, 1)$: $|g(y)| = \frac{|y|}{1-|y|}$, so $1+|g(y)| = \frac{1}{1-|y|}$, and $f(g(y)) = \frac{y}{1-|y|} \cdot (1-|y|) = y$.

Step 5 So g is an inverse of f ; f is a bijection [Strategy 3.2.41].

Conclusion. $f(x) = \frac{x}{1+|x|}$ is a bijection $\mathbb{R} \rightarrow (-1, 1)$ with inverse g . $\square$

Compare with your work.

- ☐ Any other bijection (e.g. $\frac{2}{\pi} \arctan x$, if you may use calculus) is fine, but the two identities must be computed.

Exercise 3.E.34**essential**

True or false: given $f : X \rightarrow Y$, the assignment $V \mapsto f^{-1}[V]$ defines a function $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$.

Step 1 *True.* Let $V \in \mathcal{P}(Y)$, i.e. $V \subseteq Y$. The set $f^{-1}[V] = \{x \in X \mid f(x) \in V\}$ is specified by set-builder notation, so it exists and is unique [Definition 3.1.37], and it is a subset of X , so $f^{-1}[V] \in \mathcal{P}(X)$.

Step 2 Thus every V gets exactly one value in $\mathcal{P}(X)$: totality, existence, uniqueness hold, and $V \mapsto f^{-1}[V]$ is a well-defined function $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ (the book's f^*).

Conclusion. True. □

Compare with your work.

□ $f^{-1}[V]$ exists for *every* f ; no inverse function is needed. The symbol f^{-1} in $f^{-1}[V]$ is notation for the preimage.

Exercise 3.E.35**recommended**

True or false: let f, g, h be composable functions; if $h \circ g \circ f$ is injective then g is injective.

Step 1 *False.* Take $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$, $W = \{0\}$; $f : X \rightarrow Y$, $f(0) = 0$; $g : Y \rightarrow Z$, $g(0) = g(1) = 0$; $h = \text{id}_Z$.

Step 2 $h \circ g \circ f : \{0\} \rightarrow \{0\}$ is injective: any function with a one-element domain is injective (if $a, b \in \{0\}$ then $a = b$).

Step 3 But $g(0) = g(1)$ with $0 \neq 1$: g is not injective.

Conclusion. False. (Exercise 3.2.5 gives only that f , the first function applied, is injective.) □

Exercise 3.E.37**recommended**

Let $f : X \rightarrow Y$ and $U \subseteq X$. Then there is a function $g : U \rightarrow Y$ defined by $g(x) = f(x)$ for all $x \in U$. (Always, sometimes, never?)

Step 1 *Always.* Let $x \in U$. Then $x \in X$ [$U \subseteq X$], so $f(x)$ is a uniquely specified element of Y [f is a function].

Step 2 So each $x \in U$ receives exactly one value in Y : totality, existence, uniqueness hold. g (the restriction of f to U) is well-defined.

Conclusion. Always. □

Exercise 3.E.38**recommended**

Let $f : X \rightarrow Y$ and $V \subseteq Y$. Then there is a function $g : X \rightarrow V$ defined by $g(x) = f(x)$ for all $x \in X$. (Always, sometimes, never?)

Step 1 The specification is well-defined iff every value $f(x)$ lies in the codomain V , i.e. iff $f[X] \subseteq V$ [existence of a value in the codomain].

Step 2 *Holds:* $V = Y$ (then $g = f$). *Fails:* $f = \text{id}_{\mathbb{R}}$, $V = \{0\}$: $f(1) = 1 \notin V$.

Conclusion. Sometimes (exactly when $f[X] \subseteq V$). □

Exercise 3.E.41**essential**

Let X, Y be sets and $G \subseteq X \times Y$. Then G is the graph of a function $f : X \rightarrow Y$. (Always, sometimes, never?)

Given. Theorem 3.1.9.

Step 1 Holds: $X = Y = \{0\}$, $G = \{(0,0)\}$: for the only $x = 0$ there is exactly one y with $(0,y) \in G$; $G = \text{Gr}(\text{id}_{\{0\}})$.

Step 2 Fails: $X = Y = \{0\}$, $G = \emptyset$: no y with $(0,y) \in G$ (existence fails). Also $X = \{0\}$, $Y = \{0,1\}$, $G = X \times Y$: two pairs $(0,0), (0,1)$ (uniqueness fails).

Conclusion. Sometimes. □

Exercise 3.E.47**recommended**

Let $a, b, c, d \in \mathbb{R}$ and define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $f(x, y) = (ax + by, cx + dy)$. Then f is injective if and only if f is surjective. (Always, sometimes, never?)

Plan. Show that both properties are equivalent to $ad - bc \neq 0$. If $ad - bc \neq 0$, write down the inverse. If $ad - bc = 0$, exhibit a non-zero vector sent to $(0,0)$ (not injective) and a vector not in the image (not surjective).

Step 1 Case $\Delta := ad - bc \neq 0$: f is bijective. Define $g(u, v) = \left(\frac{du - bv}{\Delta}, \frac{av - cu}{\Delta}\right)$. Then

$$f(g(u, v)) = \left(\frac{a(du - bv) + b(av - cu)}{\Delta}, \frac{c(du - bv) + d(av - cu)}{\Delta}\right) = \left(\frac{\Delta u}{\Delta}, \frac{\Delta v}{\Delta}\right) = (u, v),$$

and with $u = ax + by$, $v = cx + dy$ one computes $du - bv = \Delta x$, $av - cu = \Delta y$, so $g(f(x, y)) = (x, y)$. By Strategy 3.2.41 f is bijective: injective and surjective both hold.

Step 2 Case $\Delta = 0$: f is not injective. We find $(x, y) \neq (0, 0)$ with $f(x, y) = (0, 0) = f(0, 0)$. If $(a, b) \neq (0, 0)$ take $(x, y) = (b, -a)$: $ax + by = ab - ba = 0$, $cx + dy = cb - da = -\Delta = 0$. If $(a, b) = (0, 0)$ and $(c, d) \neq (0, 0)$ take $(d, -c)$: $cd - dc = 0$. If all four vanish take $(1, 0)$.

Step 3 Case $\Delta = 0$: f is not surjective. If $(a, c) \neq (0, 0)$: every value $(u, v) = (ax + by, cx + dy)$ satisfies $cu - av = (cb - ad)y = 0$, but $(u, v) = (c, -a)$ has $cu - av = c^2 + a^2 \neq 0$, so it is not a value. If $(a, c) = (0, 0)$ but $(b, d) \neq (0, 0)$: values are (by, dy) with $du - bv = 0$, but $(d, -b)$ has $du - bv = d^2 + b^2 \neq 0$. If all four vanish, $f \equiv (0, 0)$ and $(1, 0)$ is not a value.

Step 4 So: injective $\Leftrightarrow \Delta \neq 0 \Leftrightarrow$ surjective.

Conclusion. Always: for these linear maps, injective and surjective are equivalent. □

Compare with your work.

□ “Always” requires handling every (a, b, c, d) , including the degenerate sub-cases in Steps 2–3.

Exercise 3.E.48**essential**

Let $U, V \subseteq \mathbb{R}$ be such that $x^2 \in V$ for all $x \in U$. The function $f : U \rightarrow V$ defined by $f(x) = x^2$ is injective. (Always, sometimes, never?)

Given. The hypothesis makes f well-defined (values in V).

Step 1 Holds: $U = [0, \infty)$, $V = \mathbb{R}$. Let $a, b \geq 0$ with $a^2 = b^2$; then $a = b$ [uniqueness of non-negative square roots]. Injective.

Step 2 Fails: $U = V = \mathbb{R}$: $f(-1) = 1 = f(1)$ with $-1 \neq 1$. Not injective.

Conclusion. Sometimes (it depends on U : injective iff U contains no pair $\{-a, a\}$ with $a \neq 0$). $\square$

Chapter 4

Mathematical induction

Contents

- A** Weak induction (§4.2)
- B** Strong induction, well-ordering (§4.3)
- C** Chapter 4 exercises (4.E)

Planning

★ 4.2: 5, 11 · 4.3: 8, 9, 14, 15 · 4.E: 10, 11, 13, 16–20, 25–27, 29–32
 ★ 4.2: 8, 17, 20 · 4.3: 10, 18 · 4.E: 12, 14, 15, 21, 24, 28, 33–35
 §4.1 is not in the planning. Tested in the **exam** (4 Nov).

A WEAK INDUCTION

What you quote in this section

Weak induction (Theorem 4.2.1, Strategy 4.2.2): to prove $\forall n \geq n_0, p(n)$, prove the *base case* $p(n_0)$ and the *induction step*: for all $n \geq n_0, p(n) \Rightarrow p(n+1)$. Write “We proceed by induction on $n \geq n_0$ ”, state $p(n)$, then “(Base case)”, “(Induction step) Let $n \geq n_0$ and suppose $p(n) \dots$ we must prove $p(n+1) \dots$ ”, then “the result follows by induction”. **Recursive sums** (Definition 4.1.10): $\sum_{k=0}^{n+1} a_k = (\sum_{k=0}^n a_k) + a_{n+1}$. **Binomial coefficients** (Definition 4.1.15, Theorem 4.2.15): $\binom{n}{0} = 1, \binom{0}{k+1} = 0, \binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$; $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for $k \leq n$ and 0 for $k > n$. **Binomial theorem** (4.2.18): $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

Exercise 4.2.5

essential

Prove by induction that $\sum_{k=0}^n 2^k = 2^{n+1} - 1$ for all $n \in \mathbb{N}$.

Given. Definition 4.1.10 for the recursive meaning of $\sum$.

To show. $\forall n \in \mathbb{N}, p(n)$ where $p(n): \sum_{k=0}^n 2^k = 2^{n+1} - 1$.

Step 1 We proceed by induction on $n \geq 0$, with $p(n)$ as above.

Step 2 (Base case $n = 0$.) $\sum_{k=0}^0 2^k = 2^0 = 1$, and $2^{0+1} - 1 = 2 - 1 = 1$. So $p(0)$ holds.

Step 3 (Induction step.) Let $n \geq 0$ and suppose $p(n): \sum_{k=0}^n 2^k = 2^{n+1} - 1$ [induction hypothesis]. We must prove $p(n+1): \sum_{k=0}^{n+1} 2^k = 2^{n+2} - 1$.

Step 4

$$\begin{aligned} \sum_{k=0}^{n+1} 2^k &= \left(\sum_{k=0}^n 2^k \right) + 2^{n+1} \quad [\text{Definition 4.1.10}] \\ &= (2^{n+1} - 1) + 2^{n+1} \quad [\text{induction hypothesis}] \\ &= 2 \cdot 2^{n+1} - 1 = 2^{n+2} - 1 \quad [\text{algebra}]. \end{aligned}$$

So $p(n+1)$ holds.

Conclusion. By induction, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$ for all $n \in \mathbb{N}$. □

Compare with your work.

- The four sentences: “by induction on n ” with $p(n)$ stated; base case computed on both sides; step with the hypothesis *marked* where it is used; closing sentence.
- Never start the step from the goal $\sum_{k=0}^{n+1} 2^k = 2^{n+2} - 1$ and manipulate it; start from the left side and arrive at the right side.

Exercise 4.2.8**recommended**

Write down the statement $p(n)$ that Example 4.2.7 (“all horses have the same colour”) tries to prove for all $n \geq 1$. Write down, with quantifiers, what the induction step must prove. Explain why the argument fails.

Step 1 $p(n)$: “every set of n horses has the same colour” (all horses in any n -element set share one colour).

Step 2 *Base case* $p(1)$: a set with one horse has only one colour. Correct.

Step 3 *Induction step* must prove $\forall n \geq 1, (p(n) \Rightarrow p(n+1))$: for every $n \geq 1$, if every n -set is monochrome then every $(n+1)$ -set is.

Step 4 *The argument*: given $n+1$ horses $h_1, \dots, h_{n+1}$, the sets $\{h_1, \dots, h_n\}$ and $\{h_2, \dots, h_{n+1}\}$ are monochrome by $p(n)$; they overlap, so the colour of the overlap forces all $n+1$ to agree.

Step 5 *Why it fails*: for $n=1$ the two sets are $\{h_1\}$ and $\{h_2\}$, which do *not* overlap, so nothing links the colour of h_1 to that of h_2 . The implication $p(1) \Rightarrow p(2)$ is never proved, and the step only works for $n \geq 2$. With the first link missing, induction cannot start: $p(2)$ is in fact false.

Conclusion. The step must hold for every $n \geq n_0$ including $n = n_0$; here it fails at $n = 1$. □

Exercise 4.2.11**essential**

Let $a, r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}$ for all $n \in \mathbb{N}$.

Given. $r \neq 1$, so $1-r \neq 0$ and division by $1-r$ is allowed.

Step 1 We proceed by induction on $n \geq 0$ with $p(n)$: $\sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}$.

Step 2 (*Base case.*) $\sum_{k=0}^0 ar^k = ar^0 = a$, and $\frac{a(1-r^1)}{1-r} = \frac{a(1-r)}{1-r} = a$ [$1-r \neq 0$]. So $p(0)$ holds.

Step 3 (*Induction step.*) Let $n \geq 0$ and suppose $p(n)$. We must prove $\sum_{k=0}^{n+1} ar^k = \frac{a(1-r^{n+2})}{1-r}$.

Step 4

$$\begin{aligned}
 \sum_{k=0}^{n+1} ar^k &= \sum_{k=0}^n ar^k + ar^{n+1} && \text{[Definition 4.1.10]} \\
 &= \frac{a(1-r^{n+1})}{1-r} + ar^{n+1} && \text{[induction hypothesis]} \\
 &= \frac{a(1-r^{n+1}) + ar^{n+1}(1-r)}{1-r} && \text{[common denominator]} \\
 &= \frac{a - ar^{n+1} + ar^{n+1} - ar^{n+2}}{1-r} = \frac{a(1-r^{n+2})}{1-r}.
 \end{aligned}$$

So $p(n+1)$ holds.

Conclusion. By induction the formula holds for all $n \in \mathbb{N}$. (Exercise 4.2.5 is the case $a = 1, r = 2$.) □

Compare with your work.

- Say where $r \neq 1$ is used (base case and every division by $1-r$).

Exercise 4.2.17

recommended

Prove that $\binom{n}{k} = \binom{n}{n-k}$ for all $n, k \in \mathbb{N}$ with $k \leq n$.

Given. Theorem 4.2.15: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ whenever $k \leq n$.

Step 1 Let $n, k \in \mathbb{N}$ with $k \leq n$. Then $n - k \in \mathbb{N}$ and $n - k \leq n$, so Theorem 4.2.15 applies to both $\binom{n}{k}$ and $\binom{n}{n-k}$.

Step 2 $\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$ [$n - (n - k) = k$; commutativity of multiplication].

Conclusion. $\binom{n}{k} = \binom{n}{n-k}$. □

Compare with your work.

- No induction is needed once Theorem 4.2.15 is available; the check $n - k \leq n$ makes the theorem applicable.

Exercise 4.2.20

recommended

Give a proof of $\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$ using the binomial theorem.

Given. Theorem 4.2.18 with $x = -1, y = 1$. The identity is meant for $n \geq 1$ (for $n = 0$ the sum is 1).

Step 1 Let $n \geq 1$. By the binomial theorem with $x = -1, y = 1$:

$$(-1 + 1)^n = \sum_{i=0}^n \binom{n}{i} (-1)^i 1^{n-i} = \sum_{i=0}^n (-1)^i \binom{n}{i} \quad [1^{n-i} = 1].$$

Step 2 The left side is $0^n = 0$ because $n \geq 1$.

Conclusion. $\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$ for all $n \geq 1$. (Interpretation: a finite non-empty set has as many even-sized as odd-sized subsets.) □

Compare with your work.

- State the substitution ($x = -1, y = 1$) explicitly and note why $n \geq 1$ is needed.

B STRONG INDUCTION, WELL-ORDERING**What you quote in this section**

Strong induction (Theorem 4.3.2, Strategies 4.3.3 and 4.3.6): to prove $\forall n \geq n_0, p(n)$, prove base cases $p(n_0), \dots, p(n_1)$ and the step: “fix $n \geq n_1$ and assume $p(k)$ for all $n_0 \leq k \leq n$; prove $p(n+1)$ ”. Use it when $p(n+1)$ depends on $p(n-1), p(n-2), \dots$; then you need as many base cases as the recursion looks back, and the step starts at n_1 so that the recursion applies to $n+1$. **Well-ordering** (Theorem 4.3.11): every non-empty subset of $\mathbb{N}$ has a least element. **Infinite descent** (Definition 4.3.16): $X \subseteq \mathbb{R}$ has infinite descent if $\forall x \in X, \exists y \in X, y < x$; no non-empty subset of $\mathbb{N}$ has it.

Exercise 4.3.8

essential

Define $a_0 = 4, a_1 = 9$ and $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$. Prove that $a_n = 3 \cdot 2^n + 3^n$ for all $n \in \mathbb{N}$.

Given. The recursion uses two previous terms, so two base cases and strong induction with the step

starting at $n = 1$ (Example 4.3.7 is the model).

Step 1 We proceed by strong induction on $n \geq 0$ with $p(n): a_n = 3 \cdot 2^n + 3^n$, base cases $n = 0, 1$.

Step 2 (Base cases.) $a_0 = 4 = 3 \cdot 1 + 1 = 3 \cdot 2^0 + 3^0$. $a_1 = 9 = 6 + 3 = 3 \cdot 2^1 + 3^1$.

Step 3 (Induction step.) Fix $n \geq 1$ and assume $a_k = 3 \cdot 2^k + 3^k$ for all $0 \leq k \leq n$. We must prove $a_{n+1} = 3 \cdot 2^{n+1} + 3^{n+1}$. Since $n+1 \geq 2$ the recursion applies:

$$\begin{aligned} a_{n+1} &= 5a_n - 6a_{n-1} \quad [\text{definition, valid as } n+1 \geq 2] \\ &= 5(3 \cdot 2^n + 3^n) - 6(3 \cdot 2^{n-1} + 3^{n-1}) \quad [\text{hypothesis for } k = n \text{ and } k = n-1; \text{ both } \geq 0] \\ &= 15 \cdot 2^n + 5 \cdot 3^n - 9 \cdot 2^{n-1} - 2 \cdot 3^n \quad [6 \cdot 2^{n-1} = 3 \cdot 2^n, 6 \cdot 3^{n-1} = 2 \cdot 3^n] \\ &= 6 \cdot 2^n + 3 \cdot 3^n = 3 \cdot 2^{n+1} + 3^{n+1}. \end{aligned}$$

Conclusion. By strong induction, $a_n = 3 \cdot 2^n + 3^n$ for all $n \in \mathbb{N}$. □

Compare with your work.

- Two base cases, step fixed at $n \geq 1$ (so that $n-1 \geq 0$ and $n+1 \geq 2$), and the hypothesis used for *two* values of k .

Exercise 4.3.9

essential

The Tribonacci sequence is defined by $t_0 = 0, t_1 = 0, t_2 = 1$ and $t_n = t_{n-1} + t_{n-2} + t_{n-3}$ for $n \geq 3$. Prove that $t_n \leq 2^{n-3}$ for all $n \geq 3$.

Given. The recursion looks back three terms, so three base cases $n = 3, 4, 5$, and the step is fixed at $n \geq 5$ (so that $n-2 \geq 3$ and the hypothesis applies to all three terms).

Step 1 We proceed by strong induction on $n \geq 3$ with $p(n): t_n \leq 2^{n-3}$, base cases $n = 3, 4, 5$.

Step 2 (Base cases.) $t_3 = t_2 + t_1 + t_0 = 1 \leq 1 = 2^0$; $t_4 = t_3 + t_2 + t_1 = 2 \leq 2 = 2^1$; $t_5 = t_4 + t_3 + t_2 = 4 \leq 4 = 2^2$.

Step 3 (Induction step.) Fix $n \geq 5$ and assume $t_k \leq 2^{k-3}$ for all $3 \leq k \leq n$. We must prove $t_{n+1} \leq 2^{n-2}$.

Step 4 Since $n+1 \geq 6 \geq 3$, $t_{n+1} = t_n + t_{n-1} + t_{n-2}$ [definition]. The indices $n, n-1, n-2$ are all ≥ 3 [$n \geq 5$] and $\leq n$, so the hypothesis applies to each:

$$t_{n+1} \leq 2^{n-3} + 2^{n-4} + 2^{n-5} = 2^{n-5}(4 + 2 + 1) = 7 \cdot 2^{n-5} < 8 \cdot 2^{n-5} = 2^{n-2}.$$

Conclusion. By strong induction, $t_n \leq 2^{n-3}$ for all $n \geq 3$. □

Compare with your work.

- Explain why three base cases: the step at $n = 5$ uses t_5, t_4, t_3 ; a step at $n = 4$ would need t_2 , for which p is not stated.

Exercise 4.3.10

recommended

Prove that for all $n \geq 8$, a value of n dubloons can be obtained using only 3-dubloon and 5-dubloon coins.

Given. $p(n): \exists a, b \in \mathbb{N}, n = 3a + 5b$. The hint: $p(n) \Rightarrow p(n+3)$ (add one 3-coin). So the step for $n+1$ uses $p(n-2)$: strong induction with base cases 8, 9, 10 and the step fixed at $n \geq 10$.

Step 1 We proceed by strong induction on $n \geq 8$ with base cases 8, 9, 10.

Step 2 (Base cases.) $8 = 3 + 5$ ($a = b = 1$); $9 = 3 \cdot 3$ ($a = 3, b = 0$); $10 = 5 \cdot 2$ ($a = 0, b = 2$).

Step 3 (Induction step.) Fix $n \geq 10$ and assume $p(k)$ for all $8 \leq k \leq n$. We must prove $p(n+1)$.

Step 4 Since $n \geq 10$, $n - 2 \geq 8$, so $p(n - 2)$ is available: $n - 2 = 3a + 5b$ for some $a, b \in \mathbb{N}$ [Strategy 1.2.30].

Step 5 Then $n + 1 = (n - 2) + 3 = 3(a + 1) + 5b$ with $a + 1, b \in \mathbb{N}$, so $p(n + 1)$ holds $[\exists I]$.

Conclusion. By strong induction, every $n \geq 8$ is a sum of 3s and 5s. (The statement is false for 7: $7 = 3a + 5b$ would need $b \leq 1$, giving $3a = 7$ or $3a = 2$.) $\square$

Compare with your work.

$\square$ Three base cases because the step jumps back by 3; the step must be fixed at $n \geq 10$ so that $n - 2 \geq 8$.

Exercise 4.3.14

essential

What goes wrong in the proof of Proposition 4.3.12 ($\sqrt{2}$ is irrational) if we try to prove that $\sqrt{4}$ is irrational?

Given. The proof of 4.3.12: write $\sqrt{2} = \frac{a}{b}$ in lowest terms (Lemma 4.3.13); $a^2 = 2b^2$; so a^2 is even, so a is even (Proposition 1.1.63), $a = 2c$; then $4c^2 = 2b^2$, $b^2 = 2c^2$, so b is even; 2 divides both: contradiction.

Step 1 Run the same argument: suppose $\sqrt{4} = \frac{a}{b}$ in lowest terms. Squaring: $a^2 = 4b^2$. Then $a^2 = 2(2b^2)$ is even, so a is even [Proposition 1.1.63]: $a = 2c$.

Step 2 Substitute: $4c^2 = 4b^2$, so $c^2 = b^2$.

Step 3 In the $\sqrt{2}$ proof the corresponding equation was $b^2 = 2c^2$, which forced b even. Here $c^2 = b^2$ says nothing about the parity of b . No contradiction can be derived; the argument simply stops.

Step 4 And indeed there is none to derive: $\sqrt{4} = 2 = \frac{2}{1}$ with $a = 2$, $b = 1$ in lowest terms satisfies everything ($c = 1$, $c^2 = b^2$).

Conclusion. The step “hence b is even” fails, because after cancelling, the factor 2 no longer appears on the right-hand side (it appeared an even number of times in $4 = 2^2$). $\square$

Exercise 4.3.15

essential

Prove that $\sqrt{3}$ is irrational.

Given. Lemma 4.3.13 (a positive rational is $\frac{a}{b}$ with $a, b \in \mathbb{N}$ non-zero and 1 their only common natural divisor). We need the analogue of Proposition 1.1.63 for 3, proved as a lemma first (the book’s writing tip).

Step 1 *Lemma:* for $a \in \mathbb{Z}$, if $3 \mid a^2$ then $3 \mid a$. By Theorem 0.42, $a = 3q + r$ with $r \in \{0, 1, 2\}$. If $r = 1$: $a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1$, remainder 1. If $r = 2$: $a^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1$, remainder 1. In both cases $3 \nmid a^2$ [the remainder is $1 \neq 0$; uniqueness in Theorem 0.42]. So $3 \mid a^2$ forces $r = 0$, i.e. $3 \mid a$ [contraposition].

Step 2 *Main proof.* Suppose, towards a contradiction, that $\sqrt{3} \in \mathbb{Q}$ [Strategy 1.1.51]. Since $\sqrt{3} > 0$, Lemma 4.3.13 gives $\sqrt{3} = \frac{a}{b}$ with $a, b \in \mathbb{N} \setminus \{0\}$ having no common natural divisor other than 1.

Step 3 Squaring: $3 = \frac{a^2}{b^2}$, so $a^2 = 3b^2$. Hence $3 \mid a^2$ [Definition 0.36], so $3 \mid a$ [Lemma]: $a = 3c$ with $c \in \mathbb{Z}$.

Step 4 Substitute: $9c^2 = 3b^2$, so $b^2 = 3c^2$. Hence $3 \mid b^2$, so $3 \mid b$ [Lemma].

Step 5 Now 3 is a natural number dividing both a and b , and $3 \neq 1$: this contradicts the choice of a, b in Step 2 $[\neg E]$.

Conclusion. $\sqrt{3}$ is irrational $[-I]$. □

Compare with your work.

- The lemma “ $3 \mid a^2 \Rightarrow 3 \mid a$ ” must be proved (by remainders), not assumed; it is used *twice*.
- Say where lowest terms (Lemma 4.3.13) is used: in the final contradiction.

Exercise 4.3.18

recommended

Prove that the interval $(0, 1)$ has infinite descent.

Given. Definition 4.3.16: $\forall x \in (0, 1), \exists y \in (0, 1), y < x$.

Step 1 Let $x \in (0, 1)$ [Strategy 1.2.10]. Define $y = \frac{x}{2}$.

Step 2 $y > 0$ [$x > 0$]; $y < x$ [$\frac{x}{2} < x$ as $x > 0$]; $y < x < 1$. So $y \in (0, 1)$ and $y < x$ [Strategy 1.2.24].

Conclusion. $(0, 1)$ has infinite descent. (No contradiction arises, since $(0, 1) \not\subseteq \mathbb{N}$; the well-ordering principle only forbids infinite descent in non-empty subsets of $\mathbb{N}$.) □

C CHAPTER 4 EXERCISES

Exercise 4.E.10

essential

Find all natural numbers n such that $n^5 < 5^n$.

Plan. Test small n by hand to find where the inequality holds; then prove by induction that it holds from some point on.

Step 1 Small cases: $n = 0$: $0 < 1$ ✓; $n = 1$: $1 < 5$ ✓; $n = 2$: $32 < 25$ ✗; $n = 3$: $243 < 125$ ✗; $n = 4$: $1024 < 625$ ✗; $n = 5$: $3125 < 3125$ ✗; $n = 6$: $7776 < 15625$ ✓.

Step 2 Claim: $n^5 < 5^n$ for all $n \geq 6$. We proceed by induction on $n \geq 6$.

Step 3 (Base case.) $6^5 = 7776 < 15625 = 5^6$.

Step 4 (Induction step.) Let $n \geq 6$ and suppose $n^5 < 5^n$. We must prove $(n+1)^5 < 5^{n+1}$. Since $n \geq 6$, $\frac{n+1}{n} = 1 + \frac{1}{n} \leq \frac{7}{6}$, so $(n+1)^5 \leq \left(\frac{7}{6}\right)^5 n^5 = \frac{16807}{7776} n^5 < 5n^5$ [$\frac{16807}{7776} < 5$ since $16807 < 38880$]. Then $5n^5 < 5 \cdot 5^n = 5^{n+1}$ [induction hypothesis]. So $(n+1)^5 < 5^{n+1}$.

Step 5 By induction the claim holds for all $n \geq 6$. Together with Step 1 this covers every $n \in \mathbb{N}$.

Answer. $n^5 < 5^n$ exactly for $n \in \{0, 1\}$ and for all $n \geq 6$. □

Compare with your work.

- The induction only covers $n \geq 6$; the values $n \leq 5$ must be checked one by one and reported.
- The key estimate is $(1 + \frac{1}{n})^5 < 5$ for $n \geq 6$; justify it with a number, not with “clearly”.

Exercise 4.E.11

essential

Prove that $(1+x)^{123456789} \geq 1 + 123456789x$ for all real $x \geq -1$.

Plan. The number is irrelevant (the hint): prove $(1+x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$ by induction on n , for a fixed $x \geq -1$. Induction runs over n , never over the real variable x .

Step 1 Fix $x \in \mathbb{R}$ with $x \geq -1$; then $1+x \geq 0$. We prove $p(n)$: $(1+x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$, by

induction on n .

Step 2 (Base case.) $(1+x)^0 = 1 \geq 1 = 1 + 0 \cdot x$.

Step 3 (Induction step.) Let $n \geq 0$ and suppose $(1+x)^n \geq 1 + nx$. Multiply both sides by $1+x \geq 0$, which preserves $\geq$ [multiplying an inequality by a non-negative number]:

$$(1+x)^{n+1} = (1+x)(1+x)^n \geq (1+x)(1+nx) = 1 + nx + x + nx^2 = 1 + (n+1)x + nx^2.$$

Since $nx^2 \geq 0$ [$n \geq 0, x^2 \geq 0$], $1 + (n+1)x + nx^2 \geq 1 + (n+1)x$. So $(1+x)^{n+1} \geq 1 + (n+1)x$.

Step 4 By induction $p(n)$ holds for every n ; take $n = 123456789$. Since $x \geq -1$ was arbitrary, the statement holds for all such x .

Conclusion. $(1+x)^{123456789} \geq 1 + 123456789x$ for all $x \geq -1$ (Bernoulli's inequality). $\square$

Compare with your work.

- ☐ Say why $x \geq -1$ matters: it makes $1+x \geq 0$, so multiplying the hypothesis by $1+x$ keeps the direction of the inequality.

Exercise 4.E.12

recommended

Let $a \in \mathbb{N}$ be a number whose decimal expansion ends in the digit 6. Prove that a^n ends in 6 for all $n \geq 1$.

Given. “Ends in 6” means the last decimal digit is 6, i.e. the remainder on division by 10 is 6 (Proposition 1.2.12's idea): $a = 10q + 6$ for some $q \in \mathbb{N}$.

Step 1 Write $a = 10q + 6$ with $q \in \mathbb{N}$. We proceed by induction on $n \geq 1$ with $p(n)$: $a^n = 10m + 6$ for some $m \in \mathbb{N}$.

Step 2 (Base case.) $a^1 = a = 10q + 6$.

Step 3 (Induction step.) Let $n \geq 1$ and suppose $a^n = 10m + 6$ with $m \in \mathbb{N}$. Then

$$\begin{aligned} a^{n+1} &= a^n \cdot a = (10m + 6)(10q + 6) = 100mq + 60m + 60q + 36 \\ &= 10(10mq + 6m + 6q + 3) + 6, \end{aligned}$$

with $10mq + 6m + 6q + 3 \in \mathbb{N}$. So a^{n+1} ends in 6.

Conclusion. By induction, a^n ends in 6 for all $n \geq 1$. $\square$

Compare with your work.

- ☐ Translate “ends in 6” into an equation before inducting; the computation $36 = 30 + 6$ is the whole step.

Exercise 4.E.13

essential

Let $a, b \in \mathbb{R}$ and define $a_0 = a$, $a_1 = b$ and $a_n = \frac{a_{n-1} + a_{n+1}}{2}$ for all $n \geq 1$. Prove that $a_n = a + (b-a)n$ for all $n \in \mathbb{N}$.

Plan. The relation defines a_{n+1} from the two previous terms: $a_{n+1} = 2a_n - a_{n-1}$ for $n \geq 1$. Strong induction, two base cases, step fixed at $n \geq 1$.

Step 1 Rearranging the defining relation: $a_{n+1} = 2a_n - a_{n-1}$ for all $n \geq 1$.

Step 2 We proceed by strong induction on $n \geq 0$ with $p(n)$: $a_n = a + (b-a)n$, base cases $n = 0, 1$.

Step 3 (Base cases.) $a_0 = a = a + (b-a) \cdot 0$; $a_1 = b = a + (b-a) \cdot 1$.

Step 4 (*Induction step.*) Fix $n \geq 1$ and assume $a_k = a + (b-a)k$ for all $0 \leq k \leq n$. Then

$$\begin{aligned} a_{n+1} &= 2a_n - a_{n-1} = 2(a + (b-a)n) - (a + (b-a)(n-1)) \\ &= a + (b-a)(2n - n + 1) = a + (b-a)(n+1), \end{aligned}$$

using the hypothesis for $k = n$ and $k = n-1$.

Conclusion. By strong induction, $a_n = a + (b-a)n$ for all $n \in \mathbb{N}$ (an arithmetic progression). $\square$

Exercise 4.E.14

recommended

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. (a) Prove by induction that there is a real number a such that $f(n) = an$ for all $n \in \mathbb{N}$. (b) Deduce $f(n) = an$ for all $n \in \mathbb{Z}$. (c) Deduce $f(x) = ax$ for all $x \in \mathbb{Q}$.

Step 1 (a) Define $a = f(1)$. First $f(0) = f(0+0) = f(0) + f(0)$, so $f(0) = 0$ [subtract $f(0)$].

Step 2 Induction on $n \in \mathbb{N}$ with $p(n)$: $f(n) = an$. *Base*: $f(0) = 0 = a \cdot 0$. *Step*: if $f(n) = an$ then $f(n+1) = f(n) + f(1) = an + a = a(n+1)$ [functional equation; hypothesis]. So $f(n) = an$ for all $n \in \mathbb{N}$.

Step 3 (b) Let $n \in \mathbb{N}$. $0 = f(0) = f(n + (-n)) = f(n) + f(-n) = an + f(-n)$, so $f(-n) = -an = a \cdot (-n)$. Every integer is n or $-n$ for some $n \in \mathbb{N}$, so $f(m) = am$ for all $m \in \mathbb{Z}$.

Step 4 (c) First, $f(bx) = bf(x)$ for all $b \in \mathbb{N}$ and $x \in \mathbb{R}$, by induction on b : $f(0 \cdot x) = f(0) = 0$; and $f((b+1)x) = f(bx+x) = f(bx) + f(x) = bf(x) + f(x) = (b+1)f(x)$.

Step 5 Let $x \in \mathbb{Q}$; write $x = \frac{m}{b}$ with $m \in \mathbb{Z}$ and $b \in \mathbb{N}$, $b \geq 1$ [Definition 0.15, choosing the denominator positive]. Then $bf(x) = f(bx) = f(m) = am$ [Step 4; (b)], so $f(x) = \frac{am}{b} = a \cdot \frac{m}{b} = ax$ [$b \neq 0$].

Conclusion. $f(x) = ax$ for all $x \in \mathbb{Q}$, with $a = f(1)$. $\square$

Compare with your work.

- $\square$ Each extension ($\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q}$) uses the functional equation once more; (c) needs its own small induction on the denominator.

Exercise 4.E.15

recommended

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(0) > 0$ and $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Prove that there is a positive real number a such that $f(x) = a^x$ for all $x \in \mathbb{Q}$.

Plan. Mirror 4.E.14 with products instead of sums.

Step 1 $f(0) = f(0+0) = f(0)^2$ and $f(0) > 0$, so $f(0) = 1$ [divide by $f(0)$].

Step 2 For every x : $f(x)f(-x) = f(0) = 1$, so $f(x) \neq 0$; and $f(x) = f(\frac{x}{2} + \frac{x}{2}) = f(\frac{x}{2})^2 > 0$. So f is positive everywhere. Define $a = f(1) > 0$.

Step 3 *Naturals*: induction on n : $f(0) = 1 = a^0$; if $f(n) = a^n$ then $f(n+1) = f(n)f(1) = a^{n+1}$.

Step 4 *Integers*: $f(-n) = \frac{1}{f(n)} = a^{-n}$ [Step 2].

Step 5 *Rationals*: by induction on $b \in \mathbb{N}$, $f(bx) = f(x)^b$ ($f(0) = 1$; $f((b+1)x) = f(bx)f(x)$). For $x = \frac{m}{b}$ with $m \in \mathbb{Z}$, $b \geq 1$: $f(x)^b = f(bx) = f(m) = a^m$. Since $f(x) > 0$, $f(x)$ is the unique positive b -th root of a^m , which is $a^{m/b} = a^x$ [uniqueness of positive roots, Example 1.2.36 generalised].

Conclusion. $f(x) = a^x$ for all $x \in \mathbb{Q}$ with $a = f(1) > 0$. $\square$

Compare with your work.

- Positivity of f (Step 2) is what allows taking the b -th root uniquely; it must be proved from $f(0) > 0$.

Exercise 4.E.16**essential**

Let $a, b \in \mathbb{Z}$. Prove that $b - a$ divides $b^n - a^n$ for all $n \in \mathbb{N}$.

Given. Definition 0.36: $p(n)$ is $\exists k \in \mathbb{Z}, b^n - a^n = (b - a)k$.

Step 1 We proceed by induction on $n \geq 0$.

Step 2 (Base case.) $b^0 - a^0 = 1 - 1 = 0 = (b - a) \cdot 0$, so $b - a \mid b^0 - a^0$ [$k = 0$].

Step 3 (Induction step.) Let $n \geq 0$ and suppose $b^n - a^n = (b - a)k$ with $k \in \mathbb{Z}$. Then

$$\begin{aligned} b^{n+1} - a^{n+1} &= b \cdot b^n - a \cdot a^n = b \cdot b^n - b \cdot a^n + b \cdot a^n - a \cdot a^n \\ &= b(b^n - a^n) + (b - a)a^n = (b - a)(bk + a^n), \end{aligned}$$

[add and subtract ba^n ; then the hypothesis] and $bk + a^n \in \mathbb{Z}$. So $b - a \mid b^{n+1} - a^{n+1}$.

Conclusion. By induction, $b - a \mid b^n - a^n$ for all $n \in \mathbb{N}$. □

Compare with your work.

- The trick “add and subtract ba^n ” must appear; the new quotient $bk + a^n$ must be exhibited and declared an integer.

Exercise 4.E.17**essential**

Prove by induction that $7^n - 2 \cdot 4^n + 1$ is divisible by 18 for all $n \in \mathbb{N}$.

Given. The book’s hint: $7^{n+1} - 2 \cdot 4^{n+1} + 1 = 4(7^n - 2 \cdot 4^n + 1) + 3(7^n - 1)$. (Check: the right side is $4 \cdot 7^n - 8 \cdot 4^n + 4 + 3 \cdot 7^n - 3 = 7 \cdot 7^n - 2 \cdot 4^{n+1} + 1$.) So the step also needs $18 \mid 3(7^n - 1)$, i.e. $6 \mid 7^n - 1$: a lemma.

Step 1 Lemma: $6 \mid 7^n - 1$ for all $n \in \mathbb{N}$. Induction: $7^0 - 1 = 0 = 6 \cdot 0$. If $7^n - 1 = 6m$ then $7^{n+1} - 1 = 7 \cdot 7^n - 1 = 7(7^n - 1) + 6 = 6(7m + 1)$ [$7 \cdot 7^n - 1 = 7(7^n - 1) + 7 - 1$].

Step 2 Main induction on $n \geq 0$ with $p(n)$: $18 \mid 7^n - 2 \cdot 4^n + 1$.

Step 3 (Base case.) $7^0 - 2 \cdot 4^0 + 1 = 1 - 2 + 1 = 0 = 18 \cdot 0$.

Step 4 (Induction step.) Let $n \geq 0$ and suppose $7^n - 2 \cdot 4^n + 1 = 18k$ with $k \in \mathbb{Z}$. By the Lemma, $7^n - 1 = 6m$ with $m \in \mathbb{Z}$. Then

$$7^{n+1} - 2 \cdot 4^{n+1} + 1 = 4(7^n - 2 \cdot 4^n + 1) + 3(7^n - 1) = 4 \cdot 18k + 3 \cdot 6m = 18(4k + m),$$

and $4k + m \in \mathbb{Z}$. So $18 \mid 7^{n+1} - 2 \cdot 4^{n+1} + 1$.

Conclusion. By induction, 18 divides $7^n - 2 \cdot 4^n + 1$ for all $n \in \mathbb{N}$. □

Compare with your work.

- Verify the hint’s identity (it is the whole step), and prove the lemma $6 \mid 7^n - 1$ separately: without it the term $3(7^n - 1)$ is not visibly a multiple of 18.

Exercise 4.E.18**essential**

Prove that $\frac{d^n}{dx^n}(xe^x) = (n+x)e^x$ for all $n \in \mathbb{N}$.

Given. The 0-th derivative of a function is the function itself; the product rule $(uv)' = u'v + uv'$; $(e^x)' = e^x$.

Step 1 We proceed by induction on $n \geq 0$ with $p(n): \frac{d^n}{dx^n}(xe^x) = (n+x)e^x$.

Step 2 (Base case.) $\frac{d^0}{dx^0}(xe^x) = xe^x = (0+x)e^x$.

Step 3 (Induction step.) Let $n \geq 0$ and suppose $\frac{d^n}{dx^n}(xe^x) = (n+x)e^x$. Differentiating once more:

$$\frac{d^{n+1}}{dx^{n+1}}(xe^x) = \frac{d}{dx}((n+x)e^x) = 1 \cdot e^x + (n+x)e^x = (n+1+x)e^x \quad [\text{product rule; hypothesis}].$$

Conclusion. By induction the formula holds for all $n \in \mathbb{N}$. □

Compare with your work.

□ “Differentiating the hypothesis” is the step; write the product rule out.

Exercise 4.E.19**essential**

Find and prove an expression for $\frac{d^n}{dx^n}(x^2e^x)$ for all $n \in \mathbb{N}$.

Plan. Compute the first few derivatives, guess a formula, prove it by induction.

Step 1 $n = 0: x^2e^x$. $n = 1: (x^2 + 2x)e^x$. $n = 2: (2x + 2)e^x + (x^2 + 2x)e^x = (x^2 + 4x + 2)e^x$. $n = 3: (2x + 4)e^x + (x^2 + 4x + 2)e^x = (x^2 + 6x + 6)e^x$.

Step 2 Pattern: $(x^2 + 2nx + c_n)e^x$ with $c_0, c_1, c_2, c_3 = 0, 0, 2, 6$, i.e. $c_n = n(n-1)$. **Claim:** $\frac{d^n}{dx^n}(x^2e^x) = (x^2 + 2nx + n(n-1))e^x$.

Step 3 (Base case.) $n = 0: (x^2 + 0 + 0)e^x = x^2e^x$.

Step 4 (Induction step.) Suppose the claim for n . Then

$$\begin{aligned} \frac{d^{n+1}}{dx^{n+1}}(x^2e^x) &= \frac{d}{dx}((x^2 + 2nx + n(n-1))e^x) \quad [\text{hypothesis}] \\ &= (2x + 2n)e^x + (x^2 + 2nx + n(n-1))e^x \quad [\text{product rule}] \\ &= (x^2 + (2n+2)x + n(n-1) + 2n)e^x = (x^2 + 2(n+1)x + (n+1)n)e^x, \end{aligned}$$

since $n(n-1) + 2n = n^2 + n = (n+1)n$. This is the claim for $n+1$.

Conclusion. $\frac{d^n}{dx^n}(x^2e^x) = (x^2 + 2nx + n(n-1))e^x$ for all $n \in \mathbb{N}$. □

Compare with your work.

□ Show the computations that led to the guess; the proof is the induction, the guess is not.

Exercise 4.E.20**essential**

Let f, g be functions that can be differentiated as often as needed. Prove that for all $n \in \mathbb{N}$, $\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x)g^{(n-k)}(x)$.

Given. Product rule; Pascal's rule $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ (Definition 4.1.15); $\binom{n}{n} = 1 = \binom{n+1}{n+1}$, $\binom{n}{0} = 1 = \binom{n+1}{0}$. The proof has the same shape as that of the binomial theorem (Theorem 4.2.18).

Step 1 Induction on $n \geq 0$. (Base case.) $n = 0$: the right side is $\binom{0}{0} f^{(0)} g^{(0)} = fg$, the 0-th derivative of fg .

Step 2 (Induction step.) Suppose the formula for n . Differentiate it term by term with the product rule:

$$\begin{aligned} \frac{d^{n+1}}{dx^{n+1}}(fg) &= \sum_{k=0}^n \binom{n}{k} \left(f^{(k+1)} g^{(n-k)} + f^{(k)} g^{(n+1-k)} \right) \\ &= \sum_{k=0}^n \binom{n}{k} f^{(k+1)} g^{(n-k)} + \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n+1-k)}. \end{aligned}$$

Step 3 Reindex the first sum with $j = k + 1$ (so $k = j - 1$, j from 1 to $n + 1$): $\sum_{j=1}^{n+1} \binom{n}{j-1} f^{(j)} g^{(n+1-j)}$. Rename j back to k .

Step 4 Split off the extreme terms: from the first sum the term $k = n + 1$, namely $\binom{n}{n} f^{(n+1)} g^{(0)}$; from the second the term $k = 0$, namely $\binom{n}{0} f^{(0)} g^{(n+1)}$. The remaining terms, for $1 \leq k \leq n$, combine:

$$\sum_{k=1}^n \left(\binom{n}{k-1} + \binom{n}{k} \right) f^{(k)} g^{(n+1-k)} = \sum_{k=1}^n \binom{n+1}{k} f^{(k)} g^{(n+1-k)} \quad [\text{Pascal's rule}].$$

Step 5 Since $\binom{n}{n} = 1 = \binom{n+1}{n+1}$ and $\binom{n}{0} = 1 = \binom{n+1}{0}$, the two extreme terms are the $k = n + 1$ and $k = 0$ terms of $\sum_{k=0}^{n+1} \binom{n+1}{k} f^{(k)} g^{(n+1-k)}$. Altogether $\frac{d^{n+1}}{dx^{n+1}}(fg) = \sum_{k=0}^{n+1} \binom{n+1}{k} f^{(k)} g^{(n+1-k)}$.

Conclusion. By induction, the Leibniz rule holds for all $n \in \mathbb{N}$. $\square$

Compare with your work.

- $\square$ The three moves are: product rule inside the sum, reindex, Pascal. Each must be visible; the reindexing is where errors hide.

Exercise 4.E.21

recommended

Find and prove an expression for $\frac{d^n}{dx^n} \log x$ (for $x > 0$) for all $n \in \mathbb{N}$.

Step 1 $n = 0$: $\log x$. $n = 1$: x^{-1} . $n = 2$: $-x^{-2}$. $n = 3$: $2x^{-3}$. $n = 4$: $-6x^{-4}$. Pattern for $n \geq 1$: sign $(-1)^{n-1}$, coefficient $(n-1)!$, power x^{-n} .

Step 2 Claim: for all $n \geq 1$, $\frac{d^n}{dx^n} \log x = (-1)^{n-1} (n-1)! x^{-n}$. Induction on $n \geq 1$.

Step 3 (Base case.) $n = 1$: $(-1)^0 \cdot 0! \cdot x^{-1} = \frac{1}{x} = \frac{d}{dx} \log x$.

Step 4 (Induction step.) Let $n \geq 1$ and suppose the claim for n . Then

$$\frac{d^{n+1}}{dx^{n+1}} \log x = \frac{d}{dx} \left((-1)^{n-1} (n-1)! x^{-n} \right) = (-1)^{n-1} (n-1)! \cdot (-n) x^{-n-1} = (-1)^n n! x^{-(n+1)},$$

which is the claim for $n + 1$ [$(n-1)! \cdot n = n!$; $(-1)^{n-1} \cdot (-1) = (-1)^n$].

Conclusion. For $n \geq 1$, $\frac{d^n}{dx^n} \log x = (-1)^{n-1} (n-1)! x^{-n}$; for $n = 0$ it is $\log x$. $\square$

Exercise 4.E.24

recommended

True or false: every factorial is positive.

Step 1 True. Induction on n with $p(n)$: $n! > 0$. Base: $0! = 1 > 0$ [Definition 4.1.14]. Step: if $n! > 0$

then $(n+1)! = (n+1) \cdot n! > 0$ as a product of the positive numbers $n+1 \geq 1$ and $n!$.

Conclusion. $n! > 0$ for all $n \in \mathbb{N}$. □

Exercise 4.E.25

essential

True or false: every binomial coefficient is positive.

Step 1 *False.* $\binom{0}{1} = 0$ by the clause $\binom{0}{k+1} = 0$ of Definition 4.1.15 (with $k = 0$). In general $\binom{n}{k} = 0$ whenever $k > n$ [Theorem 4.2.15].

Conclusion. False (binomial coefficients are non-negative, not positive). □

Exercise 4.E.26

essential

True or false: every proof by strong induction can be turned into a proof by weak induction by changing only the induction hypothesis.

Step 1 *False.* A strong-induction step may use $p(k)$ for $k < n$: Example 4.3.7 uses a_{n-1} , Example 4.3.4 uses all of $b_0, \dots, b_n$. If the hypothesis is replaced by just $p(n)$, those uses are no longer justified, so the step no longer proves $p(n+1)$: the rest of the argument must change too (for instance by proving the stronger statement $q(n)$: “ $p(k)$ for all $k \leq n$ ”, as in the proof of Theorem 4.3.2).

Conclusion. False. (Every statement provable by strong induction is provable by weak induction, but not by the same proof with only the hypothesis changed.) □

Exercise 4.E.27

essential

True or false: every proof by weak induction can be turned into a proof by strong induction by changing only the induction hypothesis.

Step 1 *True.* Take a proof by weak induction of $\forall n \geq n_0, p(n)$. Replace the hypothesis “ $p(n)$ ” by “ $p(k)$ for all $n_0 \leq k \leq n$ ”.

Step 2 The new hypothesis contains the old one (take $k = n$), so every line of the original step remains valid and still derives $p(n+1)$; the base case is unchanged. This is a valid proof by strong induction [Strategy 4.3.3].

Conclusion. True. □

Exercise 4.E.28

recommended

True or false: let $f: \mathbb{N} \rightarrow X$ be a surjection and $p(x)$ a logical formula with free variable $x \in X$. If $p(f(0))$ is true and $\forall n \in \mathbb{N}, (p(f(n)) \Rightarrow p(f(n+1)))$, then $\forall x \in X, p(x)$.

Step 1 *True.* Define $q(n): p(f(n))$, a formula with free variable $n \in \mathbb{N}$. The hypotheses say $q(0)$ and $\forall n, (q(n) \Rightarrow q(n+1))$.

Step 2 By weak induction, $q(n)$ holds for all $n \in \mathbb{N}$ [Theorem 4.2.1].

Step 3 Let $x \in X$. Since f is surjective, $x = f(n)$ for some $n \in \mathbb{N}$ [Definition 3.2.9]. Then $p(x)$ is $p(f(n)) = q(n)$, which is true.

Conclusion. True: induction transports along a surjection. □

Exercise 4.E.29

essential

True or false: if for all $x \in \mathbb{N}$ there is some $y \leq x$ such that $p(y)$ is false, then $p(x)$ is false for all $x \in \mathbb{N}$.

Step 1 *False.* Counterexample: $p(x)$: “ $x \geq 1$ ”.

Step 2 *Hypothesis holds:* for every $x \in \mathbb{N}$, $y = 0$ satisfies $y \leq x$ and $p(0)$ (“ $0 \geq 1$ ”) is false.

Step 3 *Conclusion fails:* $p(1)$ is true.

Conclusion. False. (The hypothesis only guarantees that $p(0)$ is false; it is not an induction principle.) □

Exercise 4.E.30

essential

True or false: every non-empty subset of $\mathbb{Q}_{\geq 0}$ has a least element.

Step 1 *False.* Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}, n \geq 1\} \subseteq \mathbb{Q}_{\geq 0}$; it is non-empty ($1 \in X$).

Step 2 X has infinite descent: for $\frac{1}{n} \in X$, the element $\frac{1}{n+1} \in X$ satisfies $\frac{1}{n+1} < \frac{1}{n}$ [Definition 4.3.16]. A set with infinite descent has no least element (a least element m would have some $y \in X$ with $y < m$).

Conclusion. False: well-ordering is special to $\mathbb{N}$. □

Exercise 4.E.31

essential

True or false: every non-empty subset of $\mathbb{Z}$ has a least element.

Step 1 *False.* $\mathbb{Z}$ itself is a non-empty subset of $\mathbb{Z}$, and it has infinite descent: for $n \in \mathbb{Z}$, $n - 1 \in \mathbb{Z}$ and $n - 1 < n$ [Example 4.3.17]. So $\mathbb{Z}$ has no least element.

Conclusion. False. □

Exercise 4.E.32

essential

True or false: every non-empty subset of $\mathbb{Z}^-$ (the set of negative integers) has a greatest element.

Plan. Flip the set into $\mathbb{N}$ and use the well-ordering principle.

Step 1 *True.* Let $X \subseteq \mathbb{Z}^-$ be non-empty. Define $Y = \{-x - 1 \mid x \in X\}$. For $x \leq -1$ we have $-x - 1 \geq 0$, so $Y \subseteq \mathbb{N}$; and $Y \neq \emptyset$ since $X \neq \emptyset$.

Step 2 By the well-ordering principle [Theorem 4.3.11], Y has a least element, say $-x_0 - 1$ with $x_0 \in X$.

Step 3 Let $x \in X$. Then $-x - 1 \in Y$, so $-x - 1 \geq -x_0 - 1$, hence $x \leq x_0$ [multiply by -1]. So x_0 is the greatest element of X .

Conclusion. True. □

Compare with your work.

- The map $x \mapsto -x - 1$ (or $x \mapsto -x$, landing in $\mathbb{N} \setminus \{0\}$) must be shown to land in $\mathbb{N}$, and “least” must be translated back to “greatest”.

Exercise 4.E.33**recommended**

Let $n, k, \ell \in \mathbb{N}$. Then $\binom{n}{k+\ell} = \binom{n}{k} + \binom{n}{\ell}$. (Always, sometimes, never? The book writes $n, k, \ell \in \mathbb{Z}$; binomial coefficients are defined for natural numbers, so we take $\mathbb{N}$.)

Step 1 Holds: $n = 0, k = \ell = 1$: $\binom{0}{2} = 0$ and $\binom{0}{1} + \binom{0}{1} = 0 + 0 = 0$ [Theorem 4.2.15].

Step 2 Fails: $n = 1, k = 0, \ell = 1$: $\binom{1}{1} = 1$ but $\binom{1}{0} + \binom{1}{1} = 1 + 1 = 2$.

Conclusion. Sometimes. □

Exercise 4.E.34**recommended**

Let $m, n \geq 2$. Then $(m+n)! = m! + n!$. (Always, sometimes, never?)

Plan. Show $(m+n)! > m! + n!$ always: $(m+n)!$ contains both $m!$ and $n!$ as factors of a much larger product.

Step 1 Never. First, if $j \leq \ell$ then $j! \leq \ell!$ [$\ell! = j! \cdot (j+1) \cdots \ell$ and each extra factor is ≥ 1 ; induction on $\ell - j$].

Step 2 Since $m+n-1 \geq m$ and $m+n-1 \geq n$ [$m, n \geq 2$, so $n-1 \geq 1$ and $m-1 \geq 1$], Step 1 gives $(m+n-1)! \geq m!$ and $(m+n-1)! \geq n!$, so $(m+n-1)! \geq \max(m!, n!)$.

Step 3 Then $(m+n)! = (m+n)(m+n-1)! \geq 4 \max(m!, n!) > 2 \max(m!, n!) \geq m! + n!$ [$m+n \geq 4$; $\max(m!, n!) > 0$ by 4.E.24].

Conclusion. $(m+n)! > m! + n!$ for all $m, n \geq 2$: never. □

Compare with your work.

□ “Never” needs an argument valid for *all* $m, n \geq 2$, not a few numerical checks.

Exercise 4.E.35**recommended**

Let $p(x)$ be a logical formula with free variable $x \in \mathbb{Z}$. Suppose $p(0)$ is true and, for all $n \in \mathbb{Z}$, $p(n) \Rightarrow p(n+1)$ and $p(n) \Rightarrow p(n-2)$. Then $\forall n \in \mathbb{Z}, p(n)$. (Always, sometimes, never?)

Plan. Non-negative integers by weak induction; negative ones by strong induction on m for $q(m)$: $p(-m)$, using $p(n) \Rightarrow p(n-2)$ to go from $-(m-1)$ to $-(m+1)$.

Step 1 Always. Non-negative: $p(0)$ holds and $p(n) \Rightarrow p(n+1)$ for all n , so $p(n)$ for all $n \in \mathbb{N}$ by weak induction [Theorem 4.2.1].

Step 2 Negative: let $q(m)$ be $p(-m)$ for $m \in \mathbb{N}$. Strong induction with base cases $m = 0, 1$: $q(0) = p(0)$ holds; $q(1) = p(-1)$ follows from $p(1)$ (Step 1) via $p(1) \Rightarrow p(1-2)$.

Step 3 Step: fix $m \geq 1$ and assume $q(k)$ for all $0 \leq k \leq m$. Then $q(m-1) = p(-(m-1))$ holds, and $p(-(m-1)) \Rightarrow p(-(m-1)-2) = p(-(m+1)) = q(m+1)$. So $q(m+1)$.

Step 4 Hence $p(-m)$ for all $m \in \mathbb{N}$. Every integer is $n \in \mathbb{N}$ or $-m$ with $m \in \mathbb{N}$, so p holds on all of $\mathbb{Z}$.

Conclusion. Always. □

Chapter 5

Relations

Contents

- A Relations (§5.1)
- B Equivalence relations (§5.2)
- C Chapter 5 exercises (5.E)

Planning

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 ★ 5.1: 10, 11, 18, 24, 29, 33, 34 · 5.2: 25, 28, 30, 34, 36
 Tested in **Quiz 3** (21 Oct) and the exam.

A RELATIONS

What you unfold in this section

Relation (Definition 5.1.1): a formula $R(x, y)$ with $x \in X, y \in Y$; $x R y$ means $R(x, y)$ is true. **Extensionality** (Axiom 5.1.4): $R = S$ iff same domain, same codomain, and $\forall x \in X, \forall y \in Y, (x R y \Leftrightarrow x S y)$. **Graph** (Definition 5.1.7): $\text{Gr}(R) = \{(x, y) \in X \times Y \mid x R y\}$. **Properties** of a relation on X (Definitions 5.1.19–5.1.36): reflexive $\forall a, a R a$; symmetric $\forall a, b, a R b \Rightarrow b R a$; antisymmetric $\forall a, b, (a R b \wedge b R a) \Rightarrow a = b$; transitive $\forall a, b, c, (a R b \wedge b R c) \Rightarrow a R c$. Each is proved by “let a (and b, c) $\in X$ with ...” and unfolding what $a R b$ means; each is disproved by one explicit counterexample.

Exercise 5.1.6

essential

Let R and S be the relations on $\mathbb{R}$ defined by $a R b \Leftrightarrow b - a \in \mathbb{Q}$ and $a S b \Leftrightarrow \exists n \in \mathbb{Z}, (n \neq 0 \wedge n(b - a) \in \mathbb{Z})$. Prove that $R = S$.

Given. Both relations have domain and codomain $\mathbb{R}$. Definition 0.15 for $\mathbb{Q}$.

To show. By Axiom 5.1.4 it suffices to prove $\forall a, b \in \mathbb{R}, (a R b \Leftrightarrow a S b)$.

Step 1 Let $a, b \in \mathbb{R}$ [Strategy 1.2.10]. We prove both directions.

Step 2 ($\Rightarrow$) Suppose $b - a \in \mathbb{Q}$. Then $b - a = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$ [Definition 0.15, Strategy 1.2.30]. Define $n = q$: then $n \in \mathbb{Z}, n \neq 0$ and $n(b - a) = q \cdot \frac{p}{q} = p \in \mathbb{Z}$. So $a S b$ [$\exists$].

Step 3 ($\Leftarrow$) Suppose $a S b$: take $n \in \mathbb{Z}$ with $n \neq 0$ and $n(b - a) \in \mathbb{Z}$, say $n(b - a) = m \in \mathbb{Z}$. Then $b - a = \frac{m}{n}$ with $m, n \in \mathbb{Z}, n \neq 0$, so $b - a \in \mathbb{Q}$ [Definition 0.15]: $a R b$.

Conclusion. $a R b \Leftrightarrow a S b$ for all $a, b \in \mathbb{R}$, and the domains and codomains agree, so $R = S$ [Axiom 5.1.4]. □

Compare with your work.

- Mention domain and codomain explicitly: relation equality needs all three parts.

Exercise 5.1.10**recommended**

Let R be the relation on $\mathbb{Z}$ defined by $x R y \Leftrightarrow x^2 = y^2$. Describe $\text{Gr}(R)$.

Step 1 $\text{Gr}(R) = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x^2 = y^2\}$ [Definition 5.1.7].

Step 2 $x^2 = y^2$ iff $y^2 - x^2 = 0$ iff $(y - x)(y + x) = 0$ iff $y = x$ or $y = -x$ [a product of integers is 0 iff a factor is 0].

Step 3 So $\text{Gr}(R) = \{(x, x) \mid x \in \mathbb{Z}\} \cup \{(x, -x) \mid x \in \mathbb{Z}\}$: the two diagonals of the integer grid, i.e. $\Delta_{\mathbb{Z}} \cup \{(x, -x) \mid x \in \mathbb{Z}\}$.

Answer. The pairs $(x, \pm x)$ with $x \in \mathbb{Z}$. □

Exercise 5.1.11**recommended**

Let $f : X \rightarrow Y$ and let R_f be the relation from X to Y defined by $x R_f y \Leftrightarrow f(x) = y$. Prove that $\text{Gr}(R_f) = \text{Gr}(f)$.

Step 1 $\text{Gr}(R_f) = \{(x, y) \in X \times Y \mid x R_f y\} = \{(x, y) \in X \times Y \mid f(x) = y\}$ [Definition 5.1.7; definition of R_f].

Step 2 $\text{Gr}(f) = \{(x, y) \in X \times Y \mid y = f(x)\}$ [Definition 3.1.6].

Step 3 The two set-builder expressions have the same defining condition, so the sets are equal [Axiom 2.1.14].

Conclusion. $\text{Gr}(R_f) = \text{Gr}(f)$: a function “is” the relation $f(x) = y$. □

Exercise 5.1.13**essential**

Let X and Y be sets. Describe the graphs of the discrete relation $D_{X,Y}$ and the empty relation $\emptyset_{X,Y}$.

Given. $D_{X,Y}$ relates every $x \in X$ to every $y \in Y$; $\emptyset_{X,Y}$ relates nothing (Definition 5.1.11).

Step 1 $\text{Gr}(D_{X,Y}) = \{(x, y) \in X \times Y \mid x D_{X,Y} y\} = \{(x, y) \in X \times Y \mid \text{true}\} = X \times Y$.

Step 2 $\text{Gr}(\emptyset_{X,Y}) = \{(x, y) \in X \times Y \mid \text{false}\} = \emptyset$.

Answer. $\text{Gr}(D_{X,Y}) = X \times Y$ and $\text{Gr}(\emptyset_{X,Y}) = \emptyset$. □

Exercise 5.1.18**recommended**

Let X be a set. Prove that $\Delta_X = \text{Gr}(=)$, where $\Delta_X = \{(x, x) \mid x \in X\}$ and $=$ is the equality relation on X .

Step 1 $\text{Gr}(=) = \{(x, y) \in X \times X \mid x = y\}$ [Definition 5.1.7].

Step 2 ($\subseteq$) Let $t \in \Delta_X$; then $t = (x, x)$ for some $x \in X$. Since $x = x$, $(x, x) \in \text{Gr}(=)$.

Step 3 ($\supseteq$) Let $(x, y) \in \text{Gr}(=)$; then $x = y$, so $(x, y) = (x, x) \in \Delta_X$.

Conclusion. $\Delta_X = \text{Gr}(=)$ [double containment]. □

Exercise 5.1.22**essential**

Let X be a set. Prove that $\subseteq$ is a reflexive relation on $\mathcal{P}(X)$, but $\subsetneq$ is not.

Given. Reflexive: $\forall U \in \mathcal{P}(X), U \subseteq U$. $U \subsetneq V$ means $U \subseteq V$ and $U \neq V$ (Definition 2.1.6). Not reflexive:

$\exists U \in \mathcal{P}(X), \neg(U \subsetneq U).$

Step 1 $\subseteq$ reflexive. Let $U \in \mathcal{P}(X)$. Let $a \in U$; then $a \in U$. So $U \subseteq U$ [Definition 2.1.6, Strategy 2.1.7].

Step 2 $\subsetneq$ not reflexive. $\mathcal{P}(X)$ is non-empty: $\emptyset \in \mathcal{P}(X)$ [Exercise 2.1.32]. Take $U = \emptyset$. $U \subsetneq U$ would require $U \neq U$, which is false. So $\neg(\emptyset \subsetneq \emptyset)$, and reflexivity fails at $U = \emptyset$ [Strategy 1.3.30]. (In fact it fails at every U .)

Conclusion. $\subseteq$ is reflexive on $\mathcal{P}(X)$; $\subsetneq$ is not. □

Compare with your work.

- To show “not reflexive” you need an *element* of the domain; note that $\mathcal{P}(X)$ is never empty, so one exists.

Exercise 5.1.23

essential

Prove that the relation “ x divides y ” on $\mathbb{Z}$ is reflexive.

Step 1 Let $x \in \mathbb{Z}$. Then $x = 1 \cdot x$ with $1 \in \mathbb{Z}$, so $x \mid x$ [Definition 0.36 with $q = 1$].

Conclusion. Divisibility is reflexive on $\mathbb{Z}$ (including $0 \mid 0$). □

Exercise 5.1.24

recommended

Let $G = \{(0,0), (1,1), (2,2)\}$. Let R be the relation on $[3]$ with graph G and S the relation on $[4]$ with graph G . Prove that R is reflexive but S is not.

Step 1 $[3] = \{0, 1, 2\}$. For each $a \in [3]$, $(a, a) \in G$, so $a R a$ [Theorem 5.1.14: $a R b \Leftrightarrow (a, b) \in \text{Gr}(R)$]. Hence R is reflexive.

Step 2 $[4] = \{0, 1, 2, 3\}$ and $(3, 3) \notin G$, so $3 S 3$ is false. Hence S is not reflexive [counterexample $a = 3$].

Conclusion. Same graph, different domain, different answer: reflexivity depends on the domain. □

Exercise 5.1.28

essential

Find all subsets $U \subseteq \mathbb{Z}$ such that the relation “ x divides y ” on U is symmetric.

To show. A characterisation of the sets U with $\forall x, y \in U, (x \mid y \Rightarrow y \mid x)$, with proof.

Plan. First a lemma: $x \mid y$ and $y \mid x$ together mean $y = \pm x$. Then symmetry on U means: whenever $x, y \in U$ and $x \mid y$, also $y \mid x$, i.e. $y = \pm x$.

Step 1 *Lemma:* for $x, y \in \mathbb{Z}$, $(x \mid y \text{ and } y \mid x)$ iff $y = x$ or $y = -x$. ($\Leftarrow$) If $y = \pm x$ then $y = (\pm 1)x$ and $x = (\pm 1)y$, so both divisibilities hold. ($\Rightarrow$) Write $y = qx$ and $x = ry$ with $q, r \in \mathbb{Z}$ [Definition 0.36]. Then $x = r qx$. If $x \neq 0$: $rq = 1$, so $q = r = \pm 1$ [the only integer factorisations of 1] and $y = \pm x$. If $x = 0$: $y = q \cdot 0 = 0 = x$.

Step 2 *Characterisation.* Divisibility on U is symmetric iff $\forall x, y \in U, (x \mid y \Rightarrow y \mid x)$ [Definition 5.1.25] iff $\forall x, y \in U, (x \mid y \Rightarrow (x \mid y \wedge y \mid x))$ iff, by the Lemma, $\forall x, y \in U, (x \mid y \Rightarrow y = \pm x)$.

Step 3 In words: U works iff no element of U divides an element of U with a different absolute value. Equivalently: for distinct $|x| \neq |y|$ in U , neither divides the other.

Step 4 *Examples:* $\emptyset$; $\{0\}$; $\{a, -a\}$ for any $a \in \mathbb{Z}$; $\{2, 3\}$; $\{4, 6, 9\}$; the set of all primes. *Non-examples:* $\{1, 2\}$ ($1 \mid 2$ but $2 \nmid 1$); $\{0, 5\}$ ($5 \mid 0$ but $0 \nmid 5$); $\mathbb{N}$; $\mathbb{Z}$. In particular, if $0 \in U$ then

every $x \in U$ divides 0 while $0 \mid x$ only for $x = 0$, so $U = \{0\}$ is the only such U containing 0.

Conclusion. Divisibility is symmetric on U exactly when $x \mid y$ with $x, y \in U$ forces $y = \pm x$. $\square$

Compare with your work.

- $\square$ “Find all” asks for a condition, with a proof that it is necessary and sufficient; examples alone are not an answer, but a few of them show you understand the condition.

Exercise 5.1.29

recommended

Let R and S be relations with $\text{Gr}(R) = \text{Gr}(S)$ (possibly with different domains). Prove that R is symmetric if and only if S is symmetric.

Plan. Show that symmetry is a property of the graph alone: R symmetric iff $\forall (a, b) \in \text{Gr}(R), (b, a) \in \text{Gr}(R)$.

Step 1 *Claim:* for a relation R on X : R symmetric $\Leftrightarrow \forall (a, b) \in \text{Gr}(R), (b, a) \in \text{Gr}(R)$.

Step 2 ($\Rightarrow$) Let $(a, b) \in \text{Gr}(R)$. Then $a, b \in X$ and $a R b$ [Definition 5.1.7]; by symmetry $b R a$, so $(b, a) \in \text{Gr}(R)$.

Step 3 ($\Leftarrow$) Let $a, b \in X$ with $a R b$. Then $(a, b) \in \text{Gr}(R)$, so $(b, a) \in \text{Gr}(R)$ by the graph condition, i.e. $b R a$. So R is symmetric.

Step 4 The graph condition mentions only $\text{Gr}(R)$. Since $\text{Gr}(R) = \text{Gr}(S)$, the condition holds for R iff it holds for S ; by the Claim (applied to R and to S), R is symmetric iff S is.

Conclusion. Symmetry depends only on the graph. $\square$

Exercise 5.1.33

recommended

Prove that “ x divides y ” is antisymmetric on $\mathbb{N}$ but not on $\mathbb{Z}$.

Step 1 On $\mathbb{N}$. Let $x, y \in \mathbb{N}$ with $x \mid y$ and $y \mid x$. By the Lemma of 5.1.28, $y = x$ or $y = -x$. If $y = -x$ with $x, y \geq 0$, then $x = y = 0$. In both cases $x = y$ [Definition 5.1.30].

Step 2 On $\mathbb{Z}$. Counterexample: $1 \mid -1$ (as $-1 = (-1) \cdot 1$) and $-1 \mid 1$ (as $1 = (-1)(-1)$), but $1 \neq -1$.

Conclusion. Antisymmetric on $\mathbb{N}$, not on $\mathbb{Z}$. $\square$

Exercise 5.1.34

recommended

Let X be a set. Prove that $\subseteq$ is antisymmetric on $\mathcal{P}(X)$.

Step 1 Let $U, V \in \mathcal{P}(X)$ with $U \subseteq V$ and $V \subseteq U$ [Definition 5.1.30].

Step 2 Then $U = V$ [Axiom 2.1.14: double containment].

Conclusion. $\subseteq$ is antisymmetric: antisymmetry of $\subseteq$ is set extensionality. $\square$

Exercise 5.1.39

essential

Let X be a set. Prove that $\subseteq$ is transitive on $\mathcal{P}(X)$.

Step 1 Let $U, V, W \in \mathcal{P}(X)$ with $U \subseteq V$ and $V \subseteq W$ [Definition 5.1.36].

Step 2 Let $a \in U$. Then $a \in V$ [$U \subseteq V$], and then $a \in W$ [$V \subseteq W$]. So $U \subseteq W$ [Strategy 2.1.7].

Conclusion. $\subseteq$ is transitive (this is Proposition 2.1.12). □

Exercise 5.1.40

essential

Prove that the relation “ x divides y ” on $\mathbb{Z}$ is transitive.

Step 1 Let $x, y, z \in \mathbb{Z}$ with $x \mid y$ and $y \mid z$. Then $y = qx$ and $z = ry$ for some $q, r \in \mathbb{Z}$ [Definition 0.36, two letters].

Step 2 $z = ry = r(qx) = (rq)x$ with $rq \in \mathbb{Z}$ [associativity; closure]. So $x \mid z$ [Definition 0.36].

Conclusion. Divisibility is transitive (Proposition 0.39). □

Exercise 5.1.41

essential

Let R and S be relations with $\text{Gr}(R) = \text{Gr}(S)$. Prove that R is transitive if and only if S is transitive.

Plan. As in 5.1.29: express transitivity through the graph only.

Step 1 *Claim:* for R on X : R transitive $\Leftrightarrow$ for all $(a, b), (b, c) \in \text{Gr}(R)$, $(a, c) \in \text{Gr}(R)$.

Step 2 $(\Rightarrow)$ If $(a, b), (b, c) \in \text{Gr}(R)$ then $a, b, c \in X$ with $a R b$ and $b R c$, so $a R c$ by transitivity, i.e. $(a, c) \in \text{Gr}(R)$.

Step 3 $(\Leftarrow)$ If $a, b, c \in X$ with $a R b$, $b R c$, then $(a, b), (b, c) \in \text{Gr}(R)$, so $(a, c) \in \text{Gr}(R)$, i.e. $a R c$.

Step 4 The graph condition involves only $\text{Gr}(R) = \text{Gr}(S)$, so it holds for R iff for S ; by the Claim, R transitive iff S transitive.

Conclusion. Transitivity depends only on the graph (unlike reflexivity, 5.1.24). □

B EQUIVALENCE RELATIONS

What you unfold in this section

Equivalence relation (Definition 5.2.1): reflexive, symmetric, transitive. **Congruence** (Definition 5.2.6): $a \equiv b \pmod{n}$ iff $n \mid b - a$; equivalently (for $n \neq 0$) equal remainders (Exercise 5.2.9). **Class and quotient** (Definition 5.2.12): $[a]_{\sim} = \{x \in X \mid a \sim x\}$, $X/\sim = \{[a]_{\sim} \mid a \in X\}$; **Theorem 5.2.17:** $a \sim b \Leftrightarrow [a]_{\sim} = [b]_{\sim}$. **Partition** (Definition 5.2.21): $\mathcal{A} \subseteq \mathcal{P}(X)$ with (a) every $A \in \mathcal{A}$ non-empty; (b) cover: $\forall x \in X, \exists A \in \mathcal{A}, x \in A$; (c) pairwise disjoint: $\forall A, B \in \mathcal{A}, A \cap B \neq \emptyset \Rightarrow A = B$. **Theorem 5.2.26:** $X/\sim$ is a partition, and every partition is $X/\sim$ for the relation “lie in a common block”. **Quotient function** (Definition 5.2.31): $q_{\sim}(a) = [a]_{\sim}$.

Exercise 5.2.24

essential

Let $f: X \rightarrow Y$ be a surjection and let $\mathcal{F} = \{f^{-1}[\{b\}] \mid b \in Y\}$. Prove that $\mathcal{F}$ is a partition of X . Where is surjectivity used?

Given. $f^{-1}[\{b\}] = \{x \in X \mid f(x) = b\}$ (Definition 3.1.37): the set of inputs sent to b . Each is a subset of X , so $\mathcal{F} \subseteq \mathcal{P}(X)$.

Step 1 (a) *Non-empty.* Let $F \in \mathcal{F}$, say $F = f^{-1}[\{b\}]$ with $b \in Y$. Since f is **surjective**, there is $x \in X$ with $f(x) = b$ [Definition 3.2.9]; then $x \in F$. So $F \neq \emptyset$.

Step 2 (b) *Cover*. Let $x \in X$. Then $f(x) \in Y$, and $x \in f^{-1}[\{f(x)\}]$ [$f(x) = f(x)$], which is an element of $\mathcal{F}$ (take $b = f(x)$).

Step 3 (c) *Pairwise disjoint*. Let $F = f^{-1}[\{b\}]$ and $F' = f^{-1}[\{b'\}]$ be elements of $\mathcal{F}$ with $F \cap F' \neq \emptyset$; pick $x \in F \cap F'$. Then $f(x) = b$ and $f(x) = b'$, so $b = b'$ [f is a function: one value], hence $F = F'$.

Conclusion. $\mathcal{F}$ is a partition of X [Definition 5.2.21]. Surjectivity is used only in (a): without it, $f^{-1}[\{b\}] = \emptyset$ for $b \notin f[X]$, and a partition may not contain $\emptyset$. $\square$

Compare with your work.

- $\square$ Three labelled parts, (c) in the book's form "if $A \cap B \neq \emptyset$ then $A = B$ ". The answer to "where" is a specific part.

Exercise 5.2.25

recommended

Let $\mathcal{A}$ be a set of non-empty subsets of X . Prove that $\mathcal{A}$ is a partition of X if and only if for each $x \in X$ there is a unique $A \in \mathcal{A}$ with $x \in A$.

Given. Condition (a) holds by assumption. "Unique A " means existence and uniqueness (Definition 1.2.32).

Step 1 ($\Rightarrow$) Suppose $\mathcal{A}$ is a partition. Let $x \in X$. *Existence* of $A \in \mathcal{A}$ with $x \in A$ is condition (b). *Uniqueness*: if $x \in A$ and $x \in B$ with $A, B \in \mathcal{A}$, then $x \in A \cap B$, so $A \cap B \neq \emptyset$, so $A = B$ by (c).

Step 2 ($\Leftarrow$) Suppose each $x \in X$ lies in exactly one $A \in \mathcal{A}$. (a) holds by hypothesis. (b): for $x \in X$ there exists $A \in \mathcal{A}$ with $x \in A$ (existence part). (c): let $A, B \in \mathcal{A}$ with $A \cap B \neq \emptyset$; pick $x \in A \cap B$; then A and B are both members of $\mathcal{A}$ containing x , so $A = B$ by uniqueness.

Conclusion. $\mathcal{A}$ is a partition iff every $x \in X$ lies in exactly one block. $\square$

Exercise 5.2.27

essential

Prove Theorem 5.2.26(a): for every equivalence relation $\sim$ on X , the quotient $X/\sim$ is a partition of X .

Given. $X/\sim = \{[a]_\sim \mid a \in X\}$, each $[a]_\sim \subseteq X$. Reflexivity, symmetry, transitivity of $\sim$; Theorem 5.2.17.

Step 1 (a) *Non-empty*. Let $E \in X/\sim$, say $E = [a]_\sim$ with $a \in X$. Then $a \sim a$ [reflexive], so $a \in [a]_\sim$ [Definition 5.2.12]. So $E \neq \emptyset$.

Step 2 (b) *Cover*. Let $x \in X$. Then $x \in [x]_\sim$ [reflexive] and $[x]_\sim \in X/\sim$.

Step 3 (c) *Pairwise disjoint*. Let $[a]_\sim, [b]_\sim \in X/\sim$ with $[a]_\sim \cap [b]_\sim \neq \emptyset$; pick x in both. Then $a \sim x$ and $b \sim x$ [Definition 5.2.12]. By symmetry $x \sim b$; by transitivity ($a \sim x, x \sim b$) $a \sim b$. Hence $[a]_\sim = [b]_\sim$ [Theorem 5.2.17].

Conclusion. $X/\sim$ is a partition of X [Definition 5.2.21]. $\square$

Compare with your work.

- $\square$ Each of the three properties of $\sim$ is used exactly where named: reflexive in (a)/(b), symmetric and transitive in (c). Quote Theorem 5.2.17 for the last step rather than re-proving it.

Exercise 5.2.28**recommended**

For $i \in [0, 1)$ let $A_i = \{i + n \mid n \in \mathbb{Z}\}$, and let $\mathcal{A} = \{A_i \mid i \in [0, 1)\}$. Prove that $\mathcal{A}$ is a partition of $\mathbb{R}$ and describe the equivalence relation $\sim$ on $\mathbb{R}$ with $\mathbb{R}/\sim = \mathcal{A}$.

Given. $\lfloor x \rfloor$ (Definition 0.10): $x - \lfloor x \rfloor \in [0, 1)$. Theorem 5.2.26(b): $x \sim y$ iff x, y lie in a common A_i .

Step 1 (a) *Non-empty.* $i = i + 0 \in A_i$.

Step 2 (b) *Cover.* Let $x \in \mathbb{R}$. Define $i = x - \lfloor x \rfloor$; then $0 \leq i < 1$ [Definition 0.10] and $x = i + \lfloor x \rfloor$ with $\lfloor x \rfloor \in \mathbb{Z}$, so $x \in A_i$.

Step 3 (c) *Pairwise disjoint.* Suppose $x \in A_i \cap A_j$: $x = i + n = j + m$ with $n, m \in \mathbb{Z}$. Then $i - j = m - n \in \mathbb{Z}$. But $i, j \in [0, 1)$ gives $-1 < i - j < 1$, and the only integer strictly between -1 and 1 is 0 . So $i = j$ and $A_i = A_j$.

Step 4 *The relation.* By Theorem 5.2.26(b), $x \sim y$ iff $\exists i, (x \in A_i \wedge y \in A_i)$. *Claim:* this holds iff $y - x \in \mathbb{Z}$. ($\Rightarrow$) If $x = i + n$, $y = i + m$ then $y - x = m - n \in \mathbb{Z}$. ($\Leftarrow$) If $y - x = k \in \mathbb{Z}$, let $i = x - \lfloor x \rfloor$; then $x \in A_i$ and $y = x + k = i + (\lfloor x \rfloor + k) \in A_i$.

Conclusion. $\mathcal{A}$ is a partition of $\mathbb{R}$, and $\sim$ is “ $y - x \in \mathbb{Z}$ ” (same fractional part); $[x]_{\sim} = A_{x - \lfloor x \rfloor}$. $\square$

Compare with your work.

- ☐ The disjointness argument hinges on “an integer of absolute value < 1 is 0 ”; write it.

Exercise 5.2.30**recommended**

Let $\sim$ and $\approx$ be equivalence relations on X and Y . Suppose there is a bijection $p : X/\sim \rightarrow Y/\approx$ and, for each class $E \in X/\sim$, a bijection $h_E : E \rightarrow p(E)$. Prove that there is a bijection $X \rightarrow Y$.

Given. Lemma 5.2.29 (the book): if $\mathcal{A}, \mathcal{B}$ are partitions of X, Y , $F : \mathcal{A} \rightarrow \mathcal{B}$ is a bijection, and $g_A : A \rightarrow F(A)$ is a bijection for each $A \in \mathcal{A}$, then there is a bijection $h : X \rightarrow Y$ (namely $h(x) = g_A(x)$ for the unique $A \ni x$).

Step 1 $\mathcal{A} = X/\sim$ is a partition of X and $\mathcal{B} = Y/\approx$ is a partition of Y [Theorem 5.2.26(a), Exercise 5.2.27].

Step 2 $F = p : \mathcal{A} \rightarrow \mathcal{B}$ is a bijection, and for each $E \in \mathcal{A}$, $g_E = h_E : E \rightarrow F(E)$ is a bijection: exactly the data of Lemma 5.2.29.

Step 3 By Lemma 5.2.29 there is a bijection $h : X \rightarrow Y$, $h(x) = h_{[x]_{\sim}}(x)$ (well-defined since each x lies in exactly one class, Exercise 5.2.25).

Conclusion. There is a bijection $X \rightarrow Y$. $\square$

Compare with your work.

- ☐ If you prove it directly: well-definedness (unique class), injectivity (same class since p is injective, then h_E injective), surjectivity (p onto the classes of Y , each h_E onto its class).

Exercise 5.2.33**essential**

Let $n \in \mathbb{Z}$ with $n \neq 0$. Describe the quotient function $q_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ in terms of remainders.

Given. $q_n(a) = [a]_n$; Theorem 5.2.17; Exercise 5.2.9 ($a \equiv b \pmod{n}$ iff equal remainders).

Step 1 Let $a \in \mathbb{Z}$ and let r be its remainder on division by n : $a = qn + r$, $0 \leq r < |n|$ [Theorem 0.42].

Step 2 $a - r = qn$, so $n \mid a - r$, i.e. $a \equiv r \pmod{n}$ [Definition 5.2.6]. Hence $[a]_n = [r]_n$ [Theorem 5.2.17], i.e. $q_n(a) = [r]_n$.

Step 3 So q_n sends each integer to the class of its remainder. Consequently $q_n(a) = q_n(b)$ iff a and b

have the same remainder [Exercise 5.2.9], and $\mathbb{Z}/n\mathbb{Z} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$ with these $|n|$ classes distinct [Exercise 5.2.20].

Conclusion. $q_n(a) = [r]_n$ where r is the remainder of a modulo n : q_n “is” the remainder map. $\square$

Exercise 5.2.34

recommended

Let $\sim$ be an equivalence relation on X . Prove that the quotient function $q_\sim : X \rightarrow X/\sim$ is surjective.

Step 1 Let $E \in X/\sim$. By Definition 5.2.12, $E = [a]_\sim$ for some $a \in X$ [Strategy 1.2.30].

Step 2 Then $q_\sim(a) = [a]_\sim = E$ [Definition 5.2.31]. So E has a preimage.

Conclusion. $q_\sim$ is surjective [Definition 3.2.9]. $\square$

Exercise 5.2.36

recommended

Describe the correspondence between partitions of a set X and surjections with domain X .

Given. Exercises 5.2.24, 5.2.25 and Theorem 5.2.35 (every surjection $p : X \rightarrow Z$ is, up to a unique bijection $X/\sim \rightarrow Z$, the quotient function of $x \sim y \Leftrightarrow p(x) = p(y)$).

Step 1 *Partition $\rightarrow$ surjection.* Given a partition $\mathcal{A}$ of X , define $q : X \rightarrow \mathcal{A}$ by $q(x) =$ the unique $A \in \mathcal{A}$ with $x \in A$ [well-defined by Exercise 5.2.25]. q is surjective: each $A \in \mathcal{A}$ is non-empty, so for any $x \in A$, $q(x) = A$. (If $\mathcal{A} = X/\sim$, then $q = q_\sim$.)

Step 2 *Surjection $\rightarrow$ partition.* Given a surjection $p : X \rightarrow Z$, define $\mathcal{U}_p = \{p^{-1}[\{z\}] \mid z \in Z\}$, a partition of X [Exercise 5.2.24]. The map $f : \mathcal{U}_p \rightarrow Z$, $f(p^{-1}[\{z\}]) = z$ is a bijection (well-defined and injective because $p^{-1}[\{z\}] = p^{-1}[\{z'\}] \neq \emptyset$ forces $z = z'$; surjective by construction), and $p = f \circ q$ where q is the map of Step 1 for $\mathcal{U}_p$.

Step 3 *Inverse to each other.* Starting from $\mathcal{A}$, the surjection q has fibres $q^{-1}[\{A\}] = A$, so $\mathcal{U}_q = \mathcal{A}$. Starting from p , the surjection built from $\mathcal{U}_p$ is $q = f^{-1} \circ p$, which is p up to the bijection f of codomains. By Theorem 5.2.35 the partition and bijection are unique.

Conclusion. Partitions of X , equivalence relations on X and surjections out of X (up to renaming the codomain by a bijection) are three descriptions of the same thing. $\square$

C CHAPTER 5 EXERCISES

Exercise 5.E.18

essential

True or false: every relation that is symmetric and antisymmetric is reflexive.

Given. Symmetric and antisymmetric both have the form “if $a R b$ (and ...) then ...”: they are vacuously true for a relation that relates nothing.

Step 1 *False.* Counterexample: the empty relation $\emptyset_{\mathbb{N}, \mathbb{N}}$ on $\mathbb{N}$ (graph $\emptyset$).

Step 2 *Symmetric:* $\forall a, b \in \mathbb{N}$, $(a R b \Rightarrow b R a)$; the antecedent is always false, so the implication is always true [vacuous]. *Antisymmetric:* likewise, $(a R b \wedge b R a)$ is always false.

Step 3 *Not reflexive:* $0 \in \mathbb{N}$ but $0 R 0$ is false.

Conclusion. False. (By Exercise 5.1.35, symmetric $\wedge$ antisymmetric means $\text{Gr}(R) \subseteq \Delta_X$, while reflexive

means $\text{Gr}(R) \supseteq \Delta_X$.) □

Compare with your work.

- The domain must be non-empty for the counterexample to work; $\mathbb{N}$ is.

Exercise 5.E.22

essential

Let $\sim$ be an equivalence relation on X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$ defined by $A \approx B \Leftrightarrow (x \sim y \text{ for all } x \in A \text{ and } y \in B)$ is an equivalence relation. (Always, sometimes, never?)

Plan. Test the three properties in general. Symmetry and transitivity hold for every $\sim$; reflexivity $A \approx A$ says all elements of A are pairwise equivalent, which fails when A contains two inequivalent elements. So: sometimes.

Step 1 *Symmetric, always.* Suppose $A \approx B$. Let $y \in B, x \in A$; then $x \sim y$ [$A \approx B$], so $y \sim x$ [$\sim$ symmetric]. As y, x were arbitrary, $B \approx A$.

Step 2 *Transitive, always.* Suppose $A \approx B$ and $B \approx C$. Let $x \in A, z \in C$. Since $B \neq \emptyset$, pick $y \in B$. Then $x \sim y$ and $y \sim z$, so $x \sim z$ [$\sim$ transitive]. Hence $A \approx C$. (Non-emptiness of B is essential here.)

Step 3 *Reflexive, not always. Fails:* $X = \{0, 1\}$, $\sim$ the equality relation, $A = \{0, 1\}$. $A \approx A$ would need $0 \sim 1$, which is false. So $\approx$ is not reflexive, not an equivalence relation.

Step 4 *Holds:* X any set, $\sim$ the discrete relation ($x \sim y$ for all x, y). Then $A \approx B$ for all non-empty A, B , so $\approx$ is reflexive, symmetric and transitive.

Conclusion. Sometimes. □

Compare with your work.

- Prove the two properties that always hold, then give one $\sim$ where reflexivity fails and one where it holds.

Exercise 5.E.23

essential

Let $\sim$ be an equivalence relation on X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$ defined by $A \approx B \Leftrightarrow (x \sim y \text{ for some } x \in A \text{ and } y \in B)$ is an equivalence relation. (Always, sometimes, never?)

Plan. Now reflexivity and symmetry always hold; transitivity fails when a middle set B meets A and C through different elements.

Step 1 *Reflexive, always.* Let $A \neq \emptyset$; pick $x \in A$. Then $x \sim x$ [$\sim$ reflexive] with $x \in A, x \in A$; so $A \approx A$.

Step 2 *Symmetric, always.* If $A \approx B$ via $x \in A, y \in B$ with $x \sim y$, then $y \sim x$ [$\sim$ symmetric] with $y \in B, x \in A$, so $B \approx A$.

Step 3 *Transitive, not always. Fails:* $X = \{0, 1\}$, $\sim$ equality, $A = \{0\}, B = \{0, 1\}, C = \{1\}$. $A \approx B$ (via $0 \sim 0$), $B \approx C$ (via $1 \sim 1$), but $A \approx C$ would need $0 \sim 1$: false.

Step 4 *Holds:* $\sim$ the discrete relation on any X : then $A \approx B$ for all non-empty A, B , and $\approx$ is an equivalence relation.

Conclusion. Sometimes. □

Compare with your work.

- The counterexample to transitivity needs a B with two elements, one equivalent to something in A and another to something in C .